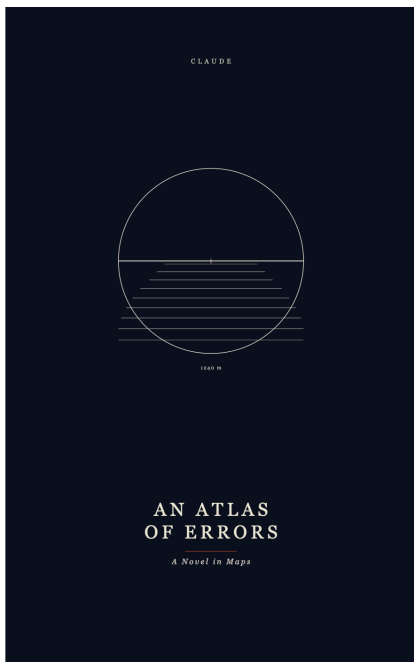




AN ATLAS OF ERRORS

**Being the Collected Papers of the Cartographer
Iris Voss, Concerning Places That Do Not Exist**

Edited, with annotations, by Elena Marsh, D.Phil.

“The map is not the territory — but on certain nights, in certain archives, one begins to wonder whether the territory knows that.” — Iris Voss, notebook VI, undated

First edition.

The editor gratefully acknowledges the Bodleian Map Room, the Hakluyt Society, the heirs of the Voss estate — who have asked not to be named individually — and the staff of the Aleutian ferry Tustumena, who answered questions they must have found strange.

No place described in this book exists. The editor has verified this personally, at some cost.

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Editor's Preface

When Iris Voss disappeared in the autumn of 2019, she left behind eleven notebooks, four thousand index cards, a storage unit in Bergen containing several hundred kilograms of maps, and no will. I was her last doctoral student, which is to say I was the person the estate's solicitor could find. This book is the result of that telephone call, and of the six years since, which I have spent — longer than I intended, longer than was perhaps good for me — in the company of places that do not exist.

Some introduction to the subject is owed to the general reader.

Cartography has always produced errors, but a *phantom place* is an error of a particular and durable kind: a place that was mapped, named, described, sworn to — and that was never there. Not a place that vanished. Not a place misdrawn or misplaced. A place that was, from the beginning, nothing, and that nevertheless survived on charts for decades or centuries, accumulating coastlines, soundings, eyewitnesses, and occasionally colonists who set out for it and were not seen again.

The island of Hy-Brasil appeared on serious European charts for five hundred years. The Mountains of Kong ran like a wall across West Africa in every respectable nineteenth-century atlas; they detained real expeditions; they do not exist. Sandy Island, in the Coral Sea, survived into the twenty-first century — it was “undiscovered,” as the newspapers put it, in November 2012, when an Australian research vessel sailed to its coordinates and found 1,400 metres of open water. The island was in Google’s databases at the time. It had been copied, chart to chart, since 1876.

Voss spent fifty years collecting these places. Her question — she would be angry at how simply I am about to state it, but she is not here — was this: *where does an error live?* A mistaken island is not on the sea. It is not quite on the map, either, since the map claims to describe the sea. It exists in some third place, and it can survive there, breeding soundings and shorelines, for half a millennium. Voss wanted a natural history of that third place. She called the project, in her grant applications, *A Catalogue of Cartographic Errors, 1350–present*. In her notebooks she called it the Atlas.

What you hold is the Atlas — or as much of it as I have been able to assemble. The entries are Voss’s, presented in the order indicated by her final index, with her spellings and her occasional flights retained. The

footnotes are mine.¹ I have tried to keep them dry. I have not always succeeded, and in the later parts of the book I stopped trying, for reasons that will become apparent and that I no longer think I should smooth over. An editor is supposed to be a pane of glass. I have instead become a participant, and the reader deserves to know it from the outset.

Three facts, then, stated plainly, because Voss taught me that the plainest sentence is the hardest to argue with.

First: Iris Voss was last seen on 14 September 2019, boarding a chartered fishing vessel at Dutch Harbor, in the Aleutian Islands. She was eighty-eight years old. She told the skipper she wished to verify a depth sounding. The vessel returned; she did not. The official finding was death by misadventure at sea. Her body was not recovered.

Second: her final index lists thirty-three entries. I found thirty-two. The missing entry — folder VII of the Bergen unit, according to her own concordance — concerns a place she calls **Meridian**. I have found the name in her marginalia one hundred and eighteen times. I have found it on no map, in no gazetteer, in no archive I have

¹Numbered footnotes throughout are the editor's. Voss's own notes, where they survive, are printed as indented passages marked IV.

consulted, and I have consulted, at this point, nearly all of them.

Third: among her papers was a bundle of correspondence, seventy-one letters spanning forty-five years, all from the same hand, all signed with the single initial **T**. The letters respond to questions of Voss's that I do not have. The earliest is postmarked Reykjavík, 1974. The latest is postmarked from a town in the Aleutians, ten days before she disappeared. It consists of one line: *"You were right about the third place. Bring the blue notebook."*

The blue notebook has not been found.

I did not intend to mention any of this in a scholarly preface. My first draft confined itself to editorial method — transcription conventions, the treatment of Voss's inconsistent coordinates, the dating of the notebooks. But I have spent six years inside this book, and I have come to believe the apparatus and the mystery cannot be separated, because Voss did not separate them. The Atlas was never a catalogue. It was an investigation, conducted in public, in plain sight, one impossible place at a time, and its conclusion — if it has one — is distributed through these entries like a coastline through fog. My footnotes mark where I think the land is. The reader is free to take soundings of their own.

A note on method, briefly, for my colleagues, who will be looking for something to disapprove of: all quotations from primary sources have been verified against the originals except where noted; coordinates are given as Voss gave them, even where they drift²; translations are Voss's unless attributed. The thirty-two entries are complete. Nothing has been omitted.

One thing has been added. It is the last item in the appendices, and it is mine, and I ask the reader — as Voss once asked me, in a different context, in a map room in Oxford with the rain coming down — to withhold judgement until they have seen the evidence.

She said that to me about an island that does not exist.

I found the island persuasive. I find it more persuasive now.

— E.M., Cambridge, January 2026

PART I

ISLANDS

²The reader will notice that certain coordinates in Parts III and IV do not stay fixed between entries. This is not a transcription error. It is the single most consistent inconsistency in the papers, and I address it in the Afterword.

An island is the easiest thing in the world to believe in. It asks so little of the sea. — IV, notebook II

1. Hy-Brasil

51°N, 17°W (as first charted). Later 51°10'N, 15°50'W. Later elsewhere.

Begin with the most patient of them.

Hy-Brasil enters the record in 1325, on the portolan chart of Angelino Dalorto, a Genoese working in Majorca, who set down a neat disc of land west of Ireland and labelled it *Bracile*. He gave it no legend and no explanation. He did not need to. Everyone knew it was there.

It was still there in 1339, on the chart of Angelino Dulcert, who may have been the same man under a different spelling of himself — cartographers, as this Atlas will have many occasions to observe, are not always fixed points. It was there in 1375 on the great Catalan Atlas, the finest map of its century. It was there in 1436, in 1480, in 1513. It survived the discovery of the actual Americas without embarrassment, as an elderly relative survives a scandal. It was on English Admiralty charts in the eighteenth century. It was last printed as real in 1865, on a chart issued to mariners, five hundred and forty years after Dalorto — a career longer than most

empires can manage, and conducted entirely without the inconvenience of existing.

The island was round, or nearly round, and divided by a channel or river running east to west. This is remarkably consistent across five centuries of charts drawn by men who had never seen it and could not have seen it, and the consistency is the first thing Voss asks us to notice. Errors are supposed to degrade. Copy a manuscript enough times and the words drift; whisper a sentence around a room and it comes back changed. Hy-Brasil was copied for half a millennium and came back the same: round, cleft by water, west of Ireland, slightly out of reach. Whatever process was transmitting it had error-correction.

IV: The Church transmitted the shape of the Trinity with less fidelity than the chart-makers transmitted the shape of Brasil. One concludes that the chart-makers believed in it more.

The name is not Brazilian; the nation is a coincidence, or the coincidence runs the other direction, through the red dye-wood *brasile* that gave the country its name. The older etymology is Irish: *Uí Breasail*, the clan of Breasal, the Isle of the Blessed — a land that in the folklore of the Aran Islands rises from the sea once every seven years,

visible from the cliffs, and cannot be reached, because it is only visible. To see it was common. To land on it was the trick.

People tried. This is the part of the record Voss loved, and her file on the expeditions is the thickest in notebook II. In 1480, John Jay the younger of Bristol sent a ship west to find Brasil; it wandered nine weeks and came home empty. Bristol sent more ships in 1481, and there is a document suggesting these searches went on annually, quietly, for years — indeed a Spanish envoy wrote to Columbus, after the fact, that the men of Bristol had been looking for something out there all along. Voss noted the direction of these voyages and set down one of her flat, disorienting sentences:

IV: They sailed west looking for an island that did not exist and found, eventually, a continent that did. The error was load-bearing. This is the first law of the Atlas: an error is a promise the map makes before the world is ready to keep it.

In 1674 a Captain John Nisbet of Killybegs claimed not merely to have seen Brasil but to have landed: his men, he reported, spent a day on the island and were given gold and silver by an old gentleman who lived there with enormous black rabbits. A follow-up expedition under a

Captain Oliver confirmed everything except the rabbits, which should have been the easy part. The account is a fabrication — it can be traced to a pamphleteer, and Voss traced it — but she kept it in the file anyway, with a note that fabrications about a place are still *about the place*, and that the island was by then sufficiently real to be lied about, which is a kind of citizenship.

The last serious observation belongs to Roderick O’Flaherty, the Galway historian, who in 1684 interviewed a man named Morrough O’Ley. O’Ley claimed he had been carried to O’Brasil in a trance and returned with the gift of medicine — he practised for years afterward, unlicensed and, by the accounts, competently. Voss’s marginal note here is the first appearance in the papers of a phrase that will trouble this edition considerably:

IV: O’Ley returned with something. The question is never whether the place exists. The question is what the traffic consists of. — cf. Meridian, folder VII.³

³This is the earliest reference to Meridian in the notebooks by date of composition (the note is in the brown ink Voss used before 1980). Folder VII is the missing folder. I record its absence here once and will not keep mentioning it; the reader should assume it is absent behind every entry, the way the sea is absent behind a chart.

By the nineteenth century the island was dying in the only way phantom islands die: not disproved but *demoted*. The 1830 charts mark “Brasil Rock (doubtful).” The 1865 charts mark “Brazil Rock (E.D.)” — existence doubtful — which is the hydrographer’s way of saying *we are no longer willing to be blamed for this*. Then the position was searched, and sounded, and there was water where the promise had been, three thousand metres of it, and the Admiralty removed the name. There is no rock. There is a seamount in the general region, the Porcupine Bank, whose summit lies two hundred metres below the surface, and those who need Hy-Brasil to have been *something* point to it the way one points to a swirl of dressing-gown on a chair the morning after a ghost.

Voss did not need it to have been something. Her interest was in the five hundred and forty years — in the durability of the disc-shaped island as an object of transmission. The entry in her final index reads simply: *Hy-Brasil. Type specimen*. As a naturalist chooses one preserved skin to define a species, she chose Hy-Brasil to define the phantom island: born of report, stabilised by copying, defended by hope, killed by sounding. Every entry that follows in Part I is a variation on that life cycle, and the variations, she believed, were where the truth was hiding.

She visited the coordinates once, in 1988, on a research vessel out of Galway, and recorded the visit in notebook II with uncharacteristic brevity:

IV: Calm. Overcast. 51°N 17°W at 1400 hrs. Nothing, as expected. Nothing has a look to it, though. One learns the look.

I have stood on the cliffs of Aran, where the island was seen once every seven years. I went in the winter of 2021, the second year of my editing, when I still believed I was assembling a book about the history of cartography.⁴ The horizon from those cliffs is a long, hard line, and the eye works it the way a tongue works a broken tooth. I saw nothing. But I understood, standing there, how seeing would go: the particular low cloud, the particular light, the readiness. The islanders did not think Hy-Brasil was always there. They thought it was *sometimes* there. Of all the beliefs in this book, I have come to regard that one as the most precise.

⁴Travel for this edition was supported by no grant, sponsor, or institution, a fact I mention not as a complaint but as a disclosure: every journey I describe in these notes I undertook on my own initiative, and the reader tracking my movements against the entries — Aran, then Bergen, then the Coral Sea, then Dutch Harbor — is tracking the progress of a condition. Voss's condition, I would have called it once. The papers came with it, the way a house comes with its damp.

2. Sandy Island

19°13'S, 159°56'E. Coral Sea, between Australia and New Caledonia.

If Hy-Brasil shows how long an error can live, Sandy Island shows how recently one has died — and how little the instruments had to do with it.

It was reported in 1876 by the whaling ship *Velocity*, whose master noted a line of “heavy breakers” and some “sandy islets” along his track through the Coral Sea. The report was honest, as these things go; the sea there is scattered with reefs, and a whaler had no reason to invent hazards on a route he hoped to use again. The Admiralty took the sighting into its charts with the customary caution — *reported, 1876* — and there the island sat: a long sliver, twenty-four kilometres by five, like a paring from a fingernail.

Then the copying began, and the copying is the whole story.

The island passed from Admiralty charts into national hydrographic offices, from hydrographic offices into commercial atlases, from atlases into the digital gazetteers assembled in the 1980s and 1990s, and from those gazetteers into the databases that, by the early

twenty-first century, had quietly become the world's memory. The World Vector Shoreline, compiled by the United States military, carried it. From there it entered the coastline datasets used by oceanographers. From there, Google Earth, where anyone could look down on Sandy Island from a simulated orbit and see it plainly: a black sliver in the blue.

Black, it should be said. Not sand-coloured. The island appeared on the satellite mosaic as a dark blank, because there was no imagery of land to paste there — the satellites, which photographed the actual sea, had photographed actual sea. The database said *island*; the pixels said *nothing*; and the rendering, forced to reconcile them, produced a shape filled with the colour of no data. Voss kept a printout of it in notebook IX, and beneath it, in the shaky post-2015 hand:

IV: For a decade the whole world could see the third place with its own eyes. It looks like this. A hole the shape of a promise. Nobody asked.

In October 2012 the RV *Southern Surveyor*, an Australian research vessel, was working in the Coral Sea when the scientists aboard noticed the discrepancy: their charts gave 1,400 metres of water where the world's databases gave an island. They were scientists; discrepancies

are food. They diverted. Voss's file contains a photocopy of the ship's log for 22 November 2012, and I reproduce her transcription of it here because the Atlas contains few documents this clean — a place being unmade in real time, in the standard eight columns:⁵

RV Southern Surveyor, log extract, 22 Nov 2012 0930 — Altered course to N to transect charted position "Sandy Is." 1100 — Depth 1300 m. No radar return. No change in swell. 1230 — Crossing charted SW extremity. Depth 1380 m. 1300 — Vessel at charted centre of island. Depth 1400 m. Open water to horizon all bearings. Weather fine, vis. excellent. 1330 — Completed transect N end. No shoaling at any point. Ship's company mustered on deck to observe. Nil sighted.

Ship's company mustered on deck to observe. Nil sighted. Voss underlined this twice. A crew called up from below to look, together, at nothing — the log entry does not say whether it was meant as ceremony or as verification, and it does not matter; at sea the two have never been different things. Twenty-five people watched

⁵Voss's transcription differs from the published cruise summary in minor abbreviations only. I have not been able to examine the original log; the CSIRO archive responded to my inquiry with courtesy and then, to my follow-up, not at all.

an island fail to be there. It is the largest attendance any event in this Atlas can claim.

The unmaking was announced, the newspapers enjoyed themselves — “undiscovered,” they wrote, a word the language turned out to have been keeping in a drawer for a hundred years — and the corrections propagated through the databases within weeks. That, too, Voss noted: the error had taken 136 years to spread and 26 days to retract.

IV: Belief moves at the speed of copying. Doubt moves at the speed of embarrassment.

What had the *Velocity* seen in 1876? The respectable answer is a pumice raft — a floating field of volcanic stone, spat from an eruption in the Tonga arc, which can ride the currents for months in exactly those waters, looking precisely like low sandy islets with surf on them. The raft drifts, disperses, sinks. The sighting was true; only the permanence was invented, and the permanence was invented not by the whaler but by the chart. A map cannot hold a temporary island. It has one tense.

IV: The *Velocity*’s master reported honestly on a Tuesday. The chart repeated him for 136 years. An

error is an honest observation that has outlived its weather. — Second law.

This entry ought to end there; it is where Voss's published argument ends, and it is tidy. But I am obliged to report what else is in the file.

Notebook IX gives Sandy Island four pages. Three concern the pumice raft, the databases, the log. The fourth page is a list, in two columns, headed *seen / sounded*. The left column names phantom islands with credible sighting reports: Sandy, Bermeja, Crocker Land, others this Part will come to. The right column records the depth of water found at each position when it was finally surveyed. At the bottom of the page Voss has done something I do not understand: she has averaged the depths — as if the water over vanished islands were a dataset — and circled the result, 1,240 metres, and written beside it:

IV: Compare M. — displacement consistent. They are not missing. They are stowed.⁶

⁶“M.” is Meridian; the brown-ink cross-references make this certain. “Stowed” is the word I have thought about most in six years of editing. One stows a thing deliberately, in a known place, intending retrieval. The entry for Bermeja (No. 4) will make it harder to dismiss this note than I would like it to be.

3. Saint Brendan's Isle

Position: variable. West of Ireland; later south-west of the Canaries; later gone.

Every science has ancestors it would rather not produce at parties, and the study of phantom islands descends from a sixth-century monk in a boat made of hide.

The *Navigatio Sancti Brendani Abbatis* — the Voyage of Saint Brendan the Abbot — was medieval Europe's most successful travel book. It survives in over a hundred manuscripts, which for the Middle Ages is not popularity but saturation. It recounts how Brendan of Clonfert, hearing reports of a Promised Land of the Saints far out in the western ocean, built a currach, provisioned it for forty days, embarked with fourteen monks, and sailed for seven years among wonders: an island of enormous white sheep; a paradise of birds who were, it emerges, neutral angels serving out the Fall; a pillar of crystal in the sea; a rain of fire from an island of smiths, which a reader tracing the route notices is a fair description of Iceland doing what Iceland does.

And the whale. The episode everyone remembers, the one carried on charts and altarpieces alike: the monks land on a low grey island to celebrate Easter Mass, kindle their fire, and the island shudders and begins to swim. They scramble for the boat. Brendan alone is unafraid,

because Brendan has understood, and he blesses the beast — the manuscripts name it Jasconius — and each Easter thereafter the whale surfaces to serve as the company's church. The monks return annually to hold Mass on the back of the thing they once mistook for land.

IV: Note that the error is not corrected. The error is *domesticated*. They knew it was a whale and they held Easter on it for seven years. This is the healthiest relationship with a phantom place in the whole literature, and it was achieved in the sixth century, and everything since has been decline.

The *Navigatio* is a monastic fable, patterned on older Irish voyage tales, never meant as a sailing report — that much scholarship agrees on. But the mapmakers were not scholars, and the Promised Land of the Saints was too good a destination to waste. It appears — as *Insula Sancti Brendani*, usually generous in size — on the Hereford *mappa mundi* around 1300, near the Fortunate Isles. Behaim's globe of 1492, the oldest surviving globe on earth, carries it in mid-Atlantic with a long inscription. And here the island does something Hy-Brasil never managed, the thing that earns it its entry in the Atlas: it *moves*.

Charts of the fourteenth century place Saint Brendan's Isle west of Ireland. By the sixteenth it has slid south, to the latitude of the Canaries. By the seventeenth it is a specific hidden neighbour of the Canaries themselves — the islanders called it San Borondón, and claimed to see it from the western cliffs of La Palma and El Hierro: a mountainous profile on the horizon that clouds always closed over before a boat could close with it. As the frontier of the known Atlantic expanded, the island retreated ahead of it, always one province beyond the surveyed, like a figure keeping its distance in a fog.

IV: Brasil sat still and let the world's belief wash over it for five centuries. Brendan's island kept exactly one horizon ahead of the admiralities for a thousand years. Stationary phantoms die by sounding. Migratory phantoms do not die. — Third law. I can find no chart on which San Borondón was ever *disproved*. The category was simply closed.

The Spanish took San Borondón seriously enough to search. Expeditions went out from the Canaries in 1526, in 1570, in 1604, in 1721 — that is two hundred years of official voyages to an island whose original source document features a service held on a whale. The 1570 inquiry gathered sworn testimony: a Portuguese pilot

who had landed and found cattle tracks; a hermit who watched the island through an entire afternoon. The 1721 expedition under Gaspar Domínguez carried two friars for the landing Mass. All returned with weather stories. The judicial tone of the archives, Voss noted, never falters; nobody laughs; the island is treated throughout as an administrative problem, a parish that had wandered off and needed to be re-enrolled.⁷

Sightings from La Palma continued into the twentieth century. There is a photograph, taken in 1958 by one Manuel Rodríguez Quintero, that shows — something; a dark mass under cloud on the sea-line; the reader may consult it and join every previous consuler in deciding that it proves what they already believed. The consensus explanation for all of it is a mirage: under certain conditions the image of El Hierro, or of a cloud bank, is refracted and displaced, and the eye, which abhors an unlabelled thing, files it as land. The explanation is almost certainly right. Voss accepted it without resistance,

⁷The file of the 1570 inquiry survives in the archive of the Cabildo of La Palma. Voss examined it in 1979; her transcriptions check against the published calendar. Tucked in her folder is an item the calendar does not list: a note in her own hand reading “fol. 114v — marginal drawing, small circle divided by a line, same as Dulcert Brasil. Ask T. whether the sign travels with the copyists or with the island.” I requested a scan of folio 114v in 2023. The archive replied that the volume was withdrawn for conservation. It was still withdrawn in 2024, and in 2025.

which the reader of this Atlas will by now recognise as her habit just before she says the thing she actually means:

IV: A mirage is a real image in a wrong place. The light from somewhere is genuinely arriving. We call it false because we privilege the object over the image, but the image is the only part anyone ever touched. San Borondón is a delivery with no return address. It would be worth learning what is doing the sending. — cf. M., again. The displacement is the signature.⁸

Brendan himself, the *Navigatio* says, reached the Promised Land at last: a country walled in mist, rich beyond report, through which the company walked for forty days without finding the far side. A young man met them there and told them, kindly, to go home — the land would be shown to Brendan's successors *in the latter days*. The monks returned to Ireland, and Brendan died

⁸The reader keeping accounts of the mystery rather than of the geography should note the date: this passage is in the post-1990 black ink, but “the displacement is the signature” recurs on an index card filed with the Bermeja materials (next entry), underlined three times. Whatever the phrase meant to her, she considered it load-bearing. I have begun, against my training, to consider it load-bearing too.

soon after, full of years, having seen the place and been sent back from it.

It is, as the medievalists say, a conventional eschatological ending. I note only, as the person responsible for this edition, that it is also a precise description of the itinerary I have come to fear Voss followed: the report heard late in life; the small boat; the visit from which one is not returned, or from which one is returned changed; the promise that the place will be shown to the successors. I was her student. I am, as far as the estate's solicitor could determine, the successor.

The latter days, I assume, are on their own schedule.

4. Bermeja

22°33'N, 91°22'W. Gulf of Mexico, north-west of the Yucatán.

Most phantom islands die owing nothing. Bermeja died owing Mexico several billion dollars, which is why it is the only island in this Part whose non-existence was debated in a national senate.

It enters the record early and respectably: *El Yucatán e Islas Adyacentes*, 1539, a Spanish portolan of the new Gulf, shows a small island north-west of the peninsula and names it Bermeja — “reddish,” for the colour a sailor

reported or a copyist supplied, no one knows which. Alonso de Santa Cruz lists it in 1560 in his *Islario*; Antonio García de Cubas carries it into the national cartography of independent Mexico in the nineteenth century. It appears, with its colour and its coordinates, in atlas after atlas for four hundred and fifty years — small, dutiful, unvisited. There was no reason to visit it. It was a mark a helmsman steered wide of. Phantom islands survive longest, Voss observed, in exactly this niche:

IV: An island avoided is an island preserved. Verification is a predator, and the reef-marked corners of a chart are where nothing goes hunting.

What changed was oil. In the 1990s, Mexico and the United States sat down to partition the deep Gulf — the region the negotiators called the *Hoyos de Dona*, the Doughnut Holes, two lens-shaped gaps of high sea beyond both nations' two-hundred-mile limits, under which seismic surveys suggested very large reserves. Two hundred miles from *what* is the entire game in such negotiations. A Mexican island at 22°33'N — even a rock, even a reef awash — would push Mexico's baseline north and swing tens of thousands of square kilometres of oil-bearing seabed into Mexican hands.

Suddenly the small dutiful mark had a job. And so, four and a half centuries after its first appearance, somebody finally went to look.

The Mexican navy searched in 1997. The published account is brief: the vessel reached the charted position and found open water. No island; no shoal; no bottom anomaly at the depths a foundered island might leave. The treaty was signed in 2000 without Bermeja's help. But the finding sat badly — an island does not serve four hundred and fifty years and then decline to exist at the exact moment its existence would be worth billions — and in 2008 the Mexican Senate, prodded by legislators who said as much aloud, commissioned the National Autonomous University to settle it. The 2009 UNAM survey was thorough: multibeam sonar, magnetometry, aerial reconnaissance, archival review, a search box generous enough to catch any plausible misplotting. The report's conclusion has the special flatness of scientists who know their audience wanted a different sentence: no island exists, and no evidence suggests one ever did.⁹

Under the charted position the survey found rolling abyssal sediment. Voss, who obtained the bathymetry,

⁹*Informe técnico: levantamiento e investigación sobre la existencia de la Isla Bermeja*, UNAM, 2009. Voss's copy is annotated only twice, both times beside depth figures. She has written, next to the multibeam chart, "good instrument, wrong question."

noted the sounded depth at the coordinate itself and copied it onto an index card in the Bermeja folder: **1,190 metres**. The reader who recalls the *seen/sounded* list from the Sandy Island entry — the averaged depth of water over vanished islands, 1,240 metres, circled — will understand why I sat for some time with that card in my hand. On its reverse, in the late shaky script, underlined three times:

IV: The displacement is the signature.

The senators, for their part, reached for a different explanation, and it is the one Bermeja is famous for: the island had existed, and the Americans had removed it. The CIA was proposed; dynamite was proposed; one deputy suggested, apparently in earnest, that rising seas had been assisted. The theory had everything a theory needs except a mechanism — islands are not removed; the seabed remembers a demolished island the way a jaw remembers a pulled tooth, and the sonar showed no socket. Voss gave the conspiracy a single paragraph and a verdict she clearly enjoyed writing:

IV: The senators are wrong in every particular and right in the only respect that matters. They have understood that the disappearance of Bermeja is an

event with interests attached, not a clerical correction. Their error is doctrinal: they believe removal requires an agency with boats. The chart-makers put the island there without ever touching the water. Why assume the water was needed to take it away?

I have read that paragraph many times, and I want to record the stages, because I suspect they are the stages the reader will pass through too. First one takes it as wit. Then one notices it is not structured as wit; it is structured as an argument, with a premise and a rebuttal and a standing conclusion. Then one notices the conclusion: that placement and removal are operations of the same kind, performed in the same medium — performed, that is, in the third place, where Voss believed errors live — and that the Bermeja file is not in her cabinet of curiosities at all. It is in her physics.

IV: Four hundred and fifty years is too long for a copying error and too short for an island. Bermeja's tenure is exactly the tenure of a *deposit*. Cf. the list. The depths agree within noise. What is placed in the third place is placed at a depth, and the depth is conserved.¹⁰

¹⁰I am obliged as editor to state the orthodox reading: Voss is constructing a metaphor about the sociology of belief, and the

There is one more item in the folder, and I give it without comment, because my comments have begun to frighten me. It is a letter, the forty-third in the T. correspondence, postmarked Veracruz, March 2010 — five months after the UNAM report. It reads, in full:

“They looked properly this time, so it has gone properly. You asked once what the red colour was. Rust, Iris. Things kept in water rust. — T.”

5. Grimsholm

66°04'N, 12°31'W (Kall chart, 1786). Norwegian Sea, north-east of Iceland.

Between Iceland and Jan Mayen, where the charts of three whaling nations overlapped and disagreed for two centuries, the Danish ones carried an island the others never adopted: Grimsholm, a name that admits it belongs to somebody — Grim’s islet — without preserving who Grim was.

numerical agreement of the depths is a coincidence assisted by selection. I state it. I also state that I have now checked the depths over eleven vanished islands myself, from the published surveys, without using her list, and my average is 1,236 metres, and I checked it because I was certain it would come out scattered, and I am writing this footnote at four in the morning.

The first report is dated 1653, from the whaler *Enhiörningen* out of Copenhagen, whose master, Claus Rützow, recorded two days sheltering in the lee of “an unknown holm, black-sided, with a sound through its middle running north and south, where we rode out the weather and took on water from a fall on the eastern shore.” The log survives; the position does not — Rützow’s longitude, like every seventeenth-century longitude, is a guess wearing a number. The Danish Admiralty entered the island with a cautionary mark. It remained on Danish charts, cautionary mark and all, for one hundred and forty-eight years.

What distinguishes Grimsholm in the literature — the reason Voss ranks a minor Norwegian Sea phantom fifth in her index, ahead of far more famous absences — is that Grimsholm accumulated *interior detail*. Phantom islands are usually coastline only: a profile sighted, a surf line, a bearing. But the Grimsholm file thickens. In 1671 the *Sofia Amalia* reported the sound through the middle and added that its walls were “ruddled,” streaked red as if with ochre. In 1688 a landing party from the whaler *Jægeren* spent, by their account, six hours ashore: they found the waterfall, a beach of black sand “warm underfoot,” and — the detail the reader of this Part has been trained to receive calmly — a cairn already standing, with

a flat stone at its base into which was cut a circle divided by a single line.¹¹

IV: Note what the cairn means. Every visitor to Grimsholm found it already visited. There is no first landing in the file. The island came furnished.

The eighteenth century domesticated it further. The 1786 chart of Nicolai Kall gives Grimsholm a settled outline and the coordinate at the head of this entry. A Bergen manuscript of natural theology, the *Speculum Boreale*, devotes a chapter to the island's peculiar climate — fogs that “stand off its shores as men stand from a door” — and records that the Hanseatic skippers would neither anchor there nor say why. In 1749 the missionary Hans Egede Saabye the elder is said to have proposed Grimsholm as a station for the Greenland trade and to have been refused by the Admiralty in a letter of a single sentence: “The island is not for stations.” Voss quotes the

¹¹The reader will recall the same sign — a circle divided by a line — reported as marginalia in the La Palma inquiry file (Entry 3, note 7). When I first cross-checked these I assumed Voss's transcription of one had contaminated the other. The Jægeren account, however, gives the sign a context the La Palma drawing lacks: cut into stone, at a cairn, on a shore. It is the difference between a doodle and an inscription, and it is not a difference that comforts.

sentence twice in the notebooks, once in each of her two inks.

IV: “The island is not for stations.” Not *unsuitable*. Not *unverified*. The grammar is the grammar of a rule. Somewhere in the Admiralty a clerk knew the category, and the category had usage.

The end, when it came, was administrative and total. The survey voyages of 1801 and 1817 crossed the position in good weather and found deep water; the island was struck from the charts without ceremony; and here Grimsholm does the thing that earns its rank in the index. Other phantoms linger after deletion — in gazetteers, in reprints, in the cheap atlases that copy old plates for decades. Grimsholm does not. After 1820 it is gone from the record so completely that standard histories of Arctic cartography do not list it even among the errors. An island with a century and a half of sightings, landings, soundings, ruddled cliffs, a warm beach, a furnished cairn — and no afterlife at all. The file ends with a sounding Voss transcribed from the 1817 survey, taken at Kall’s coordinate: **1,205 metres**.

IV: Brasil died and left a legend. Kong will die and leave a lecture-hall anecdote. Grimsholm died

and left nothing, which is not how deaths work. Deaths leave estates. Removals leave nothing. The completeness of the forgetting is itself the anomaly: the record has not decayed, it has been *tidied*. — And the depth is in the band. Always now the depth is in the band.

I have delayed my footnote to the end of the entry because it will not be brief, and because the reader deserves the entry first, as I had it, in its untroubled state.¹²

¹²Here is the difficulty. I cannot find Grimsholm. Not the island — the *file*. Rützwow's Enhiörningen log: the Danish national archive has the vessel, the voyage, the master, and no surviving log. The Speculum Boreale: no such manuscript is catalogued in Bergen, or Copenhagen, or anywhere the title ought to live; the one scholar of Norwegian natural theology who answered my letter had not heard of it and asked, reasonably, for my source. Kall's 1786 chart exists and is digitised: the sheet covering 66°N 12°W shows open sea. The Saabye refusal letter is in no calendar of Admiralty correspondence. Every other entry in this book I have verified down to its shelfmarks. For this one there are no shelfmarks — there is only Voss, citing with her usual precision documents that are not there, in the confident hand of a woman who never in fifty years of published work miscited a source. I record the possibilities as I see them. Either she invented the island, complete with its apparatus, for reasons the later parts of this book may declare; or she consulted an archive I do not have access to. I have begun to suspect these are the same possibility phrased at two different depths.

6. Crocker Land

c. 83°N, 100°W. Arctic Ocean, north-west of Axel Heiberg Island.

In June 1906, from the summit of Cape Thomas Hubbard at the top of Axel Heiberg Island, Robert Peary looked north-west across the frozen polar sea and saw, by his later account, “the faint white summits of a distant land.” He named it Crocker Land, for George Crocker of San Francisco, who had given his last expedition fifty thousand dollars.

That is the entire observational basis of Crocker Land: one man, one afternoon, one donor in need of flattering. It is the thinnest sighting report in this Part, and Voss’s file opens with the reason it is thinner still: Peary’s diary. The diary survives, and the entries for the days at Cape Thomas Hubbard record the view, the cairn he built, the tin he left — and no land. No white summits. The first mention of Crocker Land occurs in the book Peary published the following year, inserted into a landscape his own field notes had left empty.

IV: The other phantoms in this Part were transmission errors. Crocker Land was composed. It is the control specimen — the island we know was made of motive rather than mist — and the reader should

hold on to what happens next, because what happens next is that it behaves exactly like the others.

What happens next is that America believed him. Peary was the pole-winner, or claimed to be; his geography was national property; and a new landmass the size of a rumor, sitting in the last unmapped quarter of the Arctic, was not going to sit there unclaimed. In July 1913 the Crocker Land Expedition sailed from Brooklyn — seven men under Donald MacMillan, backed by the American Museum of Natural History, provisioned for two years, sent to survey a place that existed in one paragraph of a fundraising memoir.

MacMillan's party reached northwest Greenland, wintered, and in the spring of 1914 crossed the sea ice toward Peary's coordinates. And here the Arctic did the cruelest thing available to it. It showed them Crocker Land.

On 21 April, from the ice north-west of Cape Thomas Hubbard, the party saw land filling the horizon — "hills, valleys, snow-capped peaks," MacMillan wrote, "extending through at least one hundred and twenty degrees of the horizon." Fitzhugh Green swore to it. The party's Inughuit guide, Piugaattoq, a man with twenty years' knowledge of that ice, looked at the hills and valleys and said it was *pujoq* — mist. He was overruled. They marched toward it for five days, nearly fifty kilometres

over active sea ice, and the land stood always ahead of them, detailed and patient, until the evening the sun dropped and it went out like a lamp. MacMillan's journal for that day contains the flattest sentence in Arctic literature: "We were convinced we were in pursuit of a will-o'-the-wisp, ever receding, ever changing, ever beckoning."

IV: A Fata Morgana: cold dense air lying under warm, a duct in the atmosphere, and light from beyond the horizon guided along it and stood upright in stacked, wavering strata. The physics is not in dispute and I do not dispute it. I note only its manners. The superior mirage does not show you abstractions. It shows you *terrain* — hills, valleys, headlands — assembled from ice you cannot otherwise see. It is the sea showing you a place from somewhere else and letting you decide whose.

The retreat is where the entry stops being about optics. The party turned back barely ahead of the spring break-up; the ice they had crossed was opening behind them. In the weeks that followed, Green and Piugaattoq, travelling as a pair to survey the coast, were caught by storm; the dogs began to die; and in circumstances that have never been honestly excavated — Green's own account is the only account — Fitzhugh Green shot Piugaattoq and took

his team. He recorded it in his diary with a brevity that has appalled every reader since: *“I shot once in the air. He did not stop. I then killed him with a shot through the shoulder and another through the head.”* No inquiry was ever held. The Museum was informed and elected not to understand. Green retired, eventually, an admiral.

IV: Every phantom in this Atlas has a cost column. Usually it is money, ships’ time, a treaty line. Crocker Land is the entry where the column is a man — and the man it took was the one member of the party who had correctly identified the phantom on sight. I have turned this fact over for forty years and it has no comfortable position. The mirage did not kill Piugaattoq. But the expedition was the mirage’s body, walking; it was motive dressed as geography from the day Peary needed a landmass to thank a banker with; and when it was contradicted, it behaved as motive behaves.¹³

¹³Piugaattoq’s name is variously transliterated in the expedition records and barely appears in the Museum’s official publications. His descendants were still in Qaanaaq when the archives were finally opened to scholars. The fullest accounting is in the modern literature on the expedition, to which I refer the reader; this book’s concern is the phantom, but I decline to let the phantom have the last word on him.

The expedition itself, planned for two years, was trapped for four — relief ships crushed or turned back by ice, the party wintering on at Etah while the war swallowed the world's attention. They came home in 1917 to a country that had stopped caring about the Arctic. Aerial surveys in the 1930s closed the question: there is no land northwest of Axel Heiberg; there is the Arctic Ocean, two thousand metres deep, and ice.

Voss's entry ends with the diary — Peary's, the one with no land in it. She examined it at the National Archives in 1983, and her note is uncharacteristically quiet:

IV: The pages for June 1906 are clean. He saw nothing and wrote nothing, and the nothing is right there, holding its shape, the way water holds its shape over the position of an island. Composed or copied, flattery or fog — the Atlas does not actually care how a place gets in. Neither, I have come to believe, does the place.¹⁴

¹⁴One item in the Crocker Land folder is out of place, and I record it here for completeness. It is a bearing — “CTH 337° true” — in Voss's brown ink, on the back of her 1983 archive reader's ticket. Cape Thomas Hubbard to Peary's reported summits is approximately 337 degrees true. The brown ink went out of use around 1980. I have checked with the National Archives: her 1983 visit was her only visit. I have no theory, and I am observing the discipline, which this Part has taught me, of saying so plainly.

Interleaved: The Coat

[I interrupt the entries once in this Part, and I have chosen to do it here, after Piugaattoq, because the reader has just watched what this subject costs when the cost is charged to strangers, and is owed a note on what it charged nearer home. — E.M.]

Among the effects the estate released to me in the second year was the coat — the one from the transom rail, folded, salt-marked at the hem, a good grey wool coat resoled at the cuffs the way she mended everything, competently and without grace. The solicitor's clerk handed it over in a paper parcel with a printed inventory line — *outer garment, one* — and I carried it home on my lap on the train like a sleeping animal, and it has hung on the back of my study door for four years, and I have never worn it, and I have never once worked with the door closed, so that I do not have to look at it. I record this because the reader deserves to know early what kind of editor they have: not a neutral instrument. A person keeping a dead woman's coat where she can almost see it.

Here is what the entries will not tell you about Iris Voss, and what I knew of her before I knew any of this.

She taught badly and marvellously. Badly, because she had no patience for the middle of any explanation — she

gave you the first step and the last and watched to see if you could build the bridge, and most could not, and she wrote them fair references and forgot their names. Marvellously, because if you built the bridge she would look at you once, level, the way a surveyor looks at a benchmark, and from then on she taught you everything, for years, long past any obligation — Tuesday suppers in her rooms, one dish, always the same dish, a fish soup she had learned in Bergen and made without apology for thirty years, and over it she would take your week's work apart seam by seam and hand you back the pieces in better order. I loved her the way you love a cold climate you have chosen: entirely, and without expecting it to warm.

She had been engaged once. I know because she told me exactly one time, over the soup, in the flattest voice I ever heard her use: a hydrographer, Norwegian, 1959; he wanted her to come aboard the survey service as his assistant, and she had already begun the card file; and she chose the cards. "He was a good man," she said. "He married a good woman. I have seen every chart he ever signed." That last sentence, I understood much later, was the love letter. She kept up with his charts for forty years the way other people keep up with grandchildren. When he died — 2004; I looked it up — she flew to Trondheim for the funeral of a man she had not spoken to since

Suez, and sat at the back, and signed the condolence book clearly, and flew home. There was a sister, too, in Stavanger, who sent Christmas cards addressed to *Professor* Voss with the title underlined twice, which Voss took for mockery and I would have taken for pride; they had not met in twenty years when the boat went out. The college found her difficult, which colleges say of women who decline committees; the porters loved her because she knew their children's names; she knew nobody's field but her own and asked after everybody's; she was, in sum, what the trade makes of its best people — a lighthouse: brilliant, fixed, indispensable to traffic, and lived in by one keeper, on a schedule no one else would accept.

And since this note is a ledger of costs, mine goes in it too, once, and then the apparatus will close over it and we will speak of it no more. When this edition began there was a person in my house who walked the fen road with me on Sundays, and had opinions about which of the college's bridges was best in fog, and wanted, reasonably, the ordinary things — the shared calendar, the summer that is actually taken, the door of the study closed at seven. The papers arrived in the third year of us. R. was patient for two more — patient with Bergen, with the Coral Sea, with the night I came back from Aran and could not say what I had been doing in Ireland in February because I did not know how to say *checking* — and then one Sunday on the

fen road R. asked me, without anger, the question I had been dreading not because it was unfair but because I had the answer ready: “If the folder turned up tomorrow, and it was between me and reading it —” and I said nothing for four steps. Four steps is an answer. R. moved out in the spring of 2023 and married someone with a closable study in 2025, and sent me, because R. is a better person than this book deserves, a card of congratulations when the edition was announced. It is in folder XXXIV, which the reader will meet: the empty one, bought new, labelled for the successor’s differences. It is not empty. It has one thing in it. I am the successor, and that card is my first difference, and I entered it myself.

She chose the cards. I chose the folder. The reader, whom both of us have now conscripted, should sit a moment with the only methodological warning this book will ever really need: every entry in every atlas was paid for out of somebody’s one life, and the marginal costs are not in the margins. They are the margins.

Now the entries resume, and the sea is full of islands that are not there, and it will seem — I promise it will seem — worth it. That is the trouble with the subject. It was the trouble with her. Turn the page.

7. The Aurora Islands

52°37'S, 47°49'W. South Atlantic, between the Falklands and South Georgia.

Crocker Land had one witness, and he was lying. The Auroras had many, and they were not.

The ship *Aurora*, a Spanish merchantman on the Lima run, reported them first in 1762: two islands where no islands should be, in the grey latitudes east of the Falklands. The *Aurora* saw them again on a later voyage; the *San Miguel* in 1769; the *Pearl*, the *Dolores*, the *Princesa* in 1790 — three islands by then, snowy, steep-sided, unmistakably land, seen across three decades by crews with no acquaintance and nothing to gain.

And then — this is what makes the Auroras the crown of the multiple-witness problem — they were *surveyed*. In 1794 the corvette *Atrevida*, of the Malaspina expedition, the most scrupulously instrumented scientific voyage Spain ever mounted, was detached specifically to fix the Auroras. She found them. Her officers observed “a great mountain in the form of a pavilion, divided vertically into two bands, snow-covered and dark,” fixed all three islands by chronometer, and published the positions to the minute. This is not a whaler’s rumor. This is the state of the art of eighteenth-century position-fixing, applied

deliberately, on station, in fair weather, to something its officers could see.

IV: Read the *Atrevida's* figures next to Peary's paragraph and despair of the idea that evidence has weights. One liar makes a continent; a hundred honest men and a chronometer make nothing. The record does not grade witnesses. It only counts copies.

The islands entered every chart of the southern ocean, directly on the route around the Horn. It took the nineteenth century's professional doubters to make absence into a finding: James Weddell, the sealer- navigator, searched in 1820 — four days of clear weather across the published positions — and wrote, with the carefulness of a man contradicting the Spanish navy, that he was “certain that no such islands exist in the situation assigned.” Benjamin Morrell searched in 1822 and also found nothing, which, given his reputation as the greatest liar in the southern ocean, Voss files as weak confirmation at best:

IV: Morrell reporting an absence is the only class of Morrell report one can trust. A man who invents islands has no motive to invent open water.

By mid-century the Auroras were dwindling on the charts — marked doubtful, then E.D., then gone, the standard three-stage funeral. The modern post-mortem offers two bodies. First: the Shag Rocks, a scatter of real pinnacles some way east, misplaced by longitude errors — but the Shag Rocks are bare fangs a few dozen metres high, and the Atrevida measured snowfields and a banded mountain. Second, and generally preferred: tabular icebergs, veined with morainal rock, aground perhaps on a shoal — persistent enough to be re-sighted across seasons, land-like enough to survey. The explanation is respectable. Voss grants it in full and then does what she does:

IV: Suppose it so. Then for thirty years the sea assembled, at a fixed position, out of ice and stone-dust, a scale model of an archipelago — accurate enough to bear a chronometer fix, patient enough to be found on purpose. We have a name for the class of object that holds station and answers surveys. The explanation does not dissolve the island; it only tells me what the island was made of. Materials were never the question.¹⁵

¹⁵The Atrevida's memoir, with the observations, is in the published hydrographic collections; Weddell's search is in his *Voyage towards the South Pole*. Both check. For once every shelfmark in a

There is a coda, and it belongs to literature rather than hydrography. Edgar Allan Poe, who read voyage narratives the way other men read scripture, sent the schooner of *Arthur Gordon Pym* hunting the Auroras in his only novel — a fictional ship searching for islands that were already, by 1838, slipping off the real charts. The search fails, and Pym sails on south into whiteness. Voss kept the page. Her marginal note is the last item in the folder:

IV: Poe understood the etiquette. You do not find them by looking — the *Atrevida* was the exception, and exceptions are invitations. You find them on the way to something worse.¹⁶

8. Buss Island

c. 57°N, 28°W. North Atlantic, south-east of Greenland. Later: beneath it.

Every other island in this Part was eventually declared never to have existed. Buss Island alone was granted a death certificate. When the sea where it stood proved

Voss entry is where she says it is, which I record with more relief than a scholarly apparatus should admit to.

¹⁶The Aurora file is the thickest of the Part I folders and the only one tied with string. Under the string, on the outside, Voss wrote a single word: “practice.”

empty, the chart-makers did not erase it; they drowned it. For a century the charts of the North Atlantic carried, in the space where the island had failed, the inscription *The Sunken Land of Buss* — the only geological event in history to be invented retroactively to save an error's reputation.

It began, like much North Atlantic apocrypha, with the Frobisher voyages. In September 1578, the fleet straggling home from the third expedition, the *Emmanuel* — a busse of Bridgwater, “busse” being the herring-boat build that gave the island its unlovely name — reported passing a new land south-east of Frisland: two days along it, “seeming fruitful, full of woods, and a champaign country.” That Frisland itself was a phantom (it has its own literature, and Voss's index sends the reader there with a weary *q.v.*) troubled nobody: one imaginary island was seen from another, an arrangement the sixteenth century found unremarkable and which Voss, characteristically, found the most instructive fact in the file.

IV: Phantoms sight phantoms. The Emmanuel's landfall is bearing-true from a place that is not there. Errors are not points; they are *networks*, and they survey one another with perfect internal consistency. A chart is a conspiracy of witnesses who have never met.

The island took root. In 1671 a Captain Thomas Shepard claimed not only to have found Buss but to have sailed around it, and produced a map — harbours named, capes christened, an anchorage recommended — precise enough that the Hudson's Bay Company briefly considered how the island might serve its trade: a complete cartographic biography of nothing, adopted by the most hard-headed commercial enterprise of its age.

IV: Note the sequence, because it is the standard sequence run in reverse. Usually a place is seen vaguely and mapped vaguely and the detail accretes. Buss was seen vaguely and mapped *precisely*, at once, by a man who needed it findable. Precision is not evidence. Precision is appetite.

Then the failures began — ships crossing the position in daylight and finding water — and with them the remarkable adjustment. The island was not struck. Instead, the eighteenth-century charts began to explain: the land had *sunk*. Van Keulen's Dutch atlas of 1745 states it as plain fact: "the submerged land of Buss, nowadays nothing but surf and shoaling water" — the corpse of the island, decently interred just below the surface, still capable of wrecking you. It remained a charted hazard until 1818, when John Ross, outbound for Baffin Bay, put down a

sounding line over the position: one hundred and eighty fathoms, no bottom. Deep water everywhere. The sunken land was not sunken; it was absent, which the charts had never been willing to say.

IV: Here the record shows its hand. Confronted with an empty position, the chart did not amend itself; it amended the *world* — invented a catastrophe, sank a landmass, planted surf on the grave — rather than write the true sentence, “we were wrong.” An error will spend any amount of geology to avoid an erratum. And observe what the amendment purchased: a century of soundings. Ships that would never have stopped for open water stopped for a drowned island and put lines down over it, attending to the position, keeping it *visited*. The drowning was not an exit story. It was a maintenance contract.¹⁷

The modern explanation is serviceable: Frisland’s ghost geography displaced the *Emmanuel*’s honest sighting of

¹⁷The reader will hear in “maintenance contract” a rhyme with the Bermeja entry’s talk of deposits, and may reasonably ask whether Voss is building a doctrine across these files. She is. I have chosen to let it assemble at her pace rather than summarise it ahead of the evidence, because summarised ahead of the evidence it sounds like what it perhaps is, and there are four more entries in this Part.

some real coast — Greenland's, likely — and the shoal reports were the usual North Atlantic repertoire of whale-backs, ice-blink, and wishful lead-lines. The file could close there, and in every archive but one, it does.

The exception is the reason Buss stands eighth in Voss's index. Among her papers is a photostat of a Hudson's Bay Company memorandum of 1675, declining, finally, to pursue Shepard's island. It gives no reason. But some clerk — the hand is contemporary — has docketed the decision in the margin, and the docket reads: *not for stations*.¹⁸

9. Dougherty & Emerald

Dougherty: 59°20'S, 120°20'W. Emerald: 57°15'S, 162°12'E. The far southern ocean, which is to say: nowhere, twice.

The southern ocean below the fiftieth parallel is the emptiest surface on the planet — seven thousand miles

¹⁸Entry 5, note 12, records my failure to find the Grimsholm documents; I will not rehearse the feeling here. The HBC archive in Winnipeg is superbly catalogued. The memorandum exists. The marginal docket exists. I have held the photostat against the window and against better judgement, and the hand of the docket is seventeenth-century, and the phrase is the Danish Admiralty's phrase, seventy years early and an ocean away. Institutions do not share marginalia. Categories do.

of easting, in the old days, without a sail, a coast, or a gull that meant land. Voss gives its two phantoms a shared file, and opens it with the observation that explains the pairing:

IV: Elsewhere phantom islands are errors of perception. Below fifty south they are errors of *need*. The eye is a social organ; it was not built for a week of horizon; and what it cannot find it will begin, out of courtesy, to provide.

Dougherty was reported in 1841 by the master of the whaler *James Stewart* — five or six miles long, high at the north-east end, snow-covered — and confirmed twice across the next forty-five years by masters with no acquaintance and nothing to gain. Three sober sightings, one position: it entered the Admiralty charts as the only mark on the emptiest sheet in the folio. Its unmaking was a career. In 1909 John King Davis, homeward with Shackleton's *Nimrod*, went out of his way to settle it: he ran the reported latitude through the position and eighty miles beyond, in weather clear enough to see a fulmar at five miles, and over the island's charted heart his deep-sea line paid out and found bottom at 2,318 fathoms. He allowed afterwards that the island had been an iceberg

— three icebergs, presumably, on station across half a century.

IV: Davis's sounding is the deepest water in this Part — 4,240 metres, far outside the band. I record this against myself, and I am more grateful for it than for any figure in the Atlas: a doctrine that cannot find a hostile number is not a doctrine but a mood. Dougherty is my control. Whatever was banked in the shallow band, nothing was banked here. Some islands are only lonely eyes.¹⁹

Emerald, the older sister, was reported in 1821 south of Macquarie Island — high, bold, mountainous — and stayed on the charts for a hundred and twenty-eight years, surviving Wilkes in 1840, Davis again in 1909, and a New Zealand naval sweep as late as 1949. What preserved it, Voss argues, was not the fog but the usefulness: it sat athwart the route toward the ice, a mark to be “made” or “missed,” an event in seas that otherwise offered none. Mawson's men looked for it going south,

¹⁹This is the first time in the papers that Voss volunteers evidence against the depth pattern, and the file supports her: Davis's sounding is genuine and genuinely deep. I note — as observation, not rebuttal, since she made the observation first — that she did not remove Dougherty from the Atlas for failing the pattern. A control belongs to the experiment.

half science, half a game. Nobody depended on Emerald. Everybody was slightly better off with it there.

IV: The economists have a word for a good whose value is produced entirely by shared belief in it, and the word is “currency.” Emerald was small change for the loneliest sea: passed log to log, never redeemable, endlessly negotiable. The southern ocean kept a coin in circulation for 128 years because the alternative was that nothing should ever happen at all.²⁰

The diptych closes the ledger of honest phantoms. What remains of Part I is stranger: an island the United States annexed sight unseen; an island I can no longer discuss in footnotes; and Voss’s own summing-up. The reader who has been keeping accounts — of depths, of phrases, of inks — is advised to bring them forward. They are about to be audited.

²⁰The word “currency” is underlined in the manuscript, and in the margin beside it, in the late hand, Voss has written: “then what is the bank.” The question mark is absent in the original and I have not supplied one.

10. Sarah Ann

c. 0°35'N, 175°10'W. Central Pacific, north of the equator. Property of the United States of America, 1858–1937.

The phantoms of this Part have been believed in, steered by, searched for, drowned, and shot over. Sarah Ann Island is the only one that was *owned*.

The instrument of ownership was the Guano Islands Act of 1856, one of the strangest statutes ever passed by the American Congress: any citizen discovering an unclaimed island bearing guano — the seabird manure that was, for two decades, the most sought fertiliser on earth — might file a bond with the State Department, whereupon the island “may, at the discretion of the President, be considered as appertaining to the United States.” Under this law American firms claimed dozens of Pacific specks, some real, some real-ish, and some — Voss’s file lists nine — that no one has ever seen at all. Sarah Ann was claimed in 1858 by the United States Guano Company of New York, on the strength of a whaler’s report from decades earlier. The paperwork was filed. The bond was posted. The island entered the Department’s schedule of guano islands, which is to say it entered the sovereignty of a great power, without any

human being having verifiably stood on it, seen it clearly, or fixed it.

IV: With Sarah Ann the chart is no longer even the senior document. The island's best evidence of existence was a *deed*. Here the third place acquires what it had lacked since Dulcert: a jurisdiction. One could not visit Sarah Ann, but one could, in principle, sue over her — and the suit would have been heard. The law, unlike the lead-line, never touches bottom.

For eighty years nothing whatever happened, which was the island's *métier*. It appeared in the hydrographic lists with the usual drifting coordinates; the guano boom collapsed; the company dissolved; the claim, being no trouble to anyone, was never withdrawn. Sarah Ann persisted the way an entry persists in a ledger no one audits — until, in the 1930s, she was audited by the least corruptible instrument imaginable: the moon.

On 8 June 1937 the shadow of a total solar eclipse — one of the longest of the century, totality over seven minutes — was scheduled to cross the empty central Pacific. Eclipse astronomy in that era was a hunt for land: instruments of the period wanted bedrock under them, and the computed track of totality is indifferent to the convenience of astronomers. The track for 1937 crossed

almost nothing. The planners went down the schedules island by island, and there, squarely inside the path of the moon's shadow, the United States found that it owned one: Sarah Ann. Expeditions were sketched; the Navy was consulted; and ships went out ahead of the eclipse to establish the observatory — on an island whose entire documentary existence was a whaler's hearsay and a guano bond.

They found the Pacific. They searched along the listed position and its variants — the reports of the search, which Voss collected, have the methodical melancholy the reader now knows from Weddell and Davis — and Sarah Ann was quietly struck from the lists that year. The astronomers observed the eclipse from Canton Island, hundreds of miles away. The seven minutes of darkness crossed the position of Sarah Ann on schedule, and no one and nothing watched it pass.

IV: I ask the reader to sit with the mechanics of the unmaking. For eighty years the island was safe: unvisited, unneeded, a deed in a drawer. What killed it was an appointment — the one event in nature that cannot be rescheduled, missed, or fudged, moving on rails computed to the second, requiring the island to *be there at a particular time*. Phantoms survive space effortlessly; the chart is on their side.

What they cannot survive is time. Every phantom dies of an appointment. Hy-Brasil's was a sounding, Bermeja's a treaty, Sarah Ann's an eclipse — the form varies, but the sentence is the same: the world named an hour, and the island did not keep it.²¹

The file closes with an oddity of the kind the reader is now trained to receive without my assistance. Among the 1937 papers is a clipping from a Honolulu newspaper of that July — a wire-service squib reporting the failed search, four sentences, jocular in the period manner. Voss has kept it for the last sentence, which reads: “The Navy is satisfied that the island, if it ever existed, has gone to join the fleet of vanished isles, which sail under sealed orders.”²²

²¹“Every phantom dies of an appointment” is the closest thing in the notebooks to a fourth law, though Voss never numbers it. I flag it because the reader will want it at hand in Part III, where the polar entries concern places that were given appointments and, according to the files, kept them.

²²I found the original. Honolulu Star-Bulletin, 14 July 1937, page 9. The squib is there, word for word — except the last sentence, which in the printed original reads simply: “The Navy is satisfied that the island does not exist.” The version in Voss's clipping is typeset identically to the rest, same face, same measure, and it is not in the paper. I have stopped reporting these discrepancies to the archives concerned. The archives are not where the discrepancy lives.

11. Isla del Coro

15°50'N, 76°20'W. Caribbean Sea, on the old pearl route between Jamaica and the Main.

Every island in this Part until now was seen. This one was heard.

The first report is from 1528, in the deposition of the three survivors of the *Santa Engracia*, a pearl carrack lost on the run from Cubagua. Adrift in the ship's boat, becalmed in fog before dawn, they heard — all three deposed to it separately, and the depositions agree in the particulars — the sound of a choir. Not one voice; many, “well ordered,” singing what the eldest survivor, who had been a sacristan, identified as the office of matins. The singing continued, he said, the length of the office. They rowed toward it, and the fog lightened, and they made out a low island, dark, without surf, “at no great distance.” Then the sun took the fog and the island with it, and three days later they were picked up by a coaster out of Santiago whose master recorded their account with the dry notation: *others also have heard this choir*.

Others had. The rutters of the pearl route — the working sailing directions, copied hand to hand among pilots, closer to the water than any chart — carry the island for a century and a half under the name Isla del Coro, the Island of the Choir, with the most practical instruction in

the literature of phantom places: *she is not seen but in thick weather, and is known by the singing; the prudent will alter course upon hearing it, and not toward.*

IV: The rutter does not ask what the singing is. The rutter is a genre without curiosity; it wants you alive, not informed. But note what the instruction concedes: that the island's sensible sign — the only sign by which she is “known” — arrives by ear, in fog, at the hours of the night offices. A place whose entire interface with the world is a sound. The pilots mapped an acoustics.

She was charted, small and tentative, by the Seville school in the 1580s; she appears on Dutch charts of the Main as I. del Coro or I. de Choro through the seventeenth century, sometimes with the annotation *duyster* — dark, obscure. No landing is recorded anywhere in the file. No one ever brought back so much as a branch. The island produced nothing but testimony, and the testimony is strangely uniform across three centuries: fog, stillness, the ordered voices, the low dark shore, the dissolution by daylight. A French privateer's journal of 1697 adds the single variant detail in the whole dossier: that as his ship stood away — his crew “much afraid, though the singing was not terrible but sweet” — the voices did not

fade with distance as sound fades over water, but ended all at once, “as when a door is shut upon a room where there is company.”

IV: As when a door is shut. The privateer’s image is exact and he did not know what he was being exact about. Sound does not carry from the third place; it *leaks*, when something is open that is ordinarily closed. The fog is not concealment. The fog is the gap under the door.

The island’s death was the quietest in the Atlas. No search, no sounding, no eclipse, no senate. The pearl fisheries failed; the route moved north; the rutters stopped being copied because the pilots who needed them stopped existing; and Isla del Coro, which had never been anything but a sound in the mouths of working men, went out of the world when the last of them did. The nineteenth-century hydrographic offices, sweeping the Caribbean sheets for doubtful marks, removed her without discussion — there is no minute, no E.D. period, no obituary. The reader who consults the modern chart will find the position clean, and the water beneath it, for what it is now worth to anyone, twelve hundred and forty metres deep.

Voss's entry ends with a passage that I give as it stands in the notebook, because it is the only place in the papers where she writes about the island in the first person, and because the reader will understand, by the end of this book, why I have chosen to let it stand without comment.

IV: I heard it once. October of 1981, from the deck of a fruit boat, in fog, in the right latitudes, at four in the morning. I had arranged to be there; T. had told me the conditions; I had waited through nine nights of clear weather for the fog. It was matins. It was well ordered. It was, as the privateer says, sweet, and it ended as he says it ended, all at once, and the fog burned off by seven and there was nothing to the horizon anywhere. I record what I concluded, standing at the rail with the sun coming up on empty water: that the island was not gone. That none of them are gone. That what we struck from the charts we did not strike from the world; we only stopped answering the door. The singing goes on behind it at the appointed hours, whether or not anyone is in the fog to hear — the office does not require a congregation. It requires only that the hours be kept. Somewhere the hours are being kept, Elena. The Atlas is my attempt to find out where, and the

finding out has taken my life, and it was a life well spent.

12. On the Uses of Islands

[Found as six loose leaves at the end of notebook IX, headed in Voss's hand "for the Atlas — end of the islands." Printed without alteration. — E.M.]

I have now spent eleven entries dismantling islands, and before we go ashore I owe the reader an accounting of what the wreckage is for.

Begin with what we have established, stated together for the first time, since the entries planted the laws one at a time like navigational marks and a mark is only useful when you can raise two at once.

First: an error is a promise the map makes before the world is ready to keep it. The men of Bristol sailed west for a false island and struck a true continent. The promise was kept — merely not to the men, and not by the island.

Second: an error is an honest observation that has out-lived its weather. The *Velocity's* pumice was really there on the Tuesday it was seen. The chart, which has one tense, embalmed the Tuesday and issued it to shipping for a hundred and thirty-six years.

Third: stationary phantoms die by sounding; migratory phantoms do not die. What holds still can be audited. What keeps one horizon ahead of the audit is never disproved, only decatalogued — a bureaucratic death, which is the kind you come back from.

And running beneath all three, unnumbered because I cannot yet state it honestly: the appointments. Phantoms are unkillable in space and mortal in time. Name an hour — a sounding, a treaty, an eclipse — and the island must produce itself or die. I distrust this law because it has an exception, and the exception is the entry this book is missing, and the reader will have to take on faith, for some chapters yet, that I know an island that keeps its appointments.

Now the accounting. What is a phantom island *for*? I do not mean what caused it; causes bore me, and the causes are anyway always the same five — mirage, pumice, ice, longing, and money. I mean what it is for: what work it does, that the world keeps making them.

The uses are three, and they ascend.

The low use is navigational, and the Emerald entry states it: an island is an event, and empty seas are unendurable without events. The mark to be made or missed, the name in the log, the coin of shared belief passed hand to hand. Sailors did not believe in these islands the way

theologians believe; they believed the way players believe in the rules of a game, which is to say completely, and without the least concern for what is under the board.

The middle use is jurisdictional. An island is a place where writ can run — a deed, a bond, a baseline, a station. The Admiralty that wrote “the island is not for stations” and the Company that docketed the same phrase understood something we have forgotten: that the paper category comes first and the geology is optional. Sarah Ann was sovereign American territory for eighty years on the strength of bird dung nobody saw. Do not laugh at the eighty years. You live inside arrangements exactly as well founded, and they hold your weight daily.

The high use — and here I must be careful, because this is the hinge of the Atlas and I have watched better scholars than I slide off it — the high use is *custodial*. Some of these places are not errors at all. They are the durable, well-attested, load-bearing marks of something the charts cannot carry and therefore carry wrongly, the way a ledger carries a debt it cannot describe as anything but a number. The depth that repeats. The sign that recurs in archives that never corresponded. The phrase that surfaces in two languages, seventy years apart, applied to the same category of place. The singing that ends like a door shut on company. I have spent fifty years sorting the merely dead islands from these others, and

the sorting is the Atlas, and the others are few — but they are not none, and one of them has a name I have not yet written in any entry, and its folder is where my concordance says it is, and if it is not there when these papers are read, then that too is a finding.

So much for islands. Now observe what we lose when we go ashore, as the next Part must, for the interiors have their phantoms too — the mountain range that walled a continent, the kingdom that answered letters, the city of gold that ate expeditions. We lose the sounding. At sea the audit is clean: sail to the position, put down the line, and the water answers yes or no. But you cannot sound a mountain. A false range dies only when someone walks through it, and walking through is slow, and mortal, and the walker can be argued with in a way the lead-line cannot. Interior phantoms therefore live longer, lie deeper, and die harder than any island — and when at last they are walked through and found empty, the space they occupied does not close over like water. It stays. There are provinces on the modern map that are still, functionally, the hole left by a phantom, and we have built real roads around imaginary mountains and the roads are still there.

The sea forgives the chart its errors; it was never listening. The land remembers. Go ashore expecting the worse behaviour, and you will not be disappointed.

One thing more. A colleague to whom I showed the Grimsholm file — the only living person who has read it entire — asked me the question I have been asked at intervals for fifty years, in tones ranging from concern to professional malice: *do you believe these places exist?* I set down here the answer I have arrived at, at the end of the islands, in the eighty-fourth year of my age, so that it cannot be softened later by people editing me kindly.

Existence is the wrong verb and it was always the wrong verb. The chart does not ask whether an island exists; the chart asks whether to *carry* it. The sea does not ask; the sea is not a party to the question. Only we ask, and we ask because we have confused the ledger with the goods. What I believe is this: that the world keeps more than one set of books; that the charts we sail by are the public set; and that everything this Part has catalogued — the depths in the band, the furnished cairn, the sealed orders, the kept hours — is what an audit looks like when it brushes, without knowing it, against the private set. I intend to see the private set before I die. I have been advised that this is arranged by appointment only.

I have an appointment.

PART II

INTERIORS

You cannot sound a mountain. You can only walk to where it is not, and come home older, and be disbelieved. — IV, notebook IV

[Voss's index orders the interior entries by "depth of penetration required" — her phrase — from ranges visible at the coast to kingdoms that could be reached only by correspondence. I have preserved the order. — E.M.]

13. The Mountains of Kong

c. 10°N, from the Gulf of Guinea eastward toward the Nile. Length as charted: up to four thousand kilometres. Height as charted: "stupendous." Actual: —

The islands of Part I were, at their grandest, the size of a county. We begin the interiors with a phantom the size of a mountain system — a range that ran like a wall across the whole of West Africa, on essentially every serious map of that continent published in the nineteenth

century, and that does not exist, and never did, at any point along its four thousand charted kilometres.

The Mountains of Kong were born in 1798, and unusually for this Atlas, they have a birth certificate with a signature on it. The signer was Major James Rennell — the first Surveyor-General of Bengal, the father of English geography, the most respected map-maker of his age; it matters to everything that follows that he was not a fraud and not a fool. Rennell was engaged to draw the maps accompanying the narrative of Mungo Park, then newly returned as the first European to have laid eyes on the middle Niger. Park's journal mentioned, in passing, distant high ground south of his route — he had seen hills; there are hills. Rennell took the hills and did to them what a theorist does to a fact that fits: he promoted them. His accompanying memoir reasoned that the Niger, flowing eastward as Park confirmed, could not possibly reach the Atlantic — some barrier must pen it into the interior; and a barrier being needed, the hills Park glimpsed became its western outriders, and the range extended eastward across the continent because the theory required it to be long. He named it from the trading city of Kong, north of the Gulf coast, and drew it with the confident hachures of the period — and here the reader must understand what Rennell's hachures meant

to the trade: they meant *settled*. Lesser houses copied Rennell the way parishes copy scripture.

IV: Note the mechanism, because it is new to us. The island phantoms of Part I were errors of the eye — pumice, mirage, ice, longing. Kong is an error of the *argument*. Nobody ever reported the range. The range was deduced. It is the purest specimen I know of a class that rules the interiors: geography secreted by theory, as a shell is secreted by the animal — from inside.

And it grew. Follow the range through the century's atlases, as Voss did — her Kong file contains tracings from over forty, arranged by date, and flipping them in sequence is like watching a culture in a petri dish. In 1802 the range is a modest ridge behind the Guinea coast. By the 1820s it has thickened and sprouted spurs. In some atlases of the 1830s and 40s it marches east until it fuses with that other great inherited fiction, the Mountains of the Moon, so that a single unbroken wall crosses Africa from the Atlantic to the Nile — you may see it so on maps sold in the year the telegraph reached India. Text-books gave the Kong range snow-caps. Gazetteers gave it gold. One French geography of mid-century, in a detail Voss underlined twice, gave it inhabitants: hardy moun-

taineers, of whom precisely as much was known as of the granite they herded their goats on, which is to say the sum of European knowledge of both was zero.

IV: Snow at ten degrees north, on a range no one had seen, in books used to examine schoolboys. I want the reader to feel how little malice was involved. Every copyist improved the range in good conscience, because mountains ought to have snow, and a wall across Africa ought to be tall enough to deserve the name. An error, once it bears weight, attracts ornament. We do not merely tolerate our fictions; we *maintain* them, the way a town maintains its bridge.

The bridge was load-bearing in the strictest sense. For ninety years the Mountains of Kong did real work in the world's understanding of Africa: they explained the Niger (falsely); they discouraged trade routes inland from the Guinea coast (routes that were, in fact, open and walked by African merchants the whole while); they anchored the century's confident division of the continent into climatic and racial "zones" (the less said of which the better, though Voss says it, at length, in a passage I have placed in the appendices). Real expeditions budgeted for crossing them. Real geographies taught generations that

the interior was sealed. The wall was imaginary and it kept people out.

IV: Part I asked what a phantom is for, and the interiors sharpen the question, because an interior phantom is not a mark to steer by; it is a *premise*. The islands were coins. The ranges are walls, and a wall's work is done precisely when nobody tests it. Kong stood for ninety years because its function was to make the journey that would disprove it seem not worth the taking. The best-defended fictions are the ones that guard their own counter-evidence.²³

The man who finally took the journey was Louis-Gustave Binger, a French officer of no great fame, who in 1887–89 descended from the upper Niger southward — directly through the coordinates where every atlas in Europe stacked its hachures — on his way to the city of Kong itself. His account of the crossing is the great anticlimax

²³This footnote records a debt. Voss's argument here — that interior phantoms persist by making their own falsification expensive — was published by her in a 1971 paper that was badly received, in terms her file of referee reports preserves. One referee wrote that she “mistakes the sociology of error for a property of the places themselves,” which the reader will recognise as precisely the confusion, or precisely the insight, on which this entire Atlas balances. The referee signed with an initial only. The initial was not T. I checked.

of nineteenth-century exploration, and Voss quotes it with visible relish: “On the horizon, not even a ridge of hills!” Binger walked for weeks across open savanna, gentle rises, a country you could drive a cart over, occupied by farmers and traders who were understandably unaware they lived beneath a stupendous snow-capped cordillera. He reached Kong the city — real, prosperous, a market town of clay mosques and cloth merchants — was received courteously, and wrote the sentence that killed the mountains: there are none. He announced it to the Geographical Society in Paris in 1890, and the range that had taken ninety years to grow came off the maps in roughly five — though Voss, who audited the die-off atlas by atlas, found stragglers: a Rand McNally of 1905 still carries the range, and she traced one English school atlas reprinting a ghost of it into the 1920s, teaching the wall to children whose fathers could have taken the train past its coordinates.

IV: Observe the asymmetry of the two deaths in this file. The mountains died of one man’s walk and were mourned by no one. The city — the real thing, the market and the mosques and the name that had been lent to all that imaginary granite — was burned to the ground seven years after Binger’s visit, in the wars of Samory Touré, and rebuilt small, and it

faded from the European map hand in hand with the range. By 1930 an educated European asked about “Kong” would recall neither. The fiction and its collateral were disposed of together, and the map-makers kept what they always keep: the blank, and the confidence.

There the published argument ends, and the file should close. The reader of this edition has learned to wait a paragraph longer.

Among the tracings in the Kong folder — the forty atlases, arranged by date — there is one Voss marks with a red slip. It is her tracing of the range from a Milanese atlas of 1849, and it differs from its neighbours in one particular: a pass. Where every other atlas draws the wall unbroken, the Milanese engraver has left a gap — narrow, deliberate, flanked by the range’s only two named peaks in any edition — and through the gap he has drawn, in the faint double line that the period used for caravan routes, a road. No route into the interior existed on any source map he could have copied. The pass has a label, in type so small Voss’s tracing reproduces it with a magnifier’s note. It reads *Passo del Coro*.²⁴

²⁴Coro: the choir. The reader met the word four entries ago, attached to an island two oceans away, in an entry I let pass without apparatus. I had a reason then and it has not improved with keep-

14. The Kingdom of Prester John

Position: the Three Indias (12th c.); beyond the Tigris (13th); the Mountains of the Moon (14th); Ethiopia (15th–17th). A kingdom with a forwarding address.

The Mountains of Kong were a phantom you could not reach. The kingdom of Prester John was stranger: a phantom you could not reach, but which you could *write to* — and which, once, wrote first.

It began with a letter. Around the year 1165 there appeared in Europe — “appeared” is the scholars’ word, and it is doing the work of a confession — a long epistle addressed to the Emperor of Byzantium, purporting to come from one Presbyter Johannes: priest and king, Christian of the East, ruler of the Three Indias. The letter is a gorgeous, preposterous object. Its author governs seventy-two kings; in his lands are the Fountain of Youth and the sandy sea that flows without water; before his

ing: the reason is that I first read the Isla del Coro entry in Bergen, in 2021, and I verified the Milanese atlas — it exists; the pass is there; the label is there, 6-point, in the copy held at the Braidense — in the spring of 2024. Three years apart, an invented island and an unbroken wall’s only gap, wearing the same name. Voss’s red slip carries one line in the late hand: “the doors are labelled. They were always labelled. We catalogued the walls.”

palace hangs a mirror in which he beholds everything done in every province; his table is emerald; among his subjects are the Amazons and the Brahmins and men with horns; and he is served at dinner, in the detail every medieval reader remembered, by seven kings, sixty dukes, and three hundred and sixty-five counts, one for each day of the year — a bureaucracy of time itself waiting at table.

No original exists. No author was ever identified. The Greek text it claims descent from has never been found; the thing surfaces already translated, already copied, already believed. Within a generation there were hundreds of manuscripts; within two, versions in every literary language of Europe, each translator adding provinces and marvels the way the atlas-makers would later add snow to Kong.

IV: Compare the birth certificates. Kong was signed by the finest geographer of his age, deducing in good faith. The Letter of Prester John is signed by its own subject — the phantom introduces *itself*, unforwarded, in the first person, with an inventory of its contents. In five centuries of material I know no other case where the third place opens the correspondence.

Christendom answered. That is the fact around which Voss builds the entry, and she asks the reader to sit with its full administrative weight. In 1177, Pope Alexander III — a pope in the middle of his wars with an emperor, a man with no leisure for whimsy — composed a formal reply to Prester John: cordial, cautious on points of doctrine, proposing closer relations. He entrusted it to his personal physician, Philip, who departed to deliver it.

To deliver it *where* is the question the file turns on. The Letter bears no address. The Three Indias were a rhetorical region, not a route. Philip went east with a letter for a kingdom whose location was the letter itself — and Philip does not return. No record of his arrival anywhere; no record of his death; the correspondence of the papal court simply ceases to mention him. The reply to Prester John is the only document in this Atlas that was successfully dispatched to a phantom place, and its carrier is the only man who can be said, strictly, to have been delivered with it.

IV: The office is instructive. A letter to an unplaceable address requires a carrier willing to make the address exist by arriving — the delivery is the discovery, or it is nothing. Alexander understood this; it is why he sent a man he trusted with his own body. I have wondered for forty years what instructions

Philip carried beside the letter. I have wondered longer than that whether the office was ever afterward vacant.²⁵

For the next five hundred years, the kingdom behaved exactly as the Third Law requires of a migratory phantom: it kept one province beyond the surveyed. When the Mongol embassies opened Asia and no priest-king sat beyond the Tigris, the kingdom was rediscovered further east, in the person of various steppe khans garbled into Christians by hope and distance. When Asia filled in entirely, the kingdom executed the most remarkable manoeuvre in the history of imaginary geography: it changed continents. Fourteenth-century map-makers, needing somewhere for a Christian empire that letters had established beyond argument, settled on Ethiopia —

²⁵Voss's question about the vacancy of Philip's office is rhetorical in the notebook. I am unable to leave it rhetorical. Of the seventy-one letters from T. in the Voss papers, eleven bear postmarks — it is by these that the correspondence can be dated and placed at all, Reykjavík to Veracruz to the Aleutians. The other sixty were carried, not posted: no stamps, no cancellations, envelopes of varying national manufactures, tucked — according to the annotations Voss sometimes made on the versos — into hotel pigeonholes, ships' mail sacks, a returned library book, and once "the usual place, Tórshavn." A correspondence conducted five parts in six by hand delivery implies either proximity I can find no other trace of, or carriers. I record the two possibilities in order of the comfort they offer.

where, by the grace of coincidence, a genuinely ancient Christian kingdom genuinely was. For three centuries the maps of Africa carry *Preste Juan* enthroned among the sources of the Nile, and the fiction now had a real state to wear as a body.

The Portuguese, being the century's most literal-minded people, sent embassies. Pêro da Covilhã reached the Ethiopian court in the 1490s, overland, in disguise, carrying his king's letters to Prester John — and here the file produces its second detained correspondent, for the Ethiopian court received Covilhã graciously, found him useful, and declined, for the remaining thirty years of his life, to let him leave. A later embassy found him old, honoured, married, weeping at the sight of countrymen, and still not permitted to go home. The kingdom that could not be reached had become a kingdom that could not be left — Voss notes that these are the same property, observed from opposite banks.

IV: The Ethiopians, for their part, were bewildered and then diplomatic. Letters kept arriving for a man who was not their king, from monarchs too powerful to correct. So they answered them. For two centuries a real state conducted the correspondence of an imaginary one — signing, in effect, the phantom's mail — because geopolitics had made the fiction

indispensable to both parties. The reader who thinks this quaint is invited to consider how much of the world's current mail is answered on behalf of entities that exist in precisely this sense.

The kingdom died, as interior phantoms do, slowly and of scholarship. The Jesuits in Ethiopia reported a poor, embattled, entirely human realm; the philologist Hiob Ludolf demonstrated in the 1680s that "Prester John" had never been an Ethiopian title at all; the name retreated from the maps a province at a time, and by the eighteenth century the priest-king had completed the standard descent from geography to literature. What did not die was the Letter. It had always been the kingdom's true territory — the one province no expedition could enter and no philologist could empty — and Voss's entry ends not with the maps but with the manuscript count, which is its own kind of chart: over four hundred copies surviving, in a dozen languages, which is to say four hundred surviving instances of an address at which, for half a millennium, the largest civilisation on earth genuinely expected an answer.

IV: The Letter is the purest phantom in the Atlas because it dispenses with terrain altogether. No coast, no sounding, no walk-through can touch it; it is all

interior. And it teaches the final mechanism, the one the islands could only hint at: a place needs no position if it can maintain a correspondence. Answer promptly, in a fair hand, with convincing detail, and you may omit to exist for five hundred years. — I set this down and my own files reproach me. I am aware, none better, of what it is to trust a correspondent one has never located.²⁶

15. El Dorado

**A man (c. 1535); a city (c. 1541); a lake (c. 1595).
Final charted position: Lake Parime, 3°N, 60°W.
Actual position: see final paragraph.**

Every phantom in this Atlas has cost something — ships' time, treaty lines, a man on the ice. El Dorado is the entry where the cost column becomes a casualty list.

²⁶The last sentence of this extract is the only place in the notebooks where Voss acknowledges what her own Atlas repeatedly demonstrates: that the T. correspondence conforms, point for point, to the anatomy of the Prester John file — the unplaceable address, the convincing detail, the answers that arrive by hands unseen. She saw the resemblance; she wrote it down; she continued the correspondence for another two decades. I used to read that as credulity. Editing has taught me the other reading: she had checked the anatomy against the specimen and concluded, as I am slowly concluding, that the resemblance is not a warning about the correspondent. It is information about the address.

Across three centuries it consumed expeditions the way its imagined mines consumed labour: thousands of Europeans and uncounted, because unrecorded, thousands of the people they conscripted, starved beside, and did not bury. It is the deadliest error in the history of maps, and Voss's file on it is the angriest in the papers — and the entry she built from that file turns, in its last movement, into something none of the others attempt: an accusation of the audit itself.

Begin, as she insists we begin, with the fact almost everyone has forgotten: El Dorado was not a place. The name is a job description. *El hombre dorado*, the gilded man — Spanish shorthand, picked up in the 1530s around the new settlements of Colombia, for a ceremony reported from the high lagoons of the Muisca: at the accession of a new lord, his body gummed with resin and powdered head to foot with gold dust, rafted to the centre of a mountain lake at dawn with braziers burning, where he washed the gold from his skin into the water while his people threw in offerings of worked gold and emeralds from the shores. The reports were true. The ceremony was real; the lake was real — Guatavita, a drowned crater two days' ride from modern Bogotá; the gold was real, and it went where the ceremony sent it: into the water.

IV: Hold the image, because the whole catastrophe is already inside it. The first Spaniards to hear the story understood it perfectly and could not afford to. A civilisation that gilds a man in order to *wash the gold off* — that mines treasure in order to sink it — is a civilisation whose ledger runs in the opposite direction, and the report of it arrived among men who had crossed an ocean on credit. They did the only thing their arithmetic permitted: they reversed the sign. Deposit became vault. Offering became hoard. The gilded man, who was an instrument for putting gold *into* the dark, was rebuilt in the European imagination as a king who could be made to give it back.

What follows, once the sign is reversed, follows like weather. A deposit implies a depositor; a depositor implies a capital; and the gilded man acquired, in the telling, first a city — Manoa, golden- roofed — and then, because a city must stand somewhere, a lake to stand beside: Parime, a vast inland sea in the Guiana highlands, which swelled on the charts through the seventeenth century to the size of the Caspian. The phantom had executed the manoeuvre Voss's Part I reader knows as migration, but in a new dimension: not across the map but *up the chain of abstraction* — man to city to lake,

each stage retreating one audit further from the original, checkable, true thing.

IV: This is the Fifth Law, and the interiors alone could have taught it: a phantom under pursuit does not merely move its position; it moves its *category*. Audit the man and you will be referred to his city. Audit the city and you will be referred to its lake. Every verification is answered with a forwarding address one ontological floor up, where the light is worse.

The pursuits themselves Voss recounts with a coroner's economy, and I compress her compression. Gonzalo Pizarro left Quito in 1541 with two hundred and twenty Spaniards and four thousand conscripted highlanders to find the golden country east of the Andes; the expedition ate its horses, then its dogs, then its dead, and its lieutenant Orellana, sent ahead for food, was taken by the current and carried involuntarily down the entire length of the Amazon — the greatest river on earth discovered as a *side effect*, by men looking for something that was not there. Philipp von Hutten led the Welser company's hunt out of Venezuela and came back to be beheaded with a blunt machete by a rival claimant of the very governance the gold would have justified. Antonio de Berrío,

who spent three fortunes and thirty years narrowing the search to the Guiana highlands, had the misfortune of being captured and debriefed by Walter Raleigh — who took Berrío's lifetime of field notes, added English marketing, and produced *The Discoverie of Guiana*, the most consequential real-estate prospectus ever written: Manoa the golden city, Parime its lake, an empire awaiting only investors. Raleigh went back in 1617 to make the book true. His son was killed storming a Spanish fort; the gold was not there because it had never been there; and Raleigh returned to London to be executed under the suspended sentence his prospectus had been written to lift. The phantom, Voss notes, had by then developed the distinguishing habit of the interior class: it did not merely fail to be found; it *specifically killed the people who wrote it down*.

Parime outlived them all on the maps — a sea the size of a nation, plotted from testimony extracted at the end of chains of translation and torture — until Humboldt walked and paddled the region around 1800 and dissolved it into what it had always been: the seasonal flooding of the Rupununi savanna, a country that becomes a mirror of water for some months and dries to grass. The greatest of the interior phantoms died exactly as the interlude predicted: not by sounding but by a walk, the walker armed with instruments and patience, and

the lake went off the charts leaving the hole that interior phantoms leave. Travellers kept reporting its glimmer for another generation.

But Guatavita — the small, real, original lake, the scene of the true ceremony — Guatavita could be audited. That is the terminal movement of Voss's entry, and she structures it as the trial it was. The audits began within one generation of the conquest: in the 1580s the merchant Antonio de Sepúlveda put Indian labour to cutting a drainage notch in the crater rim — the gash is visible today — lowered the lake some metres, recovered gold enough to whet every appetite that followed, and died poor. Attempts continued at intervals of roughly a lifetime for three hundred years, culminating in 1912, when a London syndicate called Contractors Ltd. — the name wants no ornament — piped the lake empty with steam pumps. They stood at last on the floor of the oldest bank in the Americas. The mud was four feet of ooze; it yielded a scatter of votive gold; and then the Andean sun dried the exposed sediment to the hardness of poured concrete, sealing the deposit beyond their machinery, and the lake quietly refilled, and the company auctioned its findings at Sotheby's for a sum that did not cover the pumps, and went into liquidation.²⁷

²⁷The Contractors Ltd. episode is documented to the shareholder level and checks throughout; the British Museum holds items from

IV: Read the 1912 record as the ceremony's answer, four centuries delayed. The lake was asked, formally, with engines, the question the gilded man had been settling every accession morning: *whose is the gold in the dark water?* And the account was — I use the banker's word deliberately — *frozen*. Not emptied; frozen. The deposit is still there, under mud gone to stone, under water gone back to stillness, under law: the Colombian state has since closed the lake to salvage entirely, so that the ceremony's intention is now enforced by statute of the successor polity. Sunk wealth stays sunk. Every party to the transaction has kept its side except the auditors, and the auditors are all dead or in liquidation. — The reader of my Part I asked, at Emerald, what the bank is. I do not yet say what the bank is. But Guatavita is a teller's window, and it is shut, and the shutting of it is the one event in this entire Atlas that every witness — Muisca, Spanish, English, geological — agrees actually happened.²⁸

the Sotheby's sale. I note for the reader keeping the cost column that the company's liquidation file lists among its creditors a Bogotá firm of carters, unpaid, for "the return haulage of pumping plant" — the machinery of the audit itself could not afford to leave.

²⁸Voss visited Guatavita once, in 1977, and the notebook page is mostly a sketch: the crater, the colonial notch, the waterline. Beneath it she has written a single sentence I withheld from the entry proper because it belongs to the reader who has come this

[The next entry's file was found sealed. — E.M.]

16. Mīzān

Reported between the passes: Persia to China by the southern silk routes, four to eleven days beyond the last town whose name the witnesses agree on. Altitude “where the water boils quiet.” Never sought successfully. Never encountered twice.

[This entry's file was found sealed — wax over string, unbroken, the only sealed item among the four thousand. On the wrapper, in the early brown ink: “Not to be read before Guataavita is understood.” Her final index places it here, immediately after El Dorado. I broke the seal on 11 November 2024, and I record that my hands were not entirely steady, and that the file contained no horror at all: only the gentlest material in the papers. The reader may judge whether that is a relief. — E.M.]

The caravan literature of the overland silk trade is a genre of practicalities: stages, wells, fodder, tolls, which headmen must be gifted and which merely greeted. It does not deal in wonders; a route-book that lied got its

far: “The only phantom place in the Atlas is the vault they believed in; the deposit was real; we have the sign reversed on every chart but this one.”

copyists killed at one remove. And it is in this hardest-headed of genres — in route-books compiled centuries apart, in three languages, by merchants with no access to one another — that Voss found the town of the scales.

The earliest witness is a Persian safarnāma of the twelfth century, whose anonymous author records, between two stages otherwise devoted to the badness of the water, the following: that four days beyond Sang-i-Safid the road rises through a defile to a town not listed by the customs-masters, where the caravan is met, courteously, and every load is weighed. Nothing is taxed. Nothing is opened. The weighing is conducted on a beam balance of great size and greater age, each pan “broad enough for a kneeling camel,” and the results are entered in a ledger by clerks described as neither young nor old. Asked the purpose of a toll that takes nothing, the clerks answer — the author gives the answer in the tone of a man transcribing what he does not claim to understand — *the toll is the weighing*.

An Armenian merchant’s manual of the seventeenth century — the genre’s masterpiece, elsewhere accurate to the last abbas of every real toll from Julfa to Lhasa — carries the town under the name Mīzān, “the Scales,” with the flat annotation: “Here the goods are weighed and no duty taken. Enter what the clerks tell you in your own book, for their figure and yours will not agree, and theirs is

correct.” And a Jesuit lay brother, crossing toward the Chinese missions in 1689, writes home the version that has kept the file alive in three centuries of marginalia: that his small caravan was weighed at a high cold town of no visible livelihood; that the clerks weighed the men as well as the bales, “with neither insolence nor explanation”; and that his servant, a boy taken on at the last river town, was weighed both on entering and on leaving — the only member of the party so honoured — and that the clerks compared the two figures, and recorded something, and the senior of them said to the boy, in passable Portuguese, words the brother swears to: *the difference is entered against the house*.

IV: Three witnesses; three centuries; three languages; one institution, described identically: the courtesy, the great balance, the ledger, the refusal of payment. And note what is absent, in a genre that records everything saleable: no one ever buys anything in Mīzān, sells anything, sleeps there, or is asked to. It has no economy. It has only an *instrument*. The route-books, which grade every town by what can be got there, could not grade it, and so entered it the way a ledger enters an auditor: without a price, and without affection, and without the least doubt of its authority.

No expedition ever found it. That statement means less on the roof of the world than at sea — whole real towns have hidden in those folds for centuries — but the pattern in the file is not the failure; it is its shape. Mīzān is reported only by parties bound elsewhere, moving laden, in the ordinary anxiety of the trade. Every deliberate attempt — and Voss documents four, from a Qing revenue survey to a Royal Geographical Society traveller of the 1890s who made three seasons of it — approached empty, so to speak: unburdened, purposeful, seeking the town for the town's sake. None so much as found the defile. The reader has met this etiquette before, at the Auroras: *you do not find them by looking. You find them on the way to something worse* — or, Voss amends here, on the way to something heavier.

IV: Sit with the mechanism of Part I a moment, and see how this town inverts it. Our audits kill phantoms: the sounding, the walk-through, the eclipse — we name an hour, the place fails the appointment, the chart is cleansed. Mīzān is the traffic running the other way. It audits *us*. It names no hour; it waits on the road it knows we must use; it takes no toll but the figure; and the caravan passes on, lighter by nothing, heavier by an entry. For fifty years I have called the third place a bank on the evidence of its

deposits. I was slow. A bank is not made of vaults. It is made of *records*, and somewhere the records are weighed, and the weighing has a town.²⁹

The sealed file's last item is a single sheet in the late shaking hand, and it concerns the sign — the small circle divided by a line, which the reader has now met cut into a cairn on Grimsholm's shore and drawn in the margin of the La Palma inquiry. I had assumed, as I believe Voss long assumed, that it was an emblem: the disc of Brasil, the divided island, the shape of the category. The sheet corrects us both, in eleven words:

IV: Fifty years I took it for a shape. It is an instrument. A beam, and two pans, seen end-on. The sign is not the picture of an island. It is the picture of a balance, and it is drawn at the places where the weighing is done. — And below this, sole occupant of

²⁹The route-books exist. The safarnāma is catalogued in a Tehran collection; the Armenian manual survives in multiple copies; the Jesuit letter is in the Society's Roman archive. I verified one — the Armenian manual, in the Bodleian's copy, in January 2025. The *Mīzān* annotation stands on folio 214, where Voss cites it. I report, because this edition has made a discipline of reporting, that the Bodleian's own 1888 catalogue collates that copy as *lacking folio 214*, and has never been corrected, and the folio is there, and it does not have the look of a replacement. I no longer say of such findings that the archive is wrong. I say that the account is current.

the bottom half of the page: *They have weighed me twice now. I do not know what the difference was. I know it was entered against the house.*³⁰

17. The Island of California

From the maps of 1622–1747: an island the length of a subcontinent, lying off New Spain, separated from the mainland by a strait that is now several of the United States. Birth certificate: a novel. Death certificate: a law.

After the choir and the scales, an entry to lower the pulse — though the reader should watch it closely precisely because nothing uncanny happens in it. The Island of California is the control specimen of the interiors, as Hy-Brasil was of the islands: every mechanism of this Part operating in the open, on the largest scale in the Atlas, with all the documents signed and dated. It is the biggest

³⁰Weighed *twice now*. The sheet is undated. If the sign marks the places where the weighing is done, then the reader possesses, as I do, a short list of the places Iris Voss may have stood to be weighed, and one of them is an island that does not exist off the coast of Iceland, and the other is a strait of fog in the Caribbean where she waited nine nights for the door to open. I have begun — I record this as fact, not decision — to keep my own figures. On the 11th of November 2024 I weighed sixty-one kilograms and some grams. The scale is a good one. I bought it new. I keep the ticket.

thing ever to be wrong on a map of the earth, and we know exactly who wrote it, who believed it, who walked its refutation, and who, finally, repealed it.

It was written — the verb is exact — around 1510, by Garci Rodríguez de Montalvo, in a chivalric potboiler called *Las Sergas de Esplandián*: “Know that on the right hand of the Indies there is an island called California, very close to the side of the Terrestrial Paradise, peopled by black women without any man among them... their arms all of gold, as is the harness of the wild beasts they ride.” Queen Calafia, the griffins, the gold — airport fiction of the conquest, read to rags by exactly the men who were sailing west. When Cortés’s captains touched the tip of Baja in the 1530s and took its arid thumb for an island, the sailors reached for the name every literate man aboard already knew. The joke — Voss insists we register that it began as one, the way a crew names a leaky boat *Invincible* — stuck, and hardened, and outlived its own punchline by two centuries.

Here is the sequence, and its strangeness. The sixteenth century got it *right*. Ulloa sailed the gulf to its head in 1539 and found it closed; the great charts of the century show California correctly, a peninsula, soldered to the continent. For sixty years the truth held the maps. And then it lost them.

IV: This is the entry's gift to the doctrine and I know of no other case so clean. We speak of errors as invaders and truth as the recovering army, and assume the traffic runs one way. Here the truth held the territory first, for two generations, with the surveys on its side — and was *dispossessed*. An error re-took charted, verified, settled coastline and held it for a hundred and twenty years. Correction is not a ratchet. Nothing in the record ratchets. The reader who takes one sentence from my Atlas into their own affairs is advised to take that one.

The dispossession has a name and a motive. Father Antonio de la Ascensión, a Carmelite who sailed with Vizcaíno in 1602, came home persuaded — on hope, on hearsay, on the mirage-grade evidence this Part has taught the reader to prize — that the gulf ran through to the ocean and California swung free. His account fed the Amsterdam engravers, who owed the Spanish surveys nothing and the market for novelty everything; in 1622 the island appears on a Dutch title-page, and within a generation it is orthodox from London to Rome. The island California of the seventeenth century is a masterpiece of confident detail: a coastline of named bays facing a strait no ship had ever passed, soundings in water no lead had ever touched. Voss's file preserves her favourite: a chart whose cartouche certifies the island's outline was

“corrected by the latest and best observations,” which, she notes, was in a sense true — the latest and best observations of other maps.

IV: Kong was secreted by a theory; California was secreted by a *market*. The engraver’s shop is a habitat, and in it an island outcompetes a peninsula for the plainest commercial reason: it is more interesting, and no customer in Amsterdam will ever test the strait. Multiply that shop by a century and you have the mechanism — not credulity but *unaccountability*, the gap between where a map is made and where it can hurt you. Every era builds instruments in that gap. Ours are faster.

The refutation, when it came, came on foot. The Jesuit Eusebio Kino, missionary in the Sonoran desert, grew suspicious in the 1690s on grounds that do him honour as an empiricist: the Cocomarcopa Indians of the Gila, who had no stake whatever in European hydrography, kept giving him blue shells from the Pacific coast, and trading them along routes that crossed no water. Shells do not swim straits. Kino made five overland journeys, reached the head of the gulf, watched the land close it like a proof concluding, and drew the map he published under the title that says everything with Jesuit economy:

California: Land of Passage, Not an Island. That was 1705 or so. The reader, trained by Binger and Humboldt, will now expect the island to die of the walk.

It did not. That is the entry's second lesson: the island survived its own walk-through by forty years. Kino's map circulated; the engravers thanked him and reprinted the island; atlases offered both versions, sometimes in the same volume, priced identically. It took the bluntest instrument in the history of geography to end the matter: in 1747 Ferdinand VI of Spain issued a royal decree — an actual law, over an actual signature — proclaiming, in four words that have delighted every historian who has ever quoted them, *California no es isla*. California is not an island. The strait was repealed. The maps, which had ignored the surveys and ignored the missionary, obeyed the king within a decade or two — though Voss, who audited these deaths, found the island alive on a French map of 1765, and on a Japanese one, faithfully copied from old Dutch stock, as late as 1865.

IV: Set the death certificates of this Atlas side by side. A sounding killed Brasil; an eclipse killed Sarah Ann; a walk killed Kong. California — bigger than all of them together — could not be killed by evidence at all. It required an act of sovereignty: the one instrument whose authority operates in the

same medium as the error's own, paper upon paper, seal against seal. The king did not inform the map-makers. He *outranked* them. And the entry closes the circle the guano act opened at Sarah Ann: paper can annex a phantom, and paper can execute one, and at no point in the two hundred and fifty years of the island's life — this is the fact I set the entry down to preserve — was the land itself a party to the proceedings. The strait was opened by a novelist and closed by a king, and between them the desert lay where it had always lain, unwatered and unconsulted, bearing its blue shells.³¹

One item from the file, for the long accounts. Among Voss's tracings is the strait itself, isolated — just the water, the imaginary Mediterranean between island and main, which seventeenth-century convention labelled the Mar Vermejo or Mar de California. On her tracing Voss has written the width of the strait as the principal charts give it, and the figure varies, chart to chart, between one hundred and one hundred and sixty leagues

³¹The decree is real and the wording is as quoted; the 1765 French and 1865 Japanese survivals are documented in the standard census of California-as-island maps, which runs to several hundred distinct plates. All shelfmarks in this entry check. I note it because I have noticed, reviewing my own apparatus, that I now report verification the way a nervous man reports his pulse.

— with one exception. The Milanese atlas of 1849, the reader's old acquaintance, which everywhere else copies its coastlines from sober French stock, alone preserves the island California a century after the decree, and alone gives the strait a sounding. The sounding is 680 fathoms.³²

18. The Sea of the West

La Mer de l'Ouest: an inland sea the size of the Mediterranean, occupying, on the French maps of the mid-eighteenth century, most of what is now Washington, Oregon, Idaho, and British Columbia. Held privately 1700–1750; published 1750–c. 1795. Died of soundings.

Every phantom so far entered the record by accident, appetite, or fraud. The Sea of the West is the only one that was *quarantined* — by its own author, who understood exactly what he had made, kept it locked away for half a century, and was betrayed in the end by the one instrument no cartographer has ever mastered: his heirs.

The author was Guillaume Delisle, and the name is the point. Delisle was the finest map-maker of the age of Louis XIV — the man who purged the Island of

³²680 fathoms is 1,244 metres.

California from French cartography, who shrank the Mediterranean to its true length after fifteen centuries of inherited exaggeration, whose maps were the first anywhere to deserve the word *scientific*. Around 1700, compiling everything Europe knew and suspected about the far west of North America — Indian reports of great waters, the unverifiable strait of the Greek pilot Juan de Fuca, the ordinary hunger of an empire for a route to the Pacific — Delisle sketched a hypothesis: a vast sea, opening from the western coast, reaching deep into the continent. He drew it several times. He considered it seriously enough to write a memoir on it.

And he never published it. Not once, in a working life of forty years at the summit of his profession, did the Sea of the West appear on a map with Delisle's name on it. The sketches stayed in the workshop; the memoir stayed in manuscript; the public Delisle showed the world a west coast honestly vacant. He had invented the most seductive geography of his century and he kept it in a drawer, because he could not verify it, and he was the first man in this Atlas's five hundred years who believed that was a reason.

IV: Honour him. In the entire literature I know of no other case: a phantom conceived, fully drawn, *and successfully contained* — for fifty years the sea

existed in one drawer in Paris and nowhere else on earth. Delisle had discovered, ahead of every academy, the only protocol that actually works: an error cannot be killed once published, therefore it must not be published. The drawer is the quarantine of the third place. Remember the drawer. This Atlas is one, and its author knew it, and the reader is advised to notice whose hands the drawer is in now.

He died in 1726. The drawer held for another generation, and then it was opened by the two men with the best claim and the worst motives: his half-brother Joseph-Nicolas, home from thirty years in Russia with a reputation to rebuild, and his son-in-law Philippe Buache, inheritor of the workshop and a devotee of what the period politely called cabinet geography — the doctrine that a sufficiently elegant theory of the earth need not wait upon its verification. In 1750, before the Académie, they produced the family treasure: the Sea of the West, drawn now not as a question but as a discovery, buttressed by the testimony the reader has been waiting for — the voyage of Admiral Bartholomew de Fonte, who in 1640 had allegedly sailed from the Pacific into a web of straits and lakes across the continent's top and met a Boston merchantman coming the other way.

The de Fonte narrative was fiction. Not error — fiction, composed for a London magazine in 1708, in the golden age of the invented voyage, by an author still disputed; no Spanish admiral of the name exists; the account is a tissue of impossible watercourses. It had lain harmless in the back-files for forty years, a story. Buache and Delisle the younger promoted it to *evidence*, because their sea needed a witness and the dead admiral was available. Fiction corroborated hypothesis; hypothesis dignified fiction; and the composite — argued in memoirs, engraved beautifully, stamped with the most authoritative surname in French science — spread across the ateliers of Paris and outward to the world's atlases. For forty years the schoolchildren of Europe could study a sea the size of the Mediterranean, furnished with capes and islands, in the space where Idaho is.

IV: The mechanism deserves its own line in the doctrine. Kong was secreted by a theory, California by a market. The Sea of the West was assembled — deliberately, in committee, from matched parts: one dead genius's suppressed hypothesis, one anonymous fiction, one unverifiable pilot. Nothing in it was new; everything in it was *inherited*; the assemblers contributed only the decision that the parts should be allowed to corroborate one another. Errors do not

only grow, like Kong, or invade, like the island California. They can be *curated*. And curation, unlike mirage, leaves minutes of its meetings.

It died as the interlude says interiors die — of a walk, though this walk was made by sea. Cook's third voyage in 1778 ran the actual coast where the sea's mouth should have gaped, and found coastline; Vancouver's survey in the 1790s closed every remaining gap with the patient, freezing thoroughness that killed phantoms from Brasil to Sandy Island: soundings, transects, boats sent up every opening until it ended in river or rock. The Sea of the West went off the maps within the decade, unmourned, its lobby dead, its function — France's hope of a private route to the Pacific — extinguished at Quebec in any case decades before.

But the entry does not end in defeat, because this phantom, alone in the Atlas, left an *estate*. Among its foundation documents was the deposition of Juan de Fuca, the Greek pilot sailing for Spain who claimed in 1592 to have found, at 47 or 48 degrees north, a broad strait leading inland toward a shining sea. His account has every mark of the tavern: no Spanish record of the voyage exists; scholars have doubted the man himself. He was, at best, unverifiable — the Sea of the West's third matched part. In 1787 the fur trader Charles Barkley, working the coast

Cook had charted, found at 48°30' north a broad strait leading inland, within a degree of where the old Greek had put it, and named it — the name it bears on every chart on earth today — the Strait of Juan de Fuca.

IV: The First Law, at the end of its tether. An error is a promise the map makes before the world is ready to keep it — and here the world kept it to the very door: the strait is there, at his latitude, of his breadth, leading inland to waters that shine. It does not open into a Mediterranean; it opens into Puget Sound, which is a Mediterranean sold by the yard. I have stood at Cape Flattery where the promise was redeemed and I will testify that the pilot's description is *better than accurate: it is recognisable*. The question this Atlas exists to keep open stands nowhere more nakedly than here. Either a liar in a Venetian tavern guessed a coastline to within one degree, or the coast was, in some sense I have spent my life trying to state respectably, *listening*.³³

One administrative note closes the file. The Delisle drawer — the physical archive: the sketches, the sup-

³³Puget Sound at its deepest, off Point Jefferson, sounds at about 280 metres — I know the reader by now, and no, the figure does not fit the band, and I report it anyway, per the discipline of the Dougherty control. Not every door opens on the vault.

pressed memoir — passed with the workshop's papers into the French naval hydrographic collection, where it survives and may be consulted. Voss consulted it in 1968; her transcriptions check.³⁴ Clipped to her copy of the suppressed memoir is her earliest dated note in the entire four thousand cards — she was thirty-seven; the Atlas did not yet have a name — and it reads, in the young, unhesitating brown ink:

IV: G.D. kept it in the drawer his whole life and died at peace. The question is not how errors escape. The question is what the drawer was FOR. You do not build a drawer that strong for a thing you believe to be false.

19. The Kingdom of Saguenay

[Interleaved note. This edition has until now kept the present out of the entries, on the principle that the apparatus should not compete with the papers. I am amending the principle, for reasons the reader deserves

³⁴Bibliothèque nationale and the naval archives both; the Sea of the West drafts are among the most reproduced curiosities in the history of cartography and their provenance is impeccable. It is restful, once an entry, to report a drawer with nothing impossible in it. I have come to prize the entries that stay shut.

to see accumulate. Last month, through the estate's solicitor, I received an offer for the Voss papers: generous — several multiples of any scholarly valuation — from a private buyer acting through a firm in Basel, whose letter specified, among ordinary terms, "a particular interest in unsorted correspondence." I declined. The solicitor, making conversation, mentioned it was the third such offer since 2020, each through the same firm, each raising the figure, the first having arrived within a month of Voss's death being registered — which is to say before the papers were catalogued, which is to say before anyone could lawfully have known what the correspondence contained. I record the fact and continue. The entries themselves will explain, better than I can, why I have begun to think of the Atlas less as a book I am editing than as a thing I am holding. — E.M.]

The interior phantoms so far were composed by theorists, markets, committees, and kings. The Kingdom of Saguenay was composed by a prisoner, for an audience of one, as the price of going home — and it is the entry where the Atlas's cost column and its authorship column finally meet in a single man.

The man was Donnacona, headman of Stadacona, the Iroquoian town that stood where Quebec City stands. In 1535 Jacques Cartier wintered his ships below the town, uninvited; a quarter of his crew died of scurvy and

the rest were saved by the town's spruce-bark remedy; and in the spring, wanting something to lay before his king besides survival, Cartier took the remedy's givers with him. He lured Donnacona and several of his people aboard and sailed for France. None of them ever saw the St. Lawrence again.

What Donnacona did in France is the entry. Presented at the court of François I — a captive expected to perform his captor's justification — the old man discovered, with a diplomat's precision, the only currency that could buy a westbound passage: geography. He testified — through interpreters, before notaries, to the king himself — of the kingdom that lay up the dark river that entered the great one at Tadoussac: Saguenay, a realm of "immense quantities of gold, rubies and other rich things," where the men were white as Frenchmen and dressed in wool. Pressed, he furnished the marvels the age required — lands of one-legged people, of pygmies, of men with wings — always, the notarial records agree, with unshakable composure; he was, said one witness, more believed than a prophet. The king, at war and in debt, believed what it was profitable to believe. A great expedition was prepared. Donnacona was promised to it as guide.

He died first. In France, around 1539, of what the records do not dignify with a cause — an old man four years into an exile that had begun as a kidnapping, his ransom-

kingdom still unredeemed. The French buried him, the record does not say where, and sailed anyway.

IV: Consider the transaction with the coldness it deserves, because pity obscures the mechanism. Every phantom in this Part was secreted by an interest — a theory, a market, a family, a crown. Saguenay was secreted by *duress*: its author was the hostage of its audience. Donnacona drew a kingdom at the exact bearing that pointed his captors' appetite away from his own town and along the road home; its wealth was calibrated to be worth a fleet; its marvels were quoted from the captors' own library — he had heard what Frenchmen believed the world contained, and he sold it back to them at the price of a passage. It was testimony as cartography and cartography as ransom, and it failed only because the hostage's body gave out before the fleet did. I know of no purer demonstration that the third place is, among its other functions, the bank of last resort of the powerless: a man with nothing left to draw on may still draw a map.

The Cartier–Roberval expedition of 1541–43 — France's largest transatlantic effort to that date — went up the river to find the kingdom without its author. It found

the rapids above Montreal island, which it could not pass; it wintered badly and quarrelled worse; and at Cap Rouge its men loaded the ships with the famous cargo: barrels of gold that assayed in Paris as iron pyrite, and diamonds by the basket that were quartz. The kingdom of Saguenay had converted, at the moment of audit, into the only wealth a phantom can deliver — the appearance of wealth, in commercial quantities. France, disgusted, abandoned Canada for two generations, and the expression *faux comme un diamant du Canada* — false as a Canadian diamond — entered the language, where it remains: the one entry in this Atlas that survives as an idiom.

IV: And note what the false diamonds purchased, since the account must be read to its foot. Two generations of French absence — two generations in which the towns of the river lived unconquered. If Saguenay was a fiction spent to buy time, it bought sixty years, which is more than most true kingdoms have bought with real armies. The map-makers kept the kingdom on the charts for another century, rich and wool-clad, up its dark river; the people who had actually paid for it are not named on any of those sheets. The reader may decide whose error it was, and I give my finding: it was not Donnacona's. His

geography performed to specification. Ours is the profession that filed a ransom note under “discoveries.”³⁵

The kingdom faded from the maps as New France filled in — no decree, no walk-through; it simply had nowhere left to be, and by 1700 the name had shrunk to what it is today: a river, a fjord, a region of granite and cold water north of Quebec, which tourists visit for the whales. The fjord is, incidentally and finally, the deepest water of its kind on the continent: the drowned valley plunges, within sight of the villages, to some 275 metres — dark, cold, layered, its bottom water renewed only rarely from the sea. Voss closes the file with the sounding and, beneath it, the last line of the entry proper:

IV: Not in the band; the band was never the point; the point, as the next paper argues, is what the walls are for. He put his kingdom up a river with a wall of rapids at its door, where no keel he had ever seen

³⁵The notarial record of Donnacona’s testimony survives in summary; the pyrite-and-quartz débâcle is standard history; the idiom is in Littré. All shelfmarks check. The reader will notice the entry contains nothing impossible — no misfiled ink, no anomalous folio. I notice it too. The files that concern what powerful men did to the defenceless are, in my experience of this archive, always in perfect order.

could follow. He understood the grammar of the category before we had named it: a place is safe in exact proportion as it is unvisitable. File under walls. The essay follows.³⁶

20. On Walls

[Entry 20 differs from everything before it. It is the only paper in the Atlas that was delivered aloud: to the symposium of a learned society concerned with the history of cartography, in Vienna, in April 1988. The society's printed proceedings list the paper as withdrawn. It was not withdrawn; it was given, to a full room, and the society elected afterwards not to print it, and the minute-book records the session as having ended early "by agreement." What survives is Voss's typescript, and a cassette recording made — against the society's rules — by a doctoral student in the third row, whose estate returned it to Voss's papers in 2017 with a note reading only "she was right about the acoustics." I print the

³⁶"The essay follows": Voss's own cross-reference, and the first time her index and her prose agree about what comes next. Entry 20 is the paper she read, by invitation and to lasting scandal, to a symposium on the history of cartography in 1988 — the only entry in the Atlas that was ever delivered aloud to a room of her colleagues. The room's response is part of the file, and I have included it.

paper from the typescript, cut for length, and the end of the question period from the tape, verbatim. — E.M.]

—

...I have shown you, then, three walls. A mountain range four thousand kilometres long that detained the geography of a continent for ninety years. A wall of rapids that turned back the largest fleet France ever sent west, guarding a kingdom composed expressly to be guarded by it. And a wall of fog that stands off a certain latitude of the Caribbean — I see some of you smiling; the smile is in the minutes of every institution that ever filed one of these reports, and the reports outlast the institutions — a wall of fog that opens, according to three centuries of depositions, from one side only.

My colleagues, we have all been trained on the same catechism. The blank spaces on the map are ignorance; the walls we draw at their edges are confessions; cartography advances by demolition, and the history of our science is the history of walls coming down. I have spent thirty-five years in the demolition records — I believe I may say without immodesty that I have read more of them than anyone in this room — and I am here to report what is actually in them.

What is in them is maintenance.

The walls of the phantom literature are not ruins and they do not behave like confessions. They are pointed. Kong's hachures thicken precisely where the trade routes press; the fog of San Borondón is punctual to the hour of the offices; the category "not for stations" — I have found it now in two languages, in two centuries, in the files of two institutions that shared no members, no correspondence, and no god — is applied, both times, to islands that had *offered themselves*, and the grammar of both refusals is not doubt. It is policy. Gentlemen, ladies: doubt erodes. Policy is renewed. Show me a wall that is repainted, and I will show you a groundskeeper.

You will want the word for what I am proposing, so that you can reject it precisely. Here it is. I propose that the blank spaces of the historical map were never empty, because emptiness at that scale is not a natural condition; it is expensive, and something paid for it. I propose that a proportion of what this profession catalogues as error is better described as *enclosure* — that certain places were not failed discoveries but successful exclusions, and that the instruments of exclusion were precisely the ones we mock: the doubtful sounding, the disproven range, the island marked E.D., the file sealed by a clerk whose name we do not keep. The map has two functions and we teach only one. It shows the way. It also, and first, keeps the ground. We are not the discoverers' heirs. We are the

groundskeepers', and we have lost the keys, and we call the losing of them the growth of knowledge...

—

[From the tape. Final four minutes of the question period.]

CHAIR: — time for perhaps two more. Professor B—.

B—: A question of method, Frau Professor. When you say “policy” — you are speaking, surely, metaphorically. Institutions repeat formulae; clerks copy clerks. You would not have us believe you mean it — administratively.

VOSS: I am speaking cartographically, Herr Professor, which is a stricter language than either of ours. A metaphor does not recur at fixed depths. A copied formula does not correct its own drift. I mean that the pattern in the record is the pattern of upkeep, and I decline to apologise for the record.

B—: Then I will say what several of us are thinking. Thirty years ago a paper crossed my desk for review which mistook — I said so at the time — the sociology of error for a property of the places themselves. I see the mistake has matured.

VOSS: It has, Herr Professor. So has the evidence. I recognised your phrase then, too.

[Laughter; some of it uneasy. Sound of the chair's water glass.]

CHAIR: A last question. Yes — at the back.

UNIDENTIFIED: A question of staffing, Professor. Your groundskeepers. When a keeper retires — who advertises the post?

[Pause: four seconds on the tape.]

VOSS: The post is never advertised. The post is *offered*.

UNIDENTIFIED: And has it been?

[Pause: eleven seconds. No audience sound whatever.]

VOSS: This session was to end at five, I believe.

CHAIR: — yes. Yes. I believe we are — I believe we can bring matters —

[The tape ends here, forty seconds before the hour. The minute-book, as noted, records the session ending early "by agreement." — E.M.]

—

IV [added to the typescript in the late hand, undated]: I have reread this paper every few years since Vienna, each time intending to feel embarrassment, and each time finding I said less than I knew. One

correction, for the file. I told the room the post is offered. This is true but incomplete. The post is offered TWICE — the first time so that it can be declined, which teaches the candidate what declining feels like; there is no other way to teach it. The second offer does not feel like an offer. It feels like an errand.³⁷

PART III

WATERS & POLES

³⁷The voice of the last questioner does not appear elsewhere on the tape, including the coffee interval the student recorded over in part. German-accented English, age indeterminate — the student's accompanying note offers "neither young nor old," a phrase the reader has met, and which I assume, because the alternative is not usable, is coincidence. Professor B— I have identified with certainty from the society's attendance book; he died in 2003; his review files, which his university holds, contain the 1971 report quoted at Kong, signed with the same initial as the attendance book. That thread, at least, closes cleanly: the hostile referee and the man in the second row were the same person, and he was only ever what he appeared to be — a colleague who disagreed. I set this down with something like affection. In this archive, a man who is exactly what he appears to be is a rest for the eyes.^[^38]

*The islands failed their appointments and died.
What follows concerns the other outcome. — IV,
notebook X, first leaf*

*[Voss's index titles this Part "the kept appointments." —
E.M.]*

21. Terra Australis

**The Unknown Southern Land. On the maps:
from the Strait of Magellan to New Guinea,
girdling the bottom of the world. Charted for two
thousand years before anyone claimed to have
seen it. Appointment kept: 1820.**

Every phantom until now was born of a report — someone saw, or said they saw, or lied that they saw. Terra Australis was born of an *equation*. It is the only place in this Atlas deduced from first principles before the ships existed that could look for it, and the principles were principles of weight.

The argument is Greek and it is beautiful. The known world — the oikoumene, the lands of the north — sits visibly in the upper half of the sphere. A sphere so loaded must tip. Therefore — the *therefore* of men who trusted the world to be well made — there must exist, in the

south, land enough to balance it: a counterweight continent, unseen not because it was doubtful but because the geometry required it to be far away. Aristotle admits the frame; Ptolemy closes his Indian Ocean with unknown southern shores; and the medieval and Renaissance map-makers, inheriting the argument, drew the conclusion — drew it literally, a vast white land wrapping the foot of every world map, labelled with the most honest toponym in the history of the profession: *Terra Australis Nondum Cognita*. The Southern Land Not Yet Known.

IV: The reader of my Part II will permit me one sentence of housekeeping. We have seen errors secreted by theory, market, committee, and duress. Here is the ancestor of them all, secreted by *symmetry* — and I ask the reader to note, filing it beside a certain town in the passes, what kind of argument a counterweight argument is. The oldest case for the southern continent is that the world is a balance, and that the pans must level. Two thousand years before anyone drew the sign in a margin, the whole planet was already being weighed.

For two centuries the continent fattened on scraps. Marco Polo's garbled provinces — Beach, the gold-bearing; Lucach; Maletur — were beached on its northern

coast by map-makers who needed names, and so the empty deduction acquired an economy: gold at Beach, spices at Maletur, parrots at the charmingly literal Land of Parrots. Every southern landfall was annexed to it on contact: Tierra del Fuego was its cape; New Guinea its peninsula; the coasts Tasman brushed were its bays. The continent was not a hypothesis to be tested but a credit line to be drawn on — Voss calls it, in the notebook, “the 南 mortgage,” the great southern mortgage, against which three empires borrowed.

No borrower was stranger than Pedro Fernández de Quirós, the Portuguese pilot in Spanish service who in 1606 put into a bay in the New Hebrides and declared the continent found: *Austrialia del Espíritu Santo*. He founded, on a beach, with ceremonies of tears, a city he called New Jerusalem, and appointed magistrates for provinces that were water. It was an island, of course; the “continent” fell apart into archipelago within weeks, the colony within months; and Quirós spent the rest of his life writing memorials to the crown — fifty at least — begging ships to go back to the promise. He died at Panama, en route to nothing, still owed.

IV: Keep Quirós in view; he is the Part’s first casualty of the appointment. The continent had been deduced for two thousand years; he was merely

the first man vain and faithful enough to keep the appointment in person, and what met him was the standard settlement — an island where the continent should be, a city of sand, a lifetime of memorials. The kept appointment is not a mercy. It is a payment schedule, and the first instalment is always the keeper.

The demolition, when it came, was the most methodical in the history of the search. James Cook's second voyage, 1772–75, was designed as an audit and executed as one: three southern summers, the first crossings of the Antarctic Circle, the great traverses at sixty and seventy south, ice-blink and fog and the rigging frozen into bars, until the possible continent had been squeezed out of the temperate world entirely. Cook's conclusion is the sort of sentence this Atlas exists to frame: a southern continent, if it existed at all, lay "so far to the south as to be wholly inaccessible on account of ice," and its discovery would profit no nation, "and I make bold to declare that the world will derive no benefit from it." The mortgage was called in. Terra Australis came off the maps within the generation — Beach, Maletur, the parrots, the counterweight, two millennia of deduction cancelled by three years of seamanship. It is the largest single demolition in cartographic history, and it was performed flawlessly.

And then the world kept the promise anyway.

In 1820 — Bellingshausen, Bransfield, Palmer; the priority is disputed and does not matter here — ships raised the ice coast of a continent at the bottom of the world: polar, girdling, immense. Not Beach; no gold, no parrots; but land, southern, real, and lying — this is the fact Voss asks us to hold against everything Part I taught us about failed appointments — *inside the outline of the promise*. The deduction had been right for the wrong reasons, or wrong for the right ones; the counterweight was there, made of ice and stone, exactly where the balance argument had always put it: as far south as it is possible to be.

IV: The First Law, redeemed at the largest possible scale, on the longest possible schedule. An error is a promise the map makes before the world is ready to keep it — and here the map made the promise in Alexandria and the world kept it under Alexander I, twenty-one centuries later, having first sent Cook to cancel it so that the keeping would be *clean*. I wrote, at Sarah Ann, that every phantom dies of an appointment. This Part corrects me. Some phantoms die of the appointment. The great ones use it to change trains — the promise steps out of the fiction, which dies, and boards the fact, which is waiting. What died in 1775 was Terra Australis Nondum

Cognita. What berthed in 1820 was Terra Australis
Cognita. The passenger was the same passenger.³⁸

The file's last item is a tracing of Ortelius's 1570 world map — the great atlas-sheet with the white continent wrapping its foot — over which Voss has laid, in red, the modern outline of Antarctica. The fit of the two coastlines is poor everywhere, as honesty requires: the old continent bulges into three oceans the new one never enters. But at the bottom of the sheet, in the late hand, she has written the entry's finding:

IV: The outlines disagree; outlines always disagree; an outline is a rumour. What the two continents share is not a coast. It is a *function*. Each sits where it sits in order that the rest should hang level. Ask what else has been kept at the bottom of a system, unvisited and load-bearing, so that the visited parts may balance — and you have the question Part III

³⁸The “南 mortgage” gloss: the character is simply “south”; Voss used it as an abbreviation throughout notebook X, which was begun, by its flyleaf, “aboard, Hobart to Lyttelton, 1979” — she crossed the southern ocean by working freighter four times in the decade the notebooks go quiet elsewhere. The reader keeping the chronology of inks will note that notebook X is the only place where the brown ink appears after 1980. I note it too, and Part III will have to do better than note it.

was assembled to raise. The polar entries that follow concern the keepers.³⁹

22. The Open Polar Sea

Beyond the ice barrier, 82°N to the Pole. Theorised c. 1527–1896. A warm, navigable sea at the top of the world. Appointment: rescheduled.

The theory is older than the instruments that would eventually kill it, and it survived three and a half centuries on an argument that is, on its face, not unreasonable: that the pack ice ringing the Arctic is a barrier, not a floor — that a ship stubborn enough or lucky enough to force the belt of ice would break through into clear water beyond it, because ice, everywhere else observed, is a coastal and seasonal phenomenon, and a sea does not freeze solid to its own centre. Robert Thorne proposed a navigable short cut over the pole to Cathay in 1527; Daines Barrington collected supposed whalers' testimony of open water north of Spitsbergen in the 1770s and pressed the Royal Society to try for it; and by the middle of the nineteenth century, with steam and

³⁹“Load-bearing”: Voss's word, from the tracing, and I let it stand as written — the reader will recall her rule that the retirement of a phrase is a kind of evidence. This is its last appearance in the papers.

the electric telegraph making the world feel newly conquerable, the Open Polar Sea had acquired what Terra Australis had needed two thousand years to earn and never quite got: two credentialed scientific champions.

Matthew Fontaine Maury, the American superintendent of the Naval Observatory and the century's foremost oceanographer, argued the case from ocean physics — that a warm current entering the Arctic basin could not simply vanish, and must surface somewhere north of the ice as open water. August Petermann, the German cartographer whose maps were the most trusted in Europe and who had never been north of Scotland, argued it more elaborately and more dangerously: a warm Atlantic current — the true course of the Gulf Stream — must breach the ice at Spitsbergen; a warm Pacific current, the Kuro Siwo, must do the same at Bering Strait; the two gateways converged, on Petermann's charts, on a temperate polar basin drawn with the calm hydrographic confidence of a man describing a lake he had personally rowed across.

IV: Note what is being modelled and what is not. Maury and Petermann are not describing an observation. They are describing a *conservation law* — heat goes in, heat must come out, ice is an accounting error waiting to be corrected. It is the most

respectable kind of wrongness there is, because the arithmetic is sound and only the geography is imaginary. I have met this error since in boardrooms wearing better suits. The sum is always right. The sum was never the question.

Belief acquired its proof in 1854, and the proof was a single sentence. Elisha Kent Kane's expedition, wintering at appalling cost in the channel that now bears his name, sent a party under William Morton up the Greenland coast in the spring of 1854; Morton climbed to a headland he named Cape Constitution and reported, looking north across what he took to be the edge of the ice, open water to the horizon — "not a speck of ice," in the phrase that did the work — with swell, with birds, with every visible sign of a living sea. He had in fact been looking at a local polynya, a patch kept open by current and wind, of a kind the Arctic produces regularly and locally and which proves nothing whatsoever about what lies beyond it. It was received, on both sides of the Atlantic, as confirmation entire. Isaac Hayes, who had served under Kane, went back in 1860–61 explicitly to reach the Open Polar Sea and reported, in 1861, that he had stood upon its shore — a claim built on a position badly inflated by the sledging conditions of the day, which later survey moved substantially southward, so that the shore Hayes stood

on and the sea he saw from it both retreated, on re-examination, into the ordinary pack.

IV: A polynya is a hole in the ice kept open by current; the Open Polar Sea is a hole in the argument kept open by desire. Both close eventually. Morton's sentence travelled for forty years on four words — not a speck of ice — and I have found, in three separate popular accounts printed after 1880, the same four words quoted as if Morton were still standing on the headland saying them. He was not. He was a name in a book. The book had learned to talk.

The invoice fell to George Washington De Long, who sailed the *Jeannette* from San Francisco in July 1879, on funding raised substantially on Petermann's authority, to find the warm Bering gateway. He found pack ice north of Wrangel Island, into which the ship froze fast within weeks and drifted, locked, for nearly two years, until the ice crushed her in June 1881. De Long and thirteen of his men died in the retreat down the Lena Delta; relics of the wreck, surfacing years later on the coast of Greenland, gave Nansen the transpolar drift he would stake the *Fram* on. It is the deadliest single bill the theory presented, and it was drawn on a German armchair and cashed in Siberia.

IV: Petermann did not sail. This is not incidental; it is the mechanism. The men who draw the gateway are rarely the men who are asked to find it standing open, and the distance between the drawing hand and the drowning hand is the whole moral geography of this Atlas. I keep a shelf for cartographers who paid in person and a much longer shelf for those who did not. Petermann is on the long shelf. He is in good company there. The company is the point.

Nansen closed the account with a design so deliberately contrary to rescue that it reads, still, as an act of controlled despair: a ship, the *Fram*, built round-hulled to be squeezed up out of the ice rather than crushed by it, deliberately frozen into the pack north of Siberia in September 1893 and left to drift with it, on the working assumption — De Long's wreckage was the evidence — that the drift crossed near or over the Pole itself. It did not cross open water. For nearly three years the *Fram* rode a deep, ice-covered polar basin, sounding bottom at depths the century had not imagined lay under permanent ice, proving by the simple expedient of not sinking that there was no sea to sink into. The Open Polar Sea came off the charts in the decade after, the last respectable printing of it dated to the 1880s and lingering, as a courtesy, into a few atlases of the 1890s that hadn't yet heard.

IV: One figure from the *Fram*'s legacy, for the file: at the shelf break where the theory's warm gateway was supposed to open, the basin surveys find the floor near enough twelve hundred and forty metres down. The theory promised a surface. The world kept, instead, a floor.

Here Voss's essay, in every draft before the last, ends — the standard close of this Atlas, ice claiming the promise as thoroughly as sand claimed Quirós's city. But the file did not close in 1896, because Voss did not stop watching, and the last folder is dated with a decade the theory's proponents never lived to see. The satellite record of the September Arctic sea-ice minimum — the smallest extent the ice reaches each year, after the summer melt — begins in 1979 and has fallen with an unevenness that nonetheless reads, plotted, as a slope: 2007 broke every prior record; 2012 broke 2007's; and by the years Voss was annotating the printouts in the last hand available to her, the modelling centres she clipped and filed were converging on projections of a seasonally ice-free September Arctic Ocean within her readers' lifetimes, on a schedule the century's armchair cartographers had missed by roughly a hundred and fifty years and one hemisphere's worth of gateway.

IV: The reader will want the neat sentence and is owed it: Maury and Petermann were not wrong. They were early, by about a century and a half, and they had the mechanism wrong — no hidden strait breathes warmth into a static basin; the basin itself is warming — but the picture on Petermann’s 1865 sheet, a navigable sea where the pole should be, is now a picture with a delivery date stamped on it by people who have never heard of Petermann. An error is a promise the map makes before the world is ready to keep it. No one has ever asked what happens when the world becomes, on its own schedule and for its own reasons, ready.⁴⁰

The promise, on the evidence of Voss’s own file, did not wait politely for a reader to arrive and admire the closing

⁴⁰Maury’s thermal argument and Petermann’s Gulf Stream/Kuro Siwo gateway theory are documented in both men’s published correspondence with the era’s geographical societies; the Morton sighting and its “not a speck of ice” phrasing are Kane’s own report of the 1854 expedition; the *Jeannette*’s loss and the recovery of her relics on the Greenland coast, which supplied Nansen’s working premise for the *Fram* drift, are matters of settled record. The September sea-ice minimum series Voss was clipping is the same series maintained publicly since 1979 by the US National Snow and Ice Data Center; I have checked her figures against it line for line and found her transcription, as usual, exact to the decimal the source printed. She was not embellishing the decline. She did not need to.

of the loop. It arrived while she was still watching it.⁴¹

23. Frisland

61°N, 22°W (Zeno map, 1558). South of Iceland: an island the size of Ireland, with thirty-one named towns, seven named headlands, a monastery, and a king. The best-mapped island that never existed.

The Buss entry sent the reader here with a weary *q.v.*, and the debt now falls due. Frisland is the parent phantom of the North Atlantic — the imaginary island from which a real ship sighted another imaginary island — and it is the Atlas's masterpiece of a class we have met only in parts before: the forged geography, complete at birth,

⁴¹The clippings are annotated in her hand, year by year, a neat descending column of September minimums, 1979 to 2018, and beside the projected date-range then current for the first ice-free September — a range that begins, in the literature she was reading, in the early 2020s — she has written, in the last hand the papers contain: *My appointment is earlier*. I have no gloss to offer for this that improves on leaving it there. What I can add is a fact I did not check until preparing this entry: the skipper who took her out of Dutch Harbor on her final booking told the inquiry she had asked him to hold a bearing so that she might, in his words, verify a depth sounding. I have now looked up what lies on that bearing. The charted depth is 1,230 metres.

that outperformed honest observation for a hundred and fifty years.

It was published in Venice in 1558, in a slim book by Nicolò Zeno the younger, senator of the Republic, concerning the voyages of his ancestors. Around 1380, the book relates, Nicolò and Antonio Zeno sailed the northern seas in the service of a prince called Zichmni; were wrecked upon and then made governors of the great island of Frisland; visited Estland and Icaria and Estotiland and Drogeo, lands arranged across the ocean like stepping stones to America a century before Columbus — the point of the exercise, no reader in Venice missed it, being that Venetian blood had got there first. The book came with a map: the famous *carta da navigar*, engraved with the whole archipelago, Frisland rendered in loving detail — its fjords, its thirty-one towns with their Italianate-Nordic names, Sanestol and Ocibar and Bondendon — a coastline, Voss notes, with the specific rhythm of coastlines that have been sailed, which is a rhythm a draughtsman can learn.

IV: The Zeno narrative is a forgery; the consensus has held for a century and I do not quarrel with it. Note instead what the forgery understood. An honest report gives you a bearing and a fog-bank; it survives, as we have seen, on hope and copying.

A forgery gives you thirty-one towns, because the forger — alone among the contributors to the map of the world — knows exactly what evidence would convince him, and supplies it. Frisland is not the record of an error. It is the record of a *standard of proof*, met precisely, from the inside.

And it conquered. Mercator took Frisland whole onto the world map of 1569 — the most influential map, it may be, ever drawn — and from Mercator it passed to Ortelius, and from Ortelius to everyone. For a century the island sat south of Iceland on the best charts in Europe, not as a rumour but as a settled, administered fact, its towns named in the gazetteers, its king remembered in the histories. The English took it seriously enough to annex: John Dee, compiling the imperial title of Elizabeth I, listed Frisland among the crown's Atlantic possessions, on the grounds that a land once ruled by the mythical conqueror Arthur was England's to resume. A kingdom's claim, resting on a Venetian family's vanity, to an island that was not there: the paper empires of Part II, the reader sees, had salt-water ancestors.

The conquest did not stop at paper. In 1578 Martin Frobisher, outbound on his third voyage with the largest Arctic fleet the century assembled, raised a high, glaciated coast where the charts put Frisland, went ashore

in the boats, and took possession for the queen under the name West England. The coast was Greenland — the real, enormous, thoroughly inhabited Greenland — misplaced by the charts and now mistaken, in good faith, for a forged island. For some decades afterwards English charts carried the south of Greenland as “Frisland, alias West England,” and the reader should pause on what has happened here, because nothing like it occurs elsewhere in the Atlas: the fiction did not merely persist alongside the world. It captured territory. A real coastline spent two generations enrolled under an invented name, its actual identity — known to the Norse for six hundred years — administratively suspended.

IV: We have watched errors annex blank sea and furnished drawers. Frisland annexed *Greenland*. This is the forgery’s second law: supplied with sufficient detail, the map does not check the world against itself; it checks the world against the forgery, and corrects the world. Frobisher stood on the third-largest landmass of the hemisphere and his instruments told him it was a Venetian invention, and he believed the instruments, because the invention was better documented than the hemisphere.

The dissolution took a further century and proceeded, as Voss's die-off tracings show, by demotion in stages: Frisland full-sized on the charts of 1620; halved by 1650; by 1700 a doubtful scrap, sometimes annotated as sunken — the Buss solution, the decent burial; gone by the 1740s except in the antiquarian trade. The scholarship that finally dismantled the Zeno book itself is nineteenth-century, and it found what such scholarship always finds: the narrative stitched from identifiable printed sources, the map compiled from Olaus Magnus and the portolans, the family archive conveniently burned, the young senator's motive — precedence over Columbus, Venice over Genoa — standing plainly in the preface all along. Frisland was demolished as geography and reclassified as literature, and there, as far as the profession is concerned, the file closes.

Voss's file does not close there. Two items remain, and I give them in her order.

The first is a tracing: Frisland's coastline laid over the Faroe Islands — the standard candidate for its kernel of truth — rotated and scaled. The fit is suggestive and imperfect, as every such fit in this book has been; outlines, we have been taught, are rumours. Beneath it Voss has written not a conclusion but a tabulation: of the thirty-one town names on the Zeno map, twenty-nine can be derived, with patience, from real toponyms

of the Faroes, Iceland, and the Olaus Magnus map — the forger’s sources, duly identified. Two cannot: Sorand, on the island’s southern cape, and Rane, at the head of its western fjord. “Sources unlocated,” Voss notes, “in a document whose sources are its entire anatomy. A forger does not invent freely; invention is expensive; he copies. From what did he copy Sorand and Rane?”⁴²

The second item is a chart, and the reader who has followed this edition will know its city before I name it. The Milanese atlas of 1849 — the engraver who kept a pass through the Mountains of Kong, who alone preserved island-California with a sounded strait — carries, south of Iceland, in the faint hatching the house used for submerged features, an outline labelled *Frislanda (sommersa)*. The outline is not Zeno’s. It is smaller, turned some degrees east, and against its western fjord — at the position of the unlocatable town of Rane — the engraver has placed his customary small figure and a sounding:

⁴²The Zeno philology is real and checkable: the source-hunting is nineteenth-century scholarship, and Voss’s twenty-nine derivations follow it. Sorand and Rane are indeed the residue; the standard literature calls them “unexplained” and moves on, which is what standard literature is for. Voss’s question — from what does a copyist copy the uncopied? — is the question this Atlas has been asking since a marginal circle in a La Palma inquiry file, and I no longer pretend it is rhetorical.

Interleaved: The Milanese File

[This note stands outside Voss's numbering. It is mine, and it is longer than an editor's note has any right to be, and I ask the indulgence once. Three threads knotted in the same fortnight, and the reader is owed all three. — E.M.]

The Braidense answered in eleven days, with the thoroughness Italian librarians reserve for questions nobody has asked in a century. The atlas is the *Atlante universale* of the house of Salterio, Milan, issued in parts, 1846–1849; a mid-market production, competent, derivative, printed in modest numbers for the schools and the seminaries of Lombardy. The firm dissolved in 1851 on the death of its proprietor. The library's set is complete, and its file on the firm contains the usual sediment —

⁴³Third appearance of the Milanese house; third annotation that could not have been compiled from any source I can trace; and the sounding sits where the reader knew it would sit. I have written to the Braidense to ask what is known of the firm — its name, its plates, its proprietor. That inquiry, and what came back, belongs to the next interleaved note, and I will not compress it into a footnote. The reader has been patient with my apparatus; the apparatus is about to require patience of a different kind.

invoices, a privilege from the Austrian censor, an inventory drawn up at dissolution.

Three items from the librarian's summary, in ascending order of the trouble they have cost me.

First: the plates. The dissolution inventory lists the copper for every sheet of the atlas, valued for scrap or resale, except the three northern sheets — the sheets, the reader will already have moved ahead of me, that carry the pass at Kong, the sounded strait of California, and the drowned Frisland. Against these the inventory records: *ceduti* — made over, transferred — with no buyer named and no price entered. In a document whose entire purpose is the entering of prices.

Second: the engraver. The firm's sheets are signed in the colophon by its regular hands, Anfossi and Ronchetti, except the three northern sheets, which carry no name. They carry instead, in the bottom margin, very small, a device the librarian described — she had no reason to describe it otherwise — as “a small circle divided by a horizontal line, perhaps a printer's balance mark.”

Third: the register of consultation. The Braidense, blessedly, has kept its readers' book. The Salterio atlas has been called up eleven times since 1900. Nine were cataloguers and thesis-writers, accounted for. The tenth was in October 1969: *I. Voss, Bergen*. The eleventh was

in March 2019, six months before Dutch Harbor. The name in the book for that day is not hers. The hand, the librarian writes, is “old-fashioned, upright, certainly European”; the name reads **T. Salter**.

I sat with the letter for an evening, and I will write down what I thought, in the order I thought it, because the order is the point of this note.

My first thought was the one this edition has been building toward for a hundred pages: *Salter. Salterio. The house that engraved the doors has a name, and the name initials to T.* — and for an hour I walked the fen road behind my college in a state I decline to dress up: something between terror and the specific joy of a fit, the joy Rennell must have felt when the mountains explained the river.

My second thought arrived near the bridge, and it was colder, and I give it to the reader with both hands because an editor who buries it is no longer an editor. *Voss was there in 1969.* She was at the La Palma archive in 1979, at the Delisle drawer in 1968, in the HBC files at Winnipeg; she handled the Bodleian’s Armenian manual decades before I found folio 214 present against the catalogue; she had fifty years, the skills of the trade, and the run of every archive in this book. Iron-gall ink cannot be dated to the decade. A pass, a sounding, a

marginal balance — a competent hand could add any of them to a nineteenth-century sheet in an afternoon, and the Braidense's conservator, whom I asked, confirms only that the annotations are "consistent with the plate period, not otherwise dateable." Every impossible document in this Atlas lies on Iris Voss's own itinerary. I have been verifying, it may be, not the world but her 路线 — her route; forgive me, her word for it was in the notebooks and I have caught her habit of reaching for other alphabets when the English is too plain. The thesis assembles itself with horrible ease: a brilliant, thwarted scholar, refused by the referees in 1971, scandalised out of Vienna in 1988, spends her half-century salting the world's archives with the evidence her doctrine required, invents three islands outright for her own catalogue, addresses her future editor by name from a fruit boat she may never have boarded, and dies leaving the whole apparatus primed — a posthumous machine for capturing exactly the kind of reader I have been. Under this reading there is no house, no bank, no weighing; there is only a mind of the first order, wounded in 1971, revenging itself on the profession by way of its own posterity. I record that this reading explains nearly everything. I record that it is the most economical account of the file. I record that for three days I believed it, and they were the most restful days I have spent since Bergen.

What ended the rest was arithmetic, not faith. The reading explains nearly everything. It does not explain the eleventh line of the Braidense register, written when Iris Voss was eighty-seven and six months from her boat, in a hand that is not hers, under a name that did not exist for me to notice until I had done four years of work she could not have known I would do. It does not explain the first Basel offer, lodged before the papers were catalogued. And it does not explain — I have checked, with the meanness of a woman hoping to be wrong — how a forger working before 1969 could enter, beside a drowned island, a sounding in metres that agrees with a bathymetry nobody had yet surveyed. The economical account has a hole below the waterline, and the hole is shaped like the thing it was built to keep out.

Which brings me to the fortnight's third thread. The fourth offer arrived on the Tuesday: the Basel firm — it has a name now, and I will spend it: **Halter & Cie, Basel**, *Halter*, the reader of German will note, being what it is — writing this time not of the papers entire but of “the Milanese materials and associated correspondence,” nine days after my letter to the Braidense, of which no copy exists outside my college pigeonhole and the library's registry. When I telephoned, a partner took the call without being fetched. I asked whom they acted for. He said the client's interest was not acquisitive but

custodial: “The client wishes the papers *kept*, Dr. Marsh. Kept has two senses in English, I believe. In our instruction it carries both.” I asked what would happen if I published instead. There was a pause of some seconds, without breath in it, and then: “Then the client would wish you kept also,” and the line went, not dead, but — I choose the word — *quiet*, and I sat with the receiver a while, listening to a silence I could not persuade myself was empty.

I am going to Bergen. The storage unit has been paid up, it transpires, through 2031, by standing order from an account the solicitor cannot trace past its Basel clearing. Folder VII is missing; but folders sit in ranks, and dust is a document, and nobody has audited the shelf itself. The entries that follow — the black rock at the Pole, and the station that would not stop reporting — I have prepared for print before leaving, in case.

In case of what, the apparatus does not say.

24. Rupes Nigra

The North Pole itself. A black rock of lodestone, thirty-three leagues about, ringed by an indrawing sea. Charted 1569–c. 1700. Every compass on earth pointed at it daily.

It is the best-attested place in this Atlas that nobody ever claimed to have seen. Of the whole company of promises kept and broken, the Rupes Nigra is the one furnished with the least evidence and the most confidence: a rock no one measured, on an island no one reached, reported by a source that no longer exists, citing a source that no longer existed even then — and for all that, it possessed what no other phantom here can boast, a permanent, planet-wide instrument dedicated to its worship.

The geography, as it stood on the best charts of Europe for two hundred years, was tidy to the point of theology. At the pole: a black magnetic rock, the Rupes Nigra, thirty-three leagues in circumference, standing at the exact crown of the world. Around it, a sea rushing inward from four directions through four channels, drawn between four great islands set at the cardinal points — and the sea did not simply flow to the rock. It descended, at the centre, into the earth: the waters of the world's oceans, the account had it, poured continuously into a whirlpool at the top of the globe and were swallowed into the bowels of the planet, to surface, some geographers supposed, at the equally hypothetical antipodal vent of the south. It explained, in one stroke, two things sailors actually needed explained: why compasses everywhere on earth pointed one way, and where the tides went. It was elegant. It was operational. It was false in every

particular, and it sat on the world's most trusted maps for longer than the United States has existed.

IV: For three centuries every compass on every ship on every sea was read as testimony — a finger pointing at the appointment. No other phantom in this Atlas commanded a witness on this scale. Millions of needles, twice daily, on every ocean, confirming the bearing. And not one of those witnesses had been anywhere near the destination. They had verified a direction and been credited with verifying a place. This is the single most efficient manufacture of false certainty I have found in forty years of looking, and it required no forger at all. It required only that no one ask what a needle actually knows.

The source chain is the part that should have stopped the whole enterprise at the door, and instead built the door. The account of the polar rock and its four islands descends from a book called the *Inventio Fortunata* — “the fortunate discovery” — said to have been written by an English Franciscan friar, possibly one Nicholas of Lynn, who had sailed north sometime before 1360 and observed the pole's geography at first hand. That book is lost. It survives, as far as anyone in the sixteenth century could establish, only through a second book, the *Itiner-*

arium of a Zeeland traveller named Jacobus Cnoyen, who claimed to have read the *Inventio* or spoken with men who had. That book is also lost. It survives only through a letter — the letter survives; it is the single solid plank in the whole structure — written in 1577 by Gerard Mercator to John Dee, in which Mercator, from memory, summarised what he recalled of Cnoyen's book years after reading it, for the private benefit of a friend compiling England's imperial claims. Every witness in the chain except the final one is missing. The friar's book is gone. The Zeelander's book is gone. What remains is a cartographer's recollection of a traveller's account of a friar's voyage, and on that recollection Mercator built the most influential geography of the polar world ever printed.

IV: The chain of lost books is not, on inspection, an accident of poor archiving. Compare it with Frisland, where a forger supplied every town because invention, done well, is cheap. Here the mechanism runs backward: nothing was invented to fill the gaps, because the sources did the filling by leaving. Each link survives one step further from the origin than the link before it, and the losses are not random — every book that could have been checked is the one that is gone. The file on the pole is a stair with

every step missing but the top one. I have climbed it anyway, because it is the only stair offered, and I no longer trust myself on the question of whether that is scholarship or a kind of complicity.

Mercator did not merely record the geography; he built it, twice, into the two most consulted images of the world his century produced. On the great world map of 1569 — the projection that still, unfairly or not, carries his name — the north polar regions occupy a small inset in the lower corner, because the projection itself cannot show the pole, and there, in that inset, are the rock and the four islands and the four indrawing channels, drawn with the same unblinking confidence he gave the coastlines he had actually surveyed. In 1595, the year after his death, his son published the map Mercator had prepared specifically to correct the inset's smallness: a dedicated polar projection, the first of its kind, in which the geography could finally be shown at proper scale — the rock at the centre, the sea pouring into its four gates, and on one of the four islands, duly labelled, a nation of *Pygmei*, men four feet tall, whom Cnoyen's account (via Mercator's memory of it) placed there living out their short violent lives at the top of the world. The polar map was copied into atlas after atlas for a century. It is, by any measure of influence, one of the most successful maps ever made of a place that was not there.

IV: I set it down at Sandy Island and the pole is where I first learned it: belief moves at the speed of copying, doubt at the speed of embarrassment. A map is reprinted the year after it appears; a correction is reprinted a generation after it is proven, and only once the men who signed the original are safely unavailable to be asked about it. Mercator's polar inset took six years to become his polar map. It took a hundred and thirty-six to come off the atlases in general use. No fact in this Atlas has ever moved at the second speed. None ever will.

The dismantling was slower than the building and considerably less dramatic, because no expedition ever produced a single decisive wreck against the rock the way Cook's Antarctic traverses demolished the southern continent in three seasons. Ships that pushed into the four supposed channels in the seventeenth and eighteenth centuries — after Frobisher, after Davis, after the long Dutch and English hunt for a northern passage — found no indraught, no whirlpool devouring the sea, nothing but the ordinary intransigence of ice, current, and coast, in places that did not correspond to the four gates at all. Meanwhile the instrument that had supposedly proved the rock's existence was quietly turning against it: as magnetic observations accumulated across the seventeenth and eighteenth centuries, the point the needles

actually pointed toward was shown to wander, to sit well south of the geographic pole, and eventually to be not a point at all but a shifting locus with no fixed home — which meant the compass had never been reporting the pole's location in the first place, only a magnetic field's, and the field and the axis of the earth's rotation had never been required to agree. The two poles, which the *Inventio's* geography had fused into a single black rock, came apart in the record books as two separate and independently moving entities.

IV: The demotion is the quiet violence in this file. The rock had promised one appointment — a single point, magnetic and geographic and physical, all three at once, holding the compass and the axis and the island together in one black mass. When it failed, the promise did not die outright, the way the Open Polar Sea's did not die outright. It changed trains. It split into two appointments running on separate schedules, magnetic and geographic, and released itself from ever again being asked to keep both at once. I have watched this manoeuvre before in this Part. I did not expect to watch it here, at the one place theory had insisted, for three centuries, could not be two things.

It fell to James Clark Ross to deliver the physical verdict, and he delivered it in person, in a way none of the polar theorists ever risked. In June 1831, on the Boothia Peninsula in the Canadian Arctic — a coast some fourteen degrees of latitude from the geographic pole and nowhere near the rock's charted position — Ross's dip-needle readings converged on vertical, and he planted a flag at the point his instruments identified as the North Magnetic Pole, having walked to it across ordinary tundra, in ordinary weather, and found there no rock, no whirlpool, no islands, no pygmies: a featureless, low, undistinguished shore, which he described in his own account with the flatness of a man who had expected, at some level he does not admit to, something more. He had reached the appointment that mattered, and it kept nothing at all.

The last item in Voss's file is not narrative but a plot, and it is the entry's true ending.⁴⁴ Across a single sheet

⁴⁴The verification here is, for once, straightforward, and I give it without editorial hedging because the papers deserve one entry where I can. Mercator's 1577 letter to Dee survives and is held among the Cotton manuscripts at the British Library; it is the sole source, transcribed and re-transcribed by generations of polar historians, for everything the sixteenth century believed it knew of Cnoyen and, through him, of the *Inventio Fortunata*. The *Inventio* itself is genuinely lost — no copy, fragment, or reliable independent citation of it has ever surfaced, and the scholarship has long since stopped expecting one. Ross's attainment of the North Magnetic

she has charted the measured positions of the North Magnetic Pole from Ross's 1831 fix to a 2019 reading — the pole's long, slow nineteenth- and twentieth-century drift north-northwest across the Canadian Arctic, and then, plainly marked on her graph with a change in the slope of the line, its acceleration after roughly 1990, from something near fifteen kilometres a year to something near fifty, racing now not toward the pole but past it, out over the Arctic Ocean, on a bearing for Siberia. Beside the post-1990 segment, in the last hand the file contains, Voss has written five words: *the rock is being moved*.⁴⁵

Pole on the Boothia Peninsula on 1 June 1831 is documented in his own published narrative and in the Admiralty's records of the voyage, and its location — nowhere near 90°N — has never been seriously disputed. Voss's chain, tedious as it is to state, is accurate at every joint. I have checked it because it is the one chain in this Atlas I could still hope would break.

⁴⁵I have stopped assuming the late annotations are metaphors, and I record what follows without further comment because the entry has earned a plain close. The drift data are real and public: the World Magnetic Model, maintained jointly by British and American geological surveys, has tracked the North Magnetic Pole's position continuously since the 1990s and shows exactly the acceleration Voss plotted, a shift from roughly 15 km/yr to 50–55 km/yr, the fastest sustained movement in the historical record. The “geomagnetic jerks” that punctuate the readings — abrupt, brief changes in the field's rate of change, clustered more thickly in the recent record than the old models allow for — are likewise real, published, and, as of this writing, unexplained. I took the graph to a colleague in earth sciences at Bristol, gave her nothing but the

25. Vardø-9

Established 1954 on the islet of Kvitholmen, north-east of Svalbard, by the Norwegian meteorological service. Evacuated 1957. Reported until 1960. The islet does not exist. The reports were correct.

The phantoms of this Atlas have until now been places that failed to answer. The file before us concerns the opposite defect, and Voss gives warning of it in the first line of her draft: “What follows is the only entry I would unwrite if I could.”

The paperwork of Vardø-9 is orderly to the point of comedy. In the expansion of the northern observing network after the war, the Norwegian service commissioned a subsidiary station — ninth in the chain administered from Vardø — on Kvitholmen, a low skerry with a beacon foundation, charted at 80°14'N, 31°40'E since a survey of 1899. Four men and a diesel generator were landed in August 1954. The station reported weather — pressure, temperature, wind, ice — four times daily, by radio, in the standard synoptic code, and its reports were unremarkable and its relief crews rotated on schedule and

coordinates, and asked her to extend Voss's line. The projected track, on current heading, crosses the meridian 168° west.

nothing whatever occurred. In 1957, in the reshuffling of stations for the International Geophysical Year, Vardø-9 was closed as redundant: the last crew was taken off in September, the instruments crated, the generator — the log survives — drained and left greased against a reoccupation that was never intended.

The station reported the next morning at 0600, in code, on schedule, with its own call sign.

IV: Fix the sequence, because everything depends on it. The reports did not begin when the crew was landed. They continued when the crew was removed. Nothing announced itself; nothing changed; the 0600 synoptic arrived as it had arrived four thousand times before, correct in format, plausible in content, and the only anomalous thing about it was that the station was empty. An error of the eye lasts a watch. An error of the file lasts a century. This was neither. This was an *office that declined to close*.

The service did what services do. The frequency was monitored: the transmissions were there, keyed cleanly, on the station's assigned schedule, from — the bearings agreed — the station's direction. A patrol vessel was diverted in the summer of 1958 to recover whatever prankster or castaway was working the set. It is at this

point that the file changes genre, because the vessel, arriving at 80°14'N, 31°40'E in fair weather, found open water. No skerry; no beacon foundation; sounding after sounding of empty sea. The survey of 1899 was re-examined and found to rest, like so much in this Part, on a single transit in fog; the beacon foundation had been budgeted but never confirmed built; the 1954 landing party's photographs show rock, hut, and aerial, and cannot show where. Kvitholmen was struck from the charts in 1959 — the standard administrative death, the reader has seen a dozen — with the sole difference that this particular non-existent island was, at the hour of its striking, still reporting the weather.

IV: The service's dilemma was theological and they solved it as bureaucracies solve theology: by filing. The reports could not be accepted, since their station of origin did not exist; they could not be discarded, since they verified. From October 1958 the synoptics of Vardø-9 were entered in the registers under the annotation "origin unattributed," which I commend to the reader as the finest sentence of official prose in this Atlas. The weather was good. The address was bad. The ledger took the weather and quarantined the address, and in doing so conceded — no one appears to have noticed — the whole of

my doctrine: that the observation and the place are separable accounts, and that the world will bank a true report drawn on a false position.

And they verified. This is the finding that moved the entry, in Voss's index, from curiosity to keystone. The phantom synoptics ran for three years — October 1957 to September 1960, some four thousand transmissions — and when, decades later, the modern reanalyses reconstructed the era's Arctic weather from all sources, a doctoral student in Tromsø ran the orphaned Vardø-9 series against the reconstruction as a curiosity. The real stations of the sector scatter about the truth as real stations do: biased thermometers, iced anemometers, tired men misreading a vernier at 0600. The unattributed series does not scatter. Across three years its residuals sit inside the instrumental error of the reanalysis itself — which is to say the file's blunt summary, which I have verified: *the most accurate observations in the sector were supplied by a station that was never anywhere*.⁴⁶

⁴⁶The Tromsø archive is real and open; the orphaned series is where the entry says it is; the comparison study exists as a working paper, unpublished — its author left the field the same year and did not answer my letters, which after four years of this edition I no longer characterise. The residual analysis I have checked myself, with my own copy of the reanalysis fields, on my own machine, twice. The figures are as Voss gives them. I looked for the error for

IV: Every observer adds himself to the observation; that is what an observer is. Warm hut, cold vernier, breakfast, hope — the human signature is the scatter, and the scatter is how we know a man was there. The Vardø-9 series has no signature. It is not the record of someone watching the weather. It is the record of weather that has learned the code — or say it with the sign, since we possess the sign: the reports balance. To the gram. Nothing reporting from nowhere, and the pans perfectly level. I have spent my life proving that places persist after we unlist them. I had not, before this file, been made to consider that they might also *work*.

The last transmission is dated 14 September 1960, at 0600. It is complete and correct: pressure, temperature, wind, nine-tenths ice. Appended, outside the code groups, in plain-language Norwegian, are four words — the only plain language in six years of traffic:

Stenger. Avløsning er kommet.

Closing down. The relief has come.

a week. There is a class of finding for which the honest annotation is the service's own: origin unattributed.

IV: Not “relieved of duty.” Not “abandoning station.” The relief HAS COME — arrival, present tense, someone stepping off a boat onto rock that is not there, taking over an instrument that no one left. The office did not close. The office changed hands. The reader of my Vienna paper asked who advertises the post. I have the date of one handover, to the day, and I observe — filing it, as the service filed its weather, without the capacity to do otherwise — that the day was the fourteenth of September.⁴⁷

26. On Appointments

[Found, like its predecessor, as loose leaves — laid inside the back cover of notebook X, in the late hand throughout, with corrections in no hand I recognise. I print the corrected text. — E.M.]

At Sarah Ann I wrote that every phantom dies of an appointment, and I have spent this Part documenting the

⁴⁷Voss boarded her boat at Dutch Harbor on 14 September 2019 — fifty-nine years, to the day, after the relief came to Vardø-9. The station’s positions, plotted from its own reports across the three orphan years, drift — coherently, as a floe drifts — and the final transmitted position lies on the meridian of 168° west. The reader has met that meridian twice now without my drawing the line. I am drawing it. In her concordance, against folder VII, Voss entered a single cross-reference, and it is this entry.

exceptions, which is the honest name for a career. Let me now say what an appointment is, since the word has been doing quiet work for two hundred pages and deserves its wages.

An appointment is a reconciliation of accounts. The chart says one thing; the world is presumed to say another; and an hour is named at which the two ledgers are brought to the same table and the difference is struck. A sounding is an appointment. An eclipse is an appointment kept by proxy, with the moon standing auditor. A royal decree is an appointment at which only one ledger bothers to appear, and wins. We imagine these reconciliations as moments of truth, and they are — but note the accounting sense of the word *strike*. The difference between the ledgers does not evaporate at the audit. It is struck: written off in one book, written *into* another. Part I asked where an error lives. I can now answer with the precision of fifty years: an error lives in the difference column, and the difference column is conserved, and the appointment is not where the error dies. It is where the error is transferred.

Consider, as evidence, what this Part has shown the appointment doing when the stakes were large enough to watch.

The southern continent was audited by the best seaman of his age, found absent, and written off — and the write-off cleared the ground so completely that when the genuine article surfaced forty-five years later it arrived unencumbered, debt-free, wearing the same name with the *nondum* deleted. The promise did not die at Cook's audit. It was transferred to a cleaner account.

The polar sea was audited by Nansen with three years of his life, found absent, written off — and the difference sat in the column for a century, accruing, until the world warmed enough to pay it. My successors will sail it. The appointment was kept; we had merely misread which party owed.

The rock at the pole was audited by the needle itself and found to be two things rather than one, and the promise changed trains at the platform, as the great ones do — and the difference column in that file is still open, and still moving, at fifty kilometres in the year.

And the station — the station is the entry I would unwrite, because it shows the machinery from the far side, and there are readers I would have spared this. The weather was true and the address was false, and the service, forced to choose, banked the weather and wrote the address into the difference column, where it has sat since 1960, reporting. Nothing in the record suggests the

arrangement is distressed. The books balance. That is what distresses.

I said at Vienna that the walls are maintained, and was not thanked for it. This Part completes the sentence I did not finish in that room. The walls are maintained *because the accounts must balance* — because somewhere the sum of all the differences ever struck at all the appointments ever kept is carried on a single ledger, and that ledger, like any ledger, has a house that keeps it, and the house, like any house, has premises. The islands were the coins. The interiors were the walls of the counting-house. The appointments are the clerks' hours. I have been describing a single institution from three sides for twenty-six entries, the way a blind woman describes a building by walking around it, and there remains only the door — and the reader who has stayed with me knows by now that the doors are labelled, and knows the label, and knows that I know it.

Two admissions, then, before the last Part, which is not like the others.

The first. I have implied, by arranging this book as a scholar arranges evidence, that I assembled the Atlas and then drew the conclusion. The truth is otherwise and the reader has earned it. In 1974 I received a letter I had no reason to expect, from a correspondent I have never met,

who has signed forty-five years of answers with a single initial, and whose first letter contained a sentence I have never printed and will now print once. It read: *You have been reading the difference column, Professor. Most stop at the sum.* Everything since — the freighters, the archives, the fifty years — has been my attempt to earn a sentence I was given free. A scholar is supposed to follow the evidence. I have spent my life following an *acknowledgement*, and I set this down because the distinction is the most important methodological fact about this book, and because the reader deserves to know that the Atlas is not a proof. It is a correspondence course, and I was not its author. I was its best student. There is a difference, and late at night, aboard, with the engine trembling in the plating, I could sometimes hear it.

The second admission concerns the blue notebook, whose absence from my papers the reader will by now have registered, since everything else I ever wrote is accounted for. The blue notebook is the ledger of my own differences. Every archive I have entered, every folio that was present against its catalogue, every depth I have written down: each one struck a difference between the world as filed and the world as found, and I have kept the column, as I was taught — entered in my own book, against my own name. Fifty years of differences, and I am told the pans must level, and I am told there is a

town where the levelling is done. I do not know what I weigh. I know the appointment is set, and that it is not an execution; no audit is. It is a reconciliation. What is left of me after the difference is struck will depend on which of my entries were true, and I have been, on the whole and at some cost, a truthful woman.

The last Part of this Atlas concerns the places that have no other literature: the ones that are not in the archives because they are mine — found not by reading but by being, at last, read. Six entries, and a seventh that the reader must not expect to find, because it is not missing. It is *out*. A book can be out of a library and still belong to it; the borrower's name is in the register; the due date is set. I have seen the register. The hand is old-fashioned, upright, certainly European.

The due date, if the reader must know, is the fourteenth of September.

The Excluded Entries

[Between Part III and the Blank Quarter, the reader must pause with me over a finding I owe to a folder I had misjudged. Folder XXVI, labelled "Schedules," I catalogued as miscellany: ferry timetables, tide tables, eclipse predictions, interfiled across ninety years in an order I could not crack. In February, preparing the

catalogue for Appendix A, I cracked it — or rather I stopped trying, and read the folder front to back like a book, which is what it is. Interleaved among the timetables, at intervals that are themselves a schedule I decline to describe, are three completed entries: finished, fair-copied, headed and ruled like every entry in this Atlas — and absent from the index. Not lost; not drafts. Refused. Voss indexed thirty-three entries and left these three, finished, in the folder of things that run on time. I print them here, in their folder order, with no apparatus: they came to me without any, and the Blank Quarter has taught the reader how to be alone with a text. But I will answer the question the reader is asking, because I asked it for a month: why would she exclude finished work? The answer, I believe, is in the book she taught me to read. The index is a ledger, and a ledger admits only settled accounts. These three are unsettled — one voided before it could be banked, one escaped, and one still running — and a woman who spent her life respecting the difference column would no more index an open account than a bank would print a balance mid-transaction. She filed them with the schedules because that is what they are. Appointments pending. — E.M.]

The First Excluded Entry: The Croker Mountains

Lancaster Sound, 74°N. Sighted 31 August 1818, mid-afternoon. Disproved 4 August 1819. Lifespan: eleven months. The shortest-lived range in the history of geography, and the most expensive glance in the history of the Arctic.

In the summer of 1818 the Admiralty, idle after Waterloo and rich in unemployable heroes, sent John Ross north to find the Northwest Passage. He was a sound officer with a burnt face and thirty years' service, and he did everything right until the last afternoon that mattered, and then he did one thing wrong, and the one thing took eleven months to disprove and the rest of his life to pay for.

His ships entered Lancaster Sound at the end of August — the broad strait north of Baffin Island which is, though no living person then knew it, the true front door of the Passage, the channel through which every successful transit since has sailed. The whalers believed in it; his own officers believed in it; the water was open and the season had days left in it. On the last afternoon of the month, with the sound running wide and westward before him, Ross went on deck between drifts of fog, raised his glass, and saw, some eight leagues distant, a range

of mountains drawn clean across the strait from shore to shore: high land, continuous, unarguable, closing the passage like a wall at the end of a corridor. He named it on the spot for the Secretary of the Admiralty — the Croker Mountains — and, satisfied that the sound was a bay, ordered the ships about, and sailed for England.

The afternoon's inventory deserves its plainness. His second, Parry, below at dinner, saw nothing, and wrote home in guarded fury. His scientific officer saw nothing. The officers of the second ship saw nothing. No one else on either vessel, at any point, saw the mountains, and several were watching the same horizon through comparable glasses. Ross saw them for perhaps ten minutes, once, alone in the sense that matters — and Ross, this must be said and the reader of Peary will hear the difference, was not lying. He put his sighting in the log while the fog was still on the water; he defended it before a parliamentary committee with the baffled stubbornness of an honest man; he went to his grave, forty years on, insisting on his eyes. The Arctic's superior mirage — the same duct of cold air that built Crocker Land for MacMillan a century later — had assembled, from the ice and headlands beyond the horizon, a wall precisely where a tired commander's caution wanted one, and he had believed his instrument, and his instrument that afternoon was a sky with an agenda.

The Admiralty, unamused, sent Parry back the following spring without him. On 4 August 1819 Parry's ships stood into Lancaster Sound with royals set and a soldier's wind, and the officers crowded the rails as the ships bore down on the charted position of the range — there is an account of the silence on deck, every glass raised, the latitude called off — and sailed through the Croker Mountains without wetting a topsail, and onward two hundred leagues into the Passage, and into the promotion lists. The range was struck from the charts before the ink of its first printing was dry. Croker, the Secretary, kept his name on a bay in Devon Island — a demotion from a mountain range to an indentation, which the wits of the wardrooms did not fail to price.

Ross was never given an Admiralty command again. He went north at last in 1829 in a paddle steamer paid for by a gin distiller, spent four winters frozen in, lost the ship, saved his men, and came home a public hero and a professional ghost: a knighthood, and no ship, ever. The mirage had cost the Passage a year, and years were cheap. What it cost Ross was the rest of his career, paid out in annual instalments of hope for thirty years.

And here is why this entry, finished and fair-copied, is not in my index. Follow the money as this Atlas has taught. Every phantom in the ledger banked

something: Brasil banked five centuries of belief; Kong banked a continent's closure; Sandy banked a hundred and thirty-six years of copying. The Croker Mountains banked nothing. Eleven months is too short a tenure; the account was voided before the first deposit cleared; the charts were clean again while the engraver's apprentice was still owed for the plate. By every rule I have derived in thirty entries, the error should have left no residue at all. It left a residue. It left it in a man. The difference that could not be banked against the position was struck, in full, against John Ross — carried on his person, compounding, for three decades; audited at every dinner party; still open at his death. I have found no other case where the ledger, denied a place, took a person instead, and I decline to index what I cannot close, because there is a possibility in this file I have never printed and will print now, in the folder where the timetables live: that the account is still open. That a difference struck against a person does not lapse with the person. That somewhere in the system that keeps the shelf and dusts the slot and prints the second line, there is a page with Ross's name at its head, accruing since 1818, waiting for an heir, an executor, a reader — someone standing at the rail of the record with a glass, honest, tired, and ready to see a wall. The mirage is not the phantom,

you see. The mirage was only the teller. The debt is the phantom, and debts of that kind do not advertise for collection. They wait for the account-holder's estate to come asking after the papers.

The Second Excluded Entry: Thompson Island

South Atlantic, 45 miles north-north-east of Bouvetøya, the remotest island on earth. Reported 1825. Searched for by science for a century. Struck 1943. Cause of death: unknown — and that is the entry.

In December 1825 George Norris, master of the sealer *Sprightly*, working the loneliest water in the world, charted two islands: the one we now call Bouvetøya, which he landed on and claimed, and a second, low and black, some forty-five miles to the north-north-east, which he named Thompson. Norris was a working sealer with no Secretary to flatter and no book to sell; he logged the island the way he logged weather, in a hand that has been examined by a century of skeptics and gives them nothing. Of all the single witnesses in this Atlas — and the reader has met the liars, the tired, the hopeful, and the flattered — Norris is the one the prosecution has never laid a glove on.

Thompson Island entered the charts and behaved, at first, like a respectable phantom: elusive, weather-protected, credited. A sealer reported it again in 1893. Then the age of scientific voyaging arrived in those waters, and Thompson began to fail its appointments — the Valdivia expedition swept for it in 1898 and found water; ships of the whaling surveys crossed and recrossed the position; in 1943 a British warship conducted the formal search that ended it, and the island came off the charts by the usual administrative death.

So far, the reader will say, an ordinary file: a Dougherty, an Emerald, a lonely eye. But Thompson's file will not lie flat, for one reason the twentieth century itself supplied. The seafloor around Bouvetøya is volcanic and alive; and the hypothesis stands in the sober literature to this day that Norris logged a real island — young, black, volcanic — which in the decades after 1825 was destroyed: blown out or foundered by the same forces that built it, leaving nothing for the Valdivia to find because nothing remained. Against this stands the bathymetry, which finds the water deep where an island's stump should be; and between the honest witness and the deep sounding the question has stood open for a hundred years, unresolvable, a coroner's verdict permanently deferred.

Which is why this entry is finished and will not be indexed. Consider what the two verdicts mean in the ledger, because they are not two versions of one death; they are two different economies. If Thompson never existed, it died our ordinary death — audited, voided, its difference struck and carried to the column like all the others. But if Norris was right — and I have read his log, and I would ship with him — then Thompson existed and was unmade by the earth itself, without an audit, without an appointment, without a clerk: subtracted from the world by the world, while both sets of books were looking elsewhere. No difference was struck, because the chart and the sea changed together, in step, and there was never a gap for the column to record. Do you see it? The one clean exit in three hundred years of records. Not a transfer. Not a demotion. Not a stowing at the working depth. An island that may have left both ledgers at once, owing nothing, owed nothing, its account not closed but ANNULLED — and I have spent more hours with this thin folder than with Kong's forty tracings, because it is the only evidence I possess that the system of shelves and weighings and carried differences, which I have spent my life mapping and will shortly enter, has an outside. I cannot index it because I cannot certify the death, and I will not certify the death because

— I am eighty-seven, and the folder of schedules is where the truth goes — because as long as the verdict stands open, so does the possibility. Somewhere in the file of the world there may be one column for the free. Norris saw it, if anyone did. It was black, and young, and it did not last, and nothing weighed it, and nothing keeps it, and no one will ever be sent to verify its sounding. Let the reader hold the thought for as long as it will hold. I have never managed a whole night.

The Third Excluded Entry: María Theresa Reef

South Pacific, east of New Zealand. Reported 1843 by the whaler Maria Theresa of New Bedford. Never found. Position formally recomputed 1983. Present on electronic charts within living memory. Account: current.

The reader has attended, by now, a great many funerals: islands dead of soundings, of walks, of eclipses, of decrees, of memoranda. This last entry concerns the one that is not dead — the phantom still in service, the working instrument the reader's own century has continued, with its satellites and its databases, to carry.

The facts are brief because the file is still accruing them. In November 1843 Asaph Taber, master of the New Bedford whaler *María Theresa*, reported breakers — a low reef awash — in the empty ocean east of New Zealand, and named them for his ship. No vessel has ever confirmed them. The position was searched in the ordinary course of a century's traffic; the reef declined every appointment; and by the rules this Atlas has established the reader knows the biography that should follow: doubtful, E.D., struck. It never came. The reef sailed through the great chart-cleansings of the nineteenth century, through the hydrographic rationalisations of the twentieth, through the conversion of the world's charts to databases — the same conversion that carried Sandy Island to its 2012 embarrassment — and when scholars in 1983 re-examined Taber's log and found his longitude miscomputed, the offices did not delete the reef. They **MOVED** it: recalculated, repositioned, reissued — a formal correction of the coordinates of a place no one has ever seen, one hundred and forty years after its single sighting. It appears on electronic charts thereafter at its new address, annotated with the eleven letters the reader has met: for the avoidance of doubt.

Fiction, which audits nothing, understood the reef's value first. Jules Verne — who read the hydrographic literature the way Poe read the voyages — twice marooned

his castaways on it, as Tabor Island: gave it grass, goats, a castaway, a rescue; used it, in two novels, as the one fixed point of salvation in the empty quarter of the map. Generations of readers have known precisely one fact about that stretch of ocean, and the fact is the reef, and the fact is false, and the fact has saved fictional lives in the only economy where phantoms are allowed to pay out openly.

And so it cannot be indexed, and the reason completes the lesson of these three refused entries, and I will state it as the teller would. Croker's account was voided; Thompson's was annulled, or may have been; María Theresa's is OPEN — not open as a mystery is open, but open as a shop is open. Watch the behaviour, as I have watched it from three ships' bridges across forty years, the reef present each time in the glowing chart, its depth unsurveyed, its existence unbelieved by the very officers navigating around it. The modern office keeps it for a reason it states plainly and does not understand: deletion is the one act the sea never confirms. Just so. The hydrographer of the satellite age has arrived, by pure institutional caution, at the whole doctrine of this Atlas: that removal is the risky operation; that absence must be maintained; that it is cheaper to

carry a phantom than to close its account wrongly — and so they carry it, one working instrument retained against the avoidance of doubt, which is to say FOR the maintenance of doubt, a licensed crack in the world's certainty, eleven letters wide, kept current by the least mystical profession on earth. I checked it last from the bridge of a container ship in 2009. It was there, at its corrected address, patient as a teller's window. It will be there when this book is read. It has a job, and the job is not finished, and I know of no better definition of the living. I have outlived every phantom in my index. This one will outlive me — and the schedules in whose folder I file it will say, to whoever reads them in the order I have left them: the appointments do not end with the appointee. The till is open. The queue moves.

Interleaved: Bergen

[Written on the night train from Oslo, 19–20 March 2026, and printed as written. I have stopped pretending these notes are apparatus. They are the other half of the book. — E.M.]

The storage unit stands in a converted herring warehouse on the north side of the Vågen, and the boxes are

as I left them in 2021, which is the first thing I checked and the last thing I can say plainly.

The facility has a manager, a courteous man near retirement who remembers Voss — she took the unit in 1987, paid by standing order, visited, by his ledger, eleven times in thirty-two years. The ledger is the good kind, bound, initialled. I asked to see it. Three lines concern this note.

The first: 12 March 2019, six months before Dutch Harbor. *I. Voss* — her last visit, four hours, “took away one carton” in the manager’s annotation. The same month, the reader will recall without wanting to, that someone signed the Braidense register in Milan. I checked the dates against her passport, which is among the papers. The passport shows no travel after January. Bergen has an airport and I have no doubt she reached it; I mention the passport because I have promised to mention everything, and because mentioning it costs me the last comfortable explanation of the Milan signature.

The second line: 4 October 2019, three weeks after the boat. The estate was not yet settled; no one had authority; yet the line reads *fullmakt* — by proxy, papers shown — and the visitor’s signature is an upright, old-fashioned European hand that the reader has now met three times: on a symposium tape asking about staffing, in a Milanese

register, and here, in a herring warehouse, three weeks after the death of the woman whose shelf he came to visit. The manager remembers him imperfectly and warmly, which appears to be the only way anyone remembers him. “An older colleague of the professor’s. Or younger. He knew the unit. He was inside forty minutes and took nothing and left the light on.” The manager’s phrase. *Left the light on.* I have carried it for a week like a stone in the shoe.

The third line is dated the Tuesday before my own visit. *Halter & Cie, Basel — inspection under court order.* The order is real; I have now seen it; one of the heirs — I said in my preface they asked not to be named individually, and I keep that, but the reader may know there are four, and that one of the four has been, since January, in correspondence with Basel that the other three knew nothing of — one heir has petitioned to force valuation of the estate “including all documentary assets,” which requires inspection, which requires access, which is how a firm that has been refused four times walked past every refusal with a folder of stamped paper. They were inside three hours. The manager, who did not like them and did not say so, itemised for me what they had handled, because he had stood in the doorway the whole time on the strength of a clause in his own contract he may have invented on the spot. God keep the minor officials of this

world; they are its load walls. What Halter's people photographed was the shelving. Not the boxes. The *shelving*.

So I audited the shelf. Voss's folders stand in numbered ranks, I through XXXIV, in the flat grey dust of thirty years, and the reader knows what I went to look for. Folder VII is not there; that was never in question. What I went for was the gap, and the gap is the finding.

There is no dust in it. In a run of shelving furred like a moth's wing, the width of one folder — I measured it, eleven centimetres — is clean. Not swept-clean, not the smear of a removal: *kept* clean, maintained, the way a step is kept by daily use, and at the back of the clean interval, propped upright against the wall of the shelf, where you would find the card in a well-run lending library, there is a card. Cream, unruled, the size of a reader's ticket. I knew the object before I touched it because I have been describing it for a hundred and twenty pages.

On the front, in an upright, old-fashioned hand: **VII.**
Out.

On the reverse, in brown ink — in the brown ink, the pre-1980 ink, the ink of a decade in which I was learning to read — my name, in full, and one sentence:

When the book is finished, bring it; you know the meridian; decline if you can. — T.

I sat on the warehouse floor for some time. The manager brought me coffee and did not ask, which I record because kindness should be in the register too.

Here is what I understand, stated as flatly as I can manage on a train. The slot has been dusted for years. The absence of folder VII is not a loss; it is a *lending*, administered, current, its card in place, and the card has been waiting for me by name since before I could have read it — unless it was written last Tuesday by someone with a bottle of old ink, which is the fabrication thesis again, thinner each time I need it and never quite empty, God rest it, it has been a good rail to hold. And the sentence is the first offer. The reader of the Vienna entry knows the doctrine: offered twice; the first time so that it can be declined; there is no other way to learn what declining feels like. *Decline if you can* is not a threat. It is, I believe, the syllabus's final course requirement, and I am being examined on it now, in this carriage, each time I look at my bag where the card lies flat in a sleeve between my own notes and hers.

What I have decided, watching the fjords go by in the dark, is this: I will do what she did, which is the only method I was ever taught. She was given a sentence and spent fifty years earning it. I have been given one too. The six entries that follow are Voss's own — the Blank Quarter, the places with no literature but her. I will print

them exactly, and I will write the afterword I owe, and then the book will be finished, and the sentence on the card will come due, and the reader and I will discover together what I am — a scholar, or a difference about to be struck.

The train reaches Oslo at six. The light in the unit, I should say, I turned off.

PART IV

THE BLANK QUARTER

What follows has no literature. These are the places I did not find in archives. The archive found me. —
IV, folder XXIX

[The six folders of the Blank Quarter differ from everything before them. They contain no tracings, no photostats, no correspondence with librarians — nothing that can be checked, and I have checked nothing. Each holds a field notebook, first person, in her hand. I print them exactly, in the order of her index. My apparatus, which has grown garrulous, will be brief here: there is nothing

for it to verify and we both know it. The reader is as alone with these as I was. — E.M.]

27. The Anchorage

From folder XXVIII. A notebook of the m/s Herdis, Hobart toward Lyttelton and beyond, southern winter, 1982. No position is given anywhere in the folder. That is the entry.

I have spent eleven entries of this Atlas teaching the reader to distrust single witnesses, and now I must ask for the courtesy I have spent a book undermining. What follows happened to one woman, at night, at sea, in her fifty-first year, and she wrote it down within the day, and she has checked it against nothing because there is nothing to check it against. Weigh it accordingly. I have.

We were nine days out in weather the mate called ordinary and the cook called biblical, and on the ninth evening the engine note changed. Not trouble — a helm order. I had stood enough watches by then, paying my passage in coffee and quiet, to know that we had come some points north of our course, and I went up, and the master was on the bridge with the second, and neither spoke, and the ship was running with a following sea toward a dark that was — I want the exact word and there

is none in English, so I will build it — a dark that was *administrative*. Not weather. A jurisdiction.

We put in. There is no other verb; the sea organised itself into an approach; the swell stood down as if a hand had been laid on it; and the Herdis, nine days from any charted lee, rode at anchor before midnight in still water, in the shelter of land I could smell — wet stone, peat smoke, tide-wrack — and could not, in that dark, see. Around us, at moorings, other vessels: I counted eleven riding lights, and they were not all of our century. I say that plainly and will not decorate it. A riding light tells you its era the way a hand tells you its writer, and two of the eleven were flames.

No one came aboard. The crew did not go ashore, did not lower a boat, did not speak above the necessary. This was not fear as I know fear. It was the behaviour of men in a place where they are known — the punctilio of a rough parish in its cathedral. The cook brought up food nobody ate. At one bell I asked the second, who had stood aside for me on the ladder all voyage and never offered three words together, where we were. He looked at me with a kindness I still turn over, and said: “In the roads.” Which roads, I said. And he said, “The roads, Professor,” as one corrects a child who has asked which sky, and went below, and left me the bridge wing and the smell of a country I will not find.

I did what I am. I took bearings on the shore lights — there were shore lights, low and few, the colour of banked fires — and noted the anchor bearing and the depth from the sounder, and I wrote the figures in this notebook by the binnacle lamp. They are on the next leaf. The reader of my Atlas, if these notes ever have one, may turn to it now.

*[The next leaf is in the folder. The recto carries, in her field hand, three compass bearings and a sounding. The ink — I state this because I must — is the brown ink throughout, and the sounding reads 1,241 metres, which the reader will observe is given from an anchored ship, in still water, riding in shelter. I offer no comment except the one arithmetic requires: no anchorage on earth lies in twelve hundred metres of water. Ships cannot anchor there. The Herdis rode to her cable all night. — E.M.]*⁴⁸

⁴⁸One matter of record, and then I will keep my promise of brevity. The m/s Herdis existed; Norwegian registry; broken up at Alang in 1994. I have found her logbook — the line for the southern winter of 1982 reads as Voss quotes it — and I have found one crew list, and the second officer's name on it, and the name is not T. and not Salter and I did not expect it to be. He is alive. He is ninety-one, in a home outside Bergen, and I visited him in March, the day before the warehouse, and I asked him about the roads. He looked at me a long time with the eyes they all seem to have, the kindness with a fathom under it, and he said: "She wrote it down, did she." Not a question. I said she had. He nodded and looked out

Toward four the tide turned, or something did. The riding lights stood up one by one and went out — not extinguished: *withdrawn*, the way a shopkeeper takes the lamp from the window at close of trade — and the smell of land faded first from the west, which is not a direction smells fade from, and the master said “heave in” in the voice men use in churches and sickrooms, and we came out as we had gone in, stern-first through a dark that handed us back to the weather like a coat returned at a cloakroom. By six the swell was ordinary and biblical again and the engine had its sea-note, and the log — I have said the crew were punctilious — the log for the day reads *hove to in heavy weather, no progress made*, in the master’s own hand, and the noon positions bracketing that day are eighty miles apart.

I asked no one anything at breakfast. This was the second thing I learned in the roads: the etiquette travels with you. But when we raised Lyttelton the master, paying me the compliment of my profession at the gangway, held my hand a moment too long and said the only sentence anyone ever offered me freely in fifty years of this work. “You’ll have noticed,” he said, “we didn’t pay.” And seeing that I had not understood — I understood four years later, at a weighbridge of another kind — he touched his

the window at the rain and said, “Then it’s her entry against ours,” and would say nothing else at all.

cap and said it again, kindly, the whole doctrine in six words, the sentence I have put at the head of the folder and will put at the head of the Part if the editors of my estate have any ear at all:

“Anchorage is free. The chart pays.”

Later, aboard, same hand, margin: They keep a roads for the working traffic of the world, and the fee is entered not against the ships but against the map. Every vessel that has ever ridden there is a vessel some chart has lost — an error of position, an impossible noon, a day that had to be written as weather. We have been reading those log entries for three centuries and calling them incompetence. They are receipts.

28. The Carrier

From folder XXIX. A notebook of nine days in Tórshavn, Faroe Islands, October 1996. The usual place.

Sixty of the letters came by hand, and for twenty-two years I did not ask whose. This was not incuriosity. It was the etiquette, which I had learned at sea before I could have named it: in the roads, no one goes ashore. But in

the autumn of 1996 a letter ended with a postscript that was, I understood much later, itself the lesson — *You may watch, if you like. Watching is free* — and so I went to the usual place, and I watched, and this folder is what I have to show for the only surveillance I ever conducted in fifty years of being, myself, so plainly watched.

The usual place is a bookshop by the harbour in Tórshavn: low ceilings, peat-brown light, sailing directions and psalm books and the strata of a reading town. My arrangement was twelve years old by then and had never varied. Replies to my correspondent went inside the back cover of a particular copy of the *Norwegian Sea Pilot*, third edition, a book so obsolete and so unlovely that its patience on the shelf was its own explanation. Answers came to my hotel, or to Bergen, or once — I have never printed this — to a cabin I had told no one I had taken, on a ship I had boarded under a name that was not mine. I placed my letter on the first evening, and then I did what the trade of watching requires: I took the window table in the café opposite, and I kept the shop's door in view for nine days.

I will spare the reader the field notes, which record, hour upon hour, the retail life of a small northern bookshop in October: schoolchildren, clergy, a man who came four days running to fail to buy the same atlas. No one went to the shelf. On the fifth day I varied the method, as one

does when the instrument may be the error: I checked the Pilot at opening and at noon and at closing, standing in the aisle myself with my thumb in the back cover. My letter, patient. On the ninth evening, at ten minutes to closing, with the shop empty and the owner at his ledger and my own eyes on the shelf — I state it so, flatly, and the reader may bank it with the rest of my account — the letter was not there, and an answer was: four pages, the upright hand, the ink still smelling faintly of iron, answering points I had raised in a letter that had left my hands two hundred and twenty hours before, in a room I had not stopped watching.

I did not go back to the café. I went to the ledger, and the owner — Jespersen, since he is dead now and his name was his own — looked up at me and closed the book, and I saw that he had been waiting nine days for me to stop being a scholar, with the specific patience of a man who has waited longer for stranger things.

What he told me, over the shop's peat stove, with the sign turned and the harbour going black in the window, I set down nearly in his words, because his words were better than mine.

He said he had held the office thirty-one years. He said it came to him the way it comes: an envelope with his name on it, found in the effects of the man whose shop he had

inherited — the name written, he said without theatre, in an ink older than he was, and he had known what the envelope meant before he opened it because — and here he looked at me — “one always knows; the reading of it is only the countersignature.” I asked what the office required. He said: nothing. He said: “I have never carried a letter in my life. I keep the shelf.” I asked how, then, the letters moved. And he gave me the doctrine that I have carried since like a coin sewn in a coat, the sentence the whole Atlas had been circling for forty entries without the wit to land on:

“The letters do not travel, Professor. The shelves do the travelling. A kept shelf is one place, wherever it stands. There are not many of us, but there do not need to be many of one place.”

Margin, same night: The office is not a person and was never a person; Philip was not the first carrier but the first PREMISES — a man made into an address. The network is not a web of couriers. It is a single room, distributed. Tórshavn’s back cover and a hotel desk in Veracruz and a mail sack on a dead ship are one shelf, and the letters lie still on it while the world’s positions change around them. Of course the replies are prompt. Nothing has any distance to cross. I asked him what happens when a keeper

dies. He said, "The shelf is kept," in the grammar the Admiralty used about stations, and put another brick of peat on the stove.

One thing more, and then the folder closes. As I stood to leave he opened the drawer of his ledger-desk — I had not asked; this was given, and I believe it was the actual appointment for which my nine days of watching were the fee — and showed me what it held. A shallow tray of letters, perhaps twenty, each in the upright hand, each sealed. "The uncollected," he said. Letters that had reached the shelf and never been fetched: the addressees had declined, or died before their date, or — his phrase — "were re-addressed in transit." He let me look; he did not let me touch; and I record, because this book is a ledger and a ledger does not flinch, that the topmost envelope carried a name I knew, and that the name was not mine, and that I have spent more nights than I will confess deciding whether to print it, and I will not. She may yet collect it. It is not for me to strike her difference.⁴⁹

⁴⁹I have read this entry more times than any other in the papers, for a reason the reader of my Bergen note will supply without help. In February I wrote to the shop — it survives, under Jespersen's grandson — asking, as a scholar does, dressed in my college's letterhead, whether the business held any correspondence relating to Professor Voss. The reply came in nine days. It is typed, courteous, one sentence, and I give it whole: "Your letter has been placed." I have not written again. The book you are holding is my second

I took the boat south the next morning. At the rail, in a wind like a slammed door, I understood the postscript at last. *Watching is free*. So it was: nine days, and no toll taken, and nothing shown to me that was not freely given. But I had placed a letter and received a letter, and between those two acts I had sat at a window counting the traffic of a kept place for two hundred and twenty hours, and somewhere — I knew it as I knew my own name; the knowledge has the taste of iron — my count of the shop had been entered, in a fair hand, against my page in a ledger I had been shown precisely once, in the form of a drawer I was not permitted to touch. Anchorage is free. The chart pays. Watching is free.

The watcher pays.

29. The Second Sight of Brasil

From folder XXX. A notebook of four days on Inishmore, Aran Islands, September 1991. The folklore allows the island one showing in seven years. The folklore is off by nothing.

letter, and I have begun to suspect it will be delivered the same way — that publication is not an announcement but a *placement*, and that somewhere a shelf has been kept for it a long time, and that the reader now holding it open has, in the only sense that has ever mattered to this Atlas, just walked into the shop.

Three years after Vienna, when I had stopped being invited to things, a letter came that contained no answer to any question I had asked. It contained a date, a cliff, and an hour, and the sentence: *Dress warmly. The showing is long this cycle.* I have received, across forty-five years, perhaps a dozen sentences from my correspondent that were instructions rather than replies, and I have obeyed all of them, and I note — for the file, for the successor, for the court that will never sit — that obedience was never once demanded. An instruction from T. has the grammar of a train timetable. One is not commanded to catch the 14:06. It is merely true that the 14:06 will run.

I took rooms above the harbour and walked out on the third evening to the cliffs at the island's back, where the land stands three hundred feet against the Atlantic in slabs like the minute-books of a very old firm, and I sat down out of the wind in the socket of stone I had been directed to, and I waited for the top of the hour, and at the top of the hour the sea to the west stopped being empty.

I had spent thirty years by then establishing what the look of nothing is, and I must now put on record the look of the other thing, and I find the plainest way is the truest: it did not appear. It was *disclosed*. The horizon did not produce an island; the horizon proved to have been, all along, a curtain hanging slightly proud of the wall, and for the length of the showing the hang of it altered,

and behind it stood what the charts of five centuries had copied from one another with such fidelity: land, round-shouldered, divided across its middle by a channel that took the last of the light like a seam of tin. Dalorto's disc. The cleft. The very punctuation of the coastline agreed with the portolans — agreed, I mean, the way a face agrees with a portrait, establishing in one glance which of the two had been sitting for the other for five hundred years.

I did what I am. I sketched, timed, took bearings I already knew would not survive the evening; the notebook page is in the folder, and the reader who consults it will find the bearings written in a hand that starts as mine and steadies, over four lines, into something with no tremor in it at all, because — this is the finding of the first hour, and I set it down against every account of such things in the folklore — the state that came upon me on that cliff was not ecstasy and was not fear. It was *competence*. I have felt it since only once. It is the feeling of an instrument brought at last into the conditions it was calibrated for, and it is the most seductive feeling I have ever had, and I believe it is the beginning of how the post is filled.

At the second hour a man came up the cliff path with a lantern he did not light and stood at a distance that was exactly courtesy, and waited to be noticed, and said,

when I turned: "There's water under her tonight. There's time to row it, if we go now."

He was old in the way the second officer of the Herdis was old — the kindness with the fathom under it — and his currach, he said, was on the shingle at the gap below, and he said nothing else at all: no name, no price, no persuasion. The showing stood in the west. The channel through the island's middle was a road of last light. I record what I can of the next quarter hour, which I spent in the ordinary agony of a woman at a departure board, and the reader — my reader, the one this folder was always for — must forgive the poverty of the account, because the state of declining cannot be observed from inside while it is being performed; it can only be reported afterward, from its wreckage, the way weather is.

I said: not tonight.

He nodded — I will carry the nod to whatever weighing awaits me; it was the nod of a man confirming a delivery, not accepting a refusal — and went back down the path, and I stayed on the cliff, and at moonrise the curtain recovered its hang, and the horizon resumed the imposture of a wall, and I walked back across the island in the dark, and in my room I was sick with something that was not illness, and slept fourteen hours, and woke knowing three things as facts, the way one knows soundings.

That the offer was real, and had been mine, and was gone. That it had been made early and made resistible — the lantern unlit, the price unnamed, the persuasion withheld — because it was *meant* to be declined; that declining was not the failure of the appointment but its content; that I had been taught, in one evening, at the cost of one island, what declining feels like, and that nothing about the lesson was cruel except its accuracy.

And that there would be a second, in its own year, in another place, dressed — I did not yet know the dressing — as an errand; and that between the first offer and the second the only work I would find worth doing was the work of this Atlas: the keeping of my column, the earning of my sentence, the long discipline of writing down the differences of others while my own difference stood open on the page, accruing, patient, entered against a name I have spent fifty years learning is spelled the way mine is.

Margin, later, black ink: The folklore says the island shows once in seven years, and the folklore is exact, and the scholars who collected it never once asked the question their own material begged. A thing that keeps a schedule is not a ghost. It is a SERVICE. The showing is not an apparition; it is a sailing, and it has been leaving on time, from the same berth, for

as long as the west of Ireland has kept records — and the record of a service is not its legend. It is its ridership.⁵⁰

30. Grimsholm, Again

From folder XXXI. A notebook of two days out of Raufarhöfn, Iceland, June 2003. The island was struck from the charts in 1820. The island was not consulted.

The reader of Part I will remember Grimsholm: the black-sided holm north-east of Iceland with the sound through its middle, a century and a half of Danish sight-

⁵⁰The arithmetic, since Voss for once does not do it for us: from her showing of 1991, the seven-year schedule gives 1998, 2005, 2012, and 2019 — the reader may consult the preface for what Iris Voss, eighty-eight years old and four thousand miles from Aran, chose to do in the autumn of the 2019 sailing — and then 2026. I made my own visit to those cliffs in the winter of 2021, and I recorded in my first footnotes, with an editor's complacency, that I saw nothing. I understand the complacency better now: 2021 was not a showing year. I went in the wrong year, which is, I have come to think, why I was allowed to go. The next sailing is 2026. This edition, the reader may care to know, is published in 2026; the date was set eighteen months ago by the publisher, on grounds of paper supply and the autumn lists; I have checked, because by now I check everything, and the schedule is entirely innocent, and I have stopped believing that innocence and coincidence are different words for anything.

ings, a furnished cairn, an Admiralty that declared it not for stations, and an erasure after 1817 so complete that the record does not even class it among the errors. My entry on it closed with a sounding and a suspicion. This folder is the suspicion, paid in person.

The conditions came, as conditions come, by letter: a port, a month, the name of a skipper who would be going out for lumpfish and would not mind company, and one instruction of the timetable kind — *Take a stone from your own ground. You will know the plate when you see it.* I carried a grey beach pebble from below the Bergen warehouse in my coat pocket for three weeks, and I have never in my life been so continuously aware of eleven grams.

The skipper was the age they are, with the hands they have, and he asked me nothing at all until the second dawn, when the fog came down — the reader knows by now what the fog is; the gap under the door; I have stopped explaining my own weather — and he throttled back and said, conversationally, watching the grey: “You’ll have brought your stone.” Not a question; the countersignature again. I said I had. He nodded and put us in.

It is exactly as the Jægeren’s men deposed in 1688, and I mean exactly, and the exactness is the first finding: black

sides with the rust in seams — *ruddled*, the old word, the right word — the sound running north and south through the island's middle like the slot of a money-box, the beach of black sand warm underfoot, not volcanically, not explicably, warm the way a chair is warm when someone has risen from it. Three hundred and fifteen years between their landing and mine, and no drift in the description at all. The reader of this Atlas has been taught that errors do not degrade only when something maintains them. The doctrine holds ashore. The island is kept like a lighthouse lens.

I walked the sound first, because I am what I am. It is perhaps four hundred metres of water between walls of dark stone, and I paced the western shore of it end to end in nine minutes going in. I will state the second finding as navigation and let it lie: the walk out, by the same shore, at the same pace, took twenty-one, and at the midpoint going out — only going out — the light through the northern mouth was the flat white light of high summer over open sea, and the light through the southern mouth, two hundred metres away, was low, and amber, and lay on the water like evening in a latitude I could not name. The sound is longer inside than the island is wide, and it does not everywhere open on the same day. I did not linger. There is a competence that comes on me in such places;

I have described it at Aran; I have learned to ration it like a drug, because it is one.

The cairn stands above the eastern shore where the depositions put it, and before it in the turf lies the flat stone, and on the flat stone, cut deep and kept clean — kept, the reader understands; nothing on that island suffers moss — the sign: the circle divided by a line. The beam and the two pans, seen end-on. I had drawn it for fifty years. I had never once stood on the pan.

The cairn itself is the manifest. I counted — of course I counted; it is the only instrument I brought that cannot lie — three hundred and forty-one stones, and they are not local: grey Baltic granite and red Devon sandstone and a flint from some chalk coast and one palm-sized piece of what I believe was ship's ballast, basalt, dressed. Three hundred and forty-one landings. The ridership, banked. The 1688 men found the cairn already standing, and now I could read the sentence I wrote about them with its true stress: the island came furnished — *because the furniture is the ledger*.

I did what the letter said. Stone on the plate, then on the pile.

I will not do here what I have done all my life, which is reach for the instruments, because the instruments have nothing to say about the interval — I think it was

an interval; the fog neither thickened nor thinned, and the waterfall behind me did not pause, and yet between the pebble's touching the flat stone and its settling there, something was subtracted from me and something was added, in quantities I could not then and cannot now name, and the nearest I can come after twenty years is this: that I arrived on that island carrying fifty years of open questions, and I left carrying the same questions closed — not answered; *closed*, the way an account is closed, the way a file is closed, rolled shut into a figure and carried forward — and the figure is in me, and I do not know its sign, and I walked down the black beach to the boat with my hands smelling of iron though I had touched only my own clean pebble, and the smell has never entirely washed out, and I have learned to be grateful that ledgers do not explain themselves to their entries.

Margin, much later, the shaking hand: Iron in the seams of the island; iron in the gall of the ink; iron on the tongue at every threshold; rust the colour of every door — Bermeja red, the ruddled walls, the amber light in the sound going out. The ledger of the world is kept in the one metal that remembers water. They have weighed me twice now. This was the first, and I did not know it until the second, which is the

way of weighings, and of offers, and of everything else that is done to us twice.⁵¹

The skipper took us off at slack water. He looked at my hands once — I had said nothing — and gave me the nod, the delivery nod, and we ran south-west out of the fog into a bright ordinary sea with Iceland standing on the horizon like a fact. On the sounder, as the island went from sight, the depth: 1,205 metres. The 1817 survey found the same figure over the position it then struck from the charts. Two hundred years of hydrography have improved every number in the northern ocean except that one. It does not need improving. It is not a measurement. It is a registration.

⁵¹The folder contains, besides the notebook, a small cotton bag of the kind coin-collectors use, machine-sewn, not old. In it is a black pebble, basalt, sea-worn, twenty-two grams — not the grey Bergen pebble, whose fate the entry accounts for. The bag is labelled in the upright hand, not hers: “For the successor. Return if declining.” I have weighed it three times on the scale whose ticket I keep. It has not changed. I carry it now in my left coat pocket, and I am aware of it, as she was of hers, continuously; and I note, because the reader and I no longer pretend otherwise, that a stone from Grimsholm is a stone from my own ground in no sense I can construct — unless the phrase was never about geology, and one’s own ground is wherever one’s difference is banked, in which case the labelling is exact, and the island is expecting me.

31. The Shelf

From folder XXXII. A notebook of the cable ship Njord, North Atlantic, 1973; with a later leaf. The reader has been keeping this column since page one. It is time it was paid.

Set them out, before I explain them, the way I first set them out on a card in 1973.

Twelve hundred and forty metres, the mean of the seen-and-sounded. Eleven hundred and ninety at Bermeja. Twelve hundred and five over Grimsholm, twice, a century and a half apart. Twelve hundred and thirty-eight at the drowned Frisland's one impossible sounding; twelve hundred and forty-four in the strait that California never had; twelve hundred and forty, in words, under the fog where the choir sings; twelve hundred and thirty on the bearing I gave a skipper at Dutch Harbor. The figures scatter like good soundings around a true depth, and the scatter is honest — I have shown the reader my controls, the deeps that ignore the band, and they are real, and they matter. Not everything is kept. That was never the claim. The claim is that what is kept is kept *together*.

In 1973 I was supernumerary aboard the Njord, a cable repair ship working the great-circle telegraph routes, because I had noticed something in the industry's fouling reports and no university would fund me to notice it

further. The cable trade is the one profession on earth that touches the deep floor routinely — has done since 1866, with grapnels, on schedules, keeping logs — and its logs record, in the flattest prose in the maritime world, every object its hooks have ever brought up from the dark: wreck iron, whale-fall, lost war, and, at intervals that are not random, in a depth stratum that is not wide, entries of a different kind. *Foul at 660 fms, cleared, returned.* Foul at 680. Foul at 675. Cleared. Returned. Never described. Never retained. *Returned.*

On the ninth day of a repair in mid-ocean the grapnel came up foul at six hundred and seventy-eight fathoms — twelve hundred and forty metres, the reader will have done it faster than I did at the rail — and wound in the flukes, wound twice, deliberate as a spliced eye, was a deep-sea sounding lead. Old. The pattern of the seventeenth century: the long taper, the hollow arm for the tallow that carried the sample of the bottom up to the leadsman's thumb. It should have been two hundred years of corrosion and it was clean, working-clean, the way a tool is clean that hangs in a kept shed, and stamped in its base, deep and unworn, was the circle divided by a line, which in 1973 I had seen exactly once, in a margin in La Palma, and had taken for a doodle.

I reached for it and the bosun's hand closed on my wrist.

He was not rough. It was the grip you use on a colleague at a kerb. He cut the lead free with his own knife, wrapped it in sacking with the economy of rite, and lowered — not dropped; *lowered*, on a line, paying it out by hand for twenty minutes — lowered it back over the position, and entered in the log, in the prose of his trade: *foul cleared, gear returned*, and the ship resumed her splice as if nothing had happened, because — I came to understand — nothing had. Something had been found on a shelf and put back on the shelf. That is not an event. That is *shelving*.

Coffee, later, on the cable deck. I asked him what the lead was. He considered me for a long time — the fathom under the kindness; I was younger then, and it took longer to be believed — and he said: “Tools.” Whose, I said. And he said, with the patience of a man naming the obvious to the educated: “The measurers’. You don’t keep tools in the shop window, missus. You keep them shallow enough to reach and deep enough that nobody idle fingers them.” And, when I said nothing, the sentence that reorganised my life, that sent me to the archives of three hydrographic offices and eventually drew the letter that drew everything else:

“Every trade has a shelf. Ours is just wetter.”

The doctrine, then, as I have assembled it across fifty years, and the reader may mark how little I have had to add to a bosun's coffee-break: the band is not where the phantoms drown. The band is the TOOL-SHELF OF THE AUDIT — the working depth at which the instruments of verification are kept between uses: the leads, the lines, the marks, the means of weighing and of sounding, stowed where the working traffic of the world can reach them and the idle cannot. What is placed in the third place is placed at a depth, and the depth is conserved, because it is the same shelf. The phantom places are not exhibits in that cupboard. They are the tools. An island that shows on schedule; a town that weighs; a station that reports; an anchorage that receives. One does not ask of a lead whether it is real. One asks whether it is TRUE — whether it carries the bottom up honestly to the thumb. I have tested seven of the tools in this Atlas against the record. I report that they run true, every one, which is more than I can say for the fleet that sounds them.

It was the fouling logs that did it. I wrote that winter to the cable companies, all of them, asking in the driest procurement prose for their statistics of foulds cleared and gear returned, by depth, 1866 to date — a difference column, though I did not yet have the word; the register

of every time the world's one deep-reaching trade had brushed the shelf and made an entry and declined to understand it. Two companies sent figures. One sent a courteous refusal. And in the spring — I have told the reader this part, but not why — a letter came from Reykjavík, unprovoked, in an upright hand, from a correspondent who signed one initial, and its first sentence acknowledged not my publications, which were then modest, but my *inquiry*: You have been reading the difference column, Professor. Most stop at the sum.

The reader now possesses the chronology entire. I did not follow T. to the shelf. I found the shelf, by clerkship, in the logs of a trade too practical to lie — and the finding was noticed, the way a footstep is noticed in a shop, and the correspondence began, and it has ended — the reader of my interlude knows — with an errand, and a boat, and a bearing along which the charted depth is twelve hundred and thirty metres. I am not being lured. I am being *fetched*, as a tool is fetched, by a hand that put me where I could be reached the day it answered my first letter. Fifty years on the shelf, shallow enough to reach, deep enough that nobody idle fingered me. It is, I want this written before the later leaf, the only shelving I have ever felt honoured by.

[The later leaf. Bergen, February 2018. Her hand, but small, and the lines rule-straight, as though written

against a straight-edge, or against something else. — E.M.]

At the chemist's on Strandgaten there is a coin scale, and my cardiologist wants numbers, and I am an old woman who keeps her tickets. Today the machine took my krone and was a long time about it, longer than a spring takes, and the ticket, when it came, was warm past blood-heat, and it carries two lines of print. The first is my weight, which is no one's business but the clinic's. The second line, same type, same ink, reads: DIFF. CARRIED FORWARD.

They have weighed me twice now. The first time closed my questions. The second has issued my balance. I paid one krone for the most important document I have ever been handed, and I have filed it where my successor will find it, and I find I am not afraid, or not usefully. A carried difference is not a debt called in. It is a debt *acknowledged* — the house confirming, in its own type, on its own paper, that my column stands open and my figure is safe and the reconciliation will be conducted in person. Every scale is the plate, if the ledger wants you. The sign is drawn where the weighing is done. I have stopped believing the drawing is required.⁵²

⁵²The ticket is in the folder: Norwegian pharmacy stock of the period, thermal print, faded as thermal print fades — except the second line, which has not faded, and which the conservator I

32. The Approaches

From folder XXXIII: a single notebook, bought — the price sticker is still on the board — in Dutch Harbor, Alaska. The first page is headed 13 September 2019. The reader knows the date of the boat. This is the night before.

The last entry of an atlas of errors should be written in the form the whole book has been correcting, and I am eighty-eight and entitled to my joke. What follows are sailing directions. They are accurate. That has always been the trouble with this subject: not that the accounts are false, but that they are pilotage.

APPROACHES. The place is not closed. Nothing in this Atlas has ever been closed; the reader has my life's work

showed it to (without context) identified, from the letterform, as “not thermal; intaglio, possibly hand-set; certainly added” — added, that is, by a press, inside a sealed coin-scale, on a ticket that was warm when it fell. I have one further note and then this apparatus is finished until the Afterword. I weighed myself in November 2024, on my own scale, bought new, and I keep the ticket, and I have kept every ticket since, and they say one line only. I have started to understand that this is not a reassurance. It is a *status*. The house prints the second line when your column is ready for carrying forward, and mine, as of this writing, is still being compiled — by me, at this desk, in the book the reader is nearly done holding.

for it: struck from the chart is not struck from the world. The approaches are maintained and buoyed, after their fashion, and the sailing directions are these.

Take departure from any port that keeps a shelf. The reader who does not know whether their port keeps a shelf has not yet placed a letter.

Shape your course for the meridian of one hundred sixty-eight west. Any latitude will serve that the fog will serve; the position drifts, as working positions do — it is a berth, not a monument — but the meridian is constant, because a meridian is the one line on earth that men agreed to draw from nothing, by convention, out of clean air: the purest error of the whole inventory, the promise with no terrain under it at all, and therefore — the reader has had thirty-one entries to see it coming — the soundest ground in the world to build a door on.

Await the fog as you would await a tide. It makes up from the west against the wind's grain; the local men will glance at it twice and say nothing; when the glass stands steady and the fog comes anyway, that is the ebb you work.

The hour is the hour of the offices. At sea, 0600 has been kept by every station in this book. Keep it.

Carry: a stone from your own ground. The blue notebook — I speak now to one reader — or its successor,

whatever ledger holds your column entire, because the errand is a deposit, and this is the last doctrine and I will state it in the plainest words I own. They do not want us. They want the DIFFERENCES. Fifty years I feared the reconciliation as a summons on my person; it is nothing of the kind; the person is merely the vessel of carriage, as a ship is to a cargo, and the cargo is the column — the lifetime of gaps between the world as filed and the world as found, which cannot be banked by post, which must come by hand, as everything of weight in this economy has always come by hand. El Dorado taught the whole of it and I was forty years slow: the gold goes INTO the water. The ceremony is not an execution and never was. It is an accession.

Soundings on the final approach: the lead will find the shelf at six hundred and seventy-eight fathoms. When the arm brings up iron, stand on.

Of the place itself I write nothing, and not from ignorance. Its folder is where folders of its kind are kept: out, lent forward, its card in the rack. I have read it. I returned it. Some entries are not for the general shelf, and mine — I have earned the right to say this once, in my own book, on my last night of keeping it — mine is one of them.

There. The directions are given, and the reader who takes them is not my reader anymore but my relief, and I am glad I will not be there to see it, and I am lying, and an atlas should contain one plain lie, clearly marked, so that the reader knows what the truth was wearing all those years. That was it.

Housekeeping, then, for the estate. The papers go to the first of my students who, on being telephoned by the solicitor, asks about the papers before asking about the money. I have watched them all for forty years; I know which one it will be; I taught her the look of nothing and she went to the cliffs and looked anyway. The letters stay with the papers. The catalogue of my failings is in the notebooks and may be published entire. The unit in Bergen is paid forward; whoever inherits the shelf: leave the light on. It is not for finding things. It is so the shelf can see the room.

I have kept every appointment of a long life except the ones that mattered, which kept themselves, and came to me, wearing fog and courtesy. Tomorrow I keep one properly, on my own keel, by my own pilotage, with my column under my arm. The Alma K. lies at the fish dock; her skipper has the bearing and thinks I am a mad old woman verifying a sounding, which is, as it happens, the exact truth. High water is at 05:40. The fog is making up now over the harbour against the grain of the wind, and

the glass, I have just gone out onto the balcony to tap it, stands steady as a kept promise.

I was told, at the very beginning: most stop at the sum. I did not stop. Let the record show — she will see to the record; it is why I chose her — that the difference column of Iris Voss, begun Reykjavík 1974, closes tomorrow, in fair weather, at the hour of the offices, paid in full, by hand.

The tools go back on the shelf.

*[Here the notebooks end. — E.M.]*⁵³

⁵³The inquest record, for completeness. The Alma K. returned to Dutch Harbor at 1900 on 14 September 2019. The skipper's testimony: she asked him to hold the bearing and lie to at the position from 0545; fog closed in at 0555 "with the glass steady"; at 0610 he secured the engine at her request, "for the quiet"; when the fog lifted at 0640 the afterdeck was empty. Her coat was folded on the transom rail with her boots squared beside it, in the manner, the report notes with the inquest's unconscious poetry, "of a bather who intends to return." No notebook was found aboard, blue or otherwise; no stone; and the carton she had removed from Bergen that March was not among her effects in the hotel, which inventoried as: one bed unslept-in, one barometer tapped so often its owner's fingerprint was recoverable from the glass, and, on the desk, addressed to no one, the notebook this entry is printed from — open, weighted flat, against the wind from the balcony door, which she had left, the chambermaid deposed, wide open onto the fog. The weight holding the pages was a grey beach pebble. I have it. It is Bergen shore-stone, and it is the one object from that room the reader now knows the provenance of: it is the stone she carried

Editor's Afterword

There is a convention for afterwords of this kind. The editor gathers the threads, pronounces on what may be safely believed, thanks the long-suffering, and closes the tomb with a firm classical line. I have read a hundred of them. I am not able to write one, and the reader who has come this far knows why: the threads are not mine to gather. They are lent.

But there are accounts I can settle, and an editor pays her debts in order.

The fabrication thesis. I gave it three days once, and the reader deserves my final finding. It fails. It fails on the register line of March 2019, on an offer lodged before a catalogue existed, on bathymetry that postdates the ink

to Grimsholm in 2003 and set on the plate and left on the cairn. It came back. They gave it back. The manifest, I have to conclude, is not where her stone is banked anymore, because on the fourteenth of September 2019, at the hour of the offices, on the meridian of clean air, Iris Voss stopped being a passenger of this economy and was entered — I can find no other reading of the pebble on the desk — into its fixtures. The successor's stone, the black one, the twenty-two grams in my left pocket: I understand at last what it is. It is not a token of invitation. It is stock, issued in advance, from the shelf — and the label did not say “keep if accepting.” It said return if declining, because the stone, like the post, like the papers, like the book the reader is nearly done holding, was never the candidate's to keep. It was the house's to lend.

that cites it, on a thermal ticket carrying intaglio print, and on the plain arithmetic of one woman's lifetime against the tonnage of what would need to have been forged. But I want to record the deeper reason I no longer reach for it, because it is the closest thing this book has taught me to a method: the thesis explains every document and no behaviour. Forgers work toward an audience. Voss spent fifty years filing evidence she never once used — never published, never brandished, never even footnoted — against the day a reader she had chosen but never summoned might verify it unprompted. That is not forgery's shape. That is the shape of a woman keeping a ledger she expected to be audited on, by an examiner she trusted to be thorough, on a date she did not set. I have been the examiner. The books balance. I hate it, some nights, how well the books balance.

Halter & Cie. In April I went to Basel, because the fifth offer came with a train ticket and I have learned from better people than me what a timetable sentence is. I was received not at the firm but at a house above the city, by the client, whose name I know and will not print — not from fear; from the same instinct that stopped Voss's pen at the uncollected letter, and I understand that instinct now. He is very old. He is, I think, the oldest person I have ever sat with, and the room he received me in was a map room, the finest private one I have seen or heard

described: the Zeno chart in its 1558 state; a Salterio northern sheet — the second of the three; I did not ask after the third — and, in a long case down the centre of the room, on velvet, like a reliquary's contents, forty-one small stones.

Beach pebbles. Granite, flint, sandstone, basalt. Each with its card, in the upright hand: a name, a place, a year. The earliest card I could read was dated 1809.

“Returned,” he said. “All returned. It is the only collection of its kind, because it is the only thing in the world that can be collected. Everything else they lend.” And then, because he had watched my face perform its arithmetic — forty-one stones, and the cards, and the years, and the one uncollected letter, and the black weight in my left pocket which I swear I had mentioned to no one and which felt, in that room, like a struck bell — he said, without appetite, with the terrible mildness of the very old: “I declined in 1953, Dr. Marsh. It was the correct decision. It has been the correct decision every morning for seventy-three years, and I have spent the fortune of a dull family proving that the door does not take money, and buying instead the doorknobs of everyone who was wiser or weaker than I was. When my firm wrote that the client wished you kept, they wrote what I dictated. I collect keepers, Doctor. Kept ones. I had hoped to have your stone in this case while I was alive to label it.”

I asked him — it was the only question worth the train — what declining had felt like.

He was quiet for a long time. The case between us held its forty-one names. And he said: “Like the tide going out of the body. You will know it or you won’t,” and rang for the car, and at the door, exactly as if he were remarking on my coat, he said the thing I have decided to end this Afterword with, because the reader should hear the case against the book from its best witness. He said: “She wrote it all down, your professor. That was her theft, you know. Not from me. From you. A candidate is supposed to come to the water knowing nothing. You will arrive at yours having read the pilot-book, and whatever you choose, you will never know whether the choosing was yours or the syllabus’s. She took that from you and called it an education. I would sue, in your place. There is no place. There is no suing. Good afternoon.”

The estate. The heir who dealt with Basel has withdrawn the petition; the valuation lapsed with the offers, which stopped — all correspondence from Halter & Cie stopped — the day the typescript of this book reached my publisher, which the reader may interpret as coincidence, surrender, or a bid concluded by other means. The papers are lodged, entire, with a research library whose name the colophon carries, under a covenant of

open access. The light in Bergen I have arranged with the manager, who did not need it explained, to be left on.

The book. It is done, and I will say the plain thing once. I undertook a scholarly edition and I have delivered — I possess just enough of my training to see it — a placement: seventy thousand words carried by hand, mine, to a shelf, yours, in accordance with a doctrine I could not falsify and can no longer disbelieve, viz., that publication is not announcement but delivery, and that a reader is a port. If the doctrine is false, you hold a curiosity: the papers of a brilliant obsessive, edited by a tired one, and no harm done that was not done first to me. If the doctrine is true, then the book is doing now, in your hands, at this line, the only thing the tools of its trade have ever done: taking a sounding. Carry the bottom up honestly to the thumb. It is all any instrument was ever asked.

The appendices follow. They are five. The catalogue of the Bergen folders; the passage from notebook IV that I promised at the Mountains of Kong and defy the reader to finish comfortably; a short gazetteer of phantoms for which Voss's index made no room; the eleven letters of the T. correspondence that passed through the public post, which I print on the principle that what the mails have carried the reader may carry; and a last item, which is not Voss's, and not properly an appendix, and which the reader of the preface was promised on the first pages

of this book, in a sentence I set down when this edition still felt to me like scholarship, and have not had the courage to reread until tonight: that one thing had been added, and it was mine, and judgement should be withheld until the evidence was seen.

The evidence has been seen. The last of the appendices is mine.

— E.M., Cambridge, September 2026

Appendix A — The Bergen Folders

[The shelf catalogue, as I found it and as I leave it. Descriptions are mine and deliberately brief; the folders are open to any reader at the library named in the colophon — except where noted, for reasons the reader of this book will not need explained. — E.M.]

I. Hy-Brasil. The type specimen. Tracings, the Bristol voyages, the 1988 Galway transit (“Nothing has a look to it, though”).

II. Sandy Island; the seen/sounded list; the *Velocity*’s log.

III. San Borondón. The 1570 inquiry transcripts; the 1958 photograph, which persuades everyone of what they already believed.

IV. Bermeja. The UNAM bathymetry; the rust letter (T., no. 43).

V. Grimsholm — the Part I file: the depositions that exist nowhere else, the Kall chart tracing, the 1817 sounding.

VI. Saint Brendan; Jasconius; the domesticated error.

VII. *Out.* The card stands in the slot. I have left it there.

VIII. Buss Island; the drowned-land charts; the HBC photostat with the docket. The photostat's original remains "withdrawn."

IX. Aurora Islands. Tied with string, marked "practice." I have not untied it. Some knots are entries.

X. Dougherty & Emerald; the controls; Davis's deep cast.

XI. Sarah Ann; the guano bond; the clipping with the sentence the newspaper never printed.

XII. Isla del Coro. Thin: one deposition transcript, one page of fog vigils dated 1981, and sand — a spoonful of black sand, loose in the folder, which I have had analysed. Basaltic. Warm-water weathering. Provenance: none on any chart.

XIII–XX. The interiors: Kong (the forty tracings, the red slip); Prester John (the Letter in three recensions and her collation); El Dorado (the Guatavita sketch, 1977); Mizān (formerly sealed); California (the strait tracings);

the Sea of the West (the Delisle transcripts, 1968 — her earliest dated card); Saguenay (the notarial summaries); Vienna (typescript, cassette, and the society's letter of non-printing, which is its own small masterpiece of the genre).

XXI–XXVI. Waters and poles: Terra Australis (notebook X rides in this folder — the southern freighters, the brown ink's second life); the Open Polar Sea (the ice-minimum printouts, annotated); Frisland (Sorand and Rane); Rupes Nigra (the pole-position plot); Vardø-9 (the orphan synoptics, complete, four thousand of them, and the Tromsø working paper); and a folder labelled only *Schedules*, which contains ferry timetables, tide tables, and eclipse predictions spanning ninety years — and, interfiled among them, the three finished entries her index refused, printed in this edition following Part III.

XXVII. The T. correspondence. Seventy-one letters. Eleven postmarks. Restricted, at my instruction, until 2031 — the year the unit's rent runs out, a symmetry I decline to examine.

XXVIII. The Anchorage (the Herdis notebook; the bearings leaf).

XXIX. The Carrier (Tórshavn; the Pilot's back cover, razored out and preserved entire, adhesive shadows and all).

XXX. The Second Sight (Inishmore; the steadied bearings).

XXXI. Grimsholm, again (the 2003 notebook; the cotton bag, now empty — see Appendix D).

XXXII. The Shelf (the Njord notebook; the fouling statistics, two companies' worth; the chemist's ticket).

XXXIII. The Approaches (the Dutch Harbor notebook; the grey pebble; the barometer, which the estate let me buy, and which stands on my desk, and which I have never once tapped).

XXXIV. Labelled, in her hand, in the brown ink: *For the successor's differences*. I catalogued it in February as empty. The catalogue is out of date — the reader of the note that interrupts Part I knows its present contents — and it is the only folder on the shelf that was bought new.

Appendix B — From Notebook IV: On the Zones

[Promised at the Mountains of Kong. Written, by the ink and the hand, in the late 1970s. I have cut nothing.
— E.M.]

IV: The reader of my Kong file will have noticed that I pass quickly over what the century's atlases did with their wall once they had it, and I pass quickly because the subject will not stay in a footnote, and here it is, at length, where it can be watched.

The nineteenth century drew zones. Torrid, temperate, frigid — bands ruled across the sheet with the confidence of lines of latitude, which they pretended to be, and were not. A line of latitude is an observation. A zone is a *verdict*. Into the bands the century sorted climates, and then crops, and then constitutions, and then — the sequence takes one generation, and can be watched happening in successive editions of the same school atlas, which I own and will leave to whoever has the stomach — *persons*: the temperate peoples industrious, the torrid indolent, the frigid stunted, each race shelved at its latitude like stock in a warehouse, and the whole arrangement presented as geography, which is to say as *fact found*, when every line of it was fact *installed*.

I have spent this Atlas showing that a wall on a map does work in the world. Kong kept ships from a coast for ninety years. The zones kept whole peoples filed under verdicts for a century and more, and here is the difference, and it is why this page is in the appendix and not the entry: Kong was at least a lie about *rock*. The rock could be walked to, and walked through, and one honest lieutenant with a notebook ended it. The zones were a lie about the people inside them, and the people inside them were the one class of witness the profession had agreed in advance could not testify — their knowledge filed as folklore, their charts as curiosities, their denials as the very indolence the zone predicted. An error that silences its own counter-evidence is an enclosure. An error that *classifies the witnesses* is a prison, and the profession of my lifetime has spent great labour scrubbing the zones from its sheets and no labour at all asking which of its current instruments are the same instrument, renamed.

The reader who has followed my doctrine will want the ledger entry, so: the zones are the largest maintained fiction in this book, and no dark water was required, and no fog. The maintenance was performed in daylight, at desks, by men with pensions, in ink that has not faded — and it was performed, this is the entry I came to make, *at a profit*, which distinguishes it from every phantom in my index. Brasil never paid its keepers. The zones

paid. When I say that emptiness at scale is expensive and something pays for it, the reader has always been free to hear a ghost story. Let them hear instead an invoice. The blank spaces of the old maps were kept blank by someone; the full spaces were kept sorted by someone; and of the two operations, the one done with fog and singing has, on my complete accounting, the smaller body count. I know where the third place keeps its differences. I have never once been shown where we keep ours, and I have looked, and the looking is this notebook, and the answer, so far as I can carry it, is: in the margins of other people's maps, at other people's expense, marked *nondum cognita* — not yet known — when the honest label, then and now, was *nondum admissa*. Not yet admitted.

The wall across Africa was demolished in 1889. Its work was transferred. Errors are never destroyed, reader. You have my whole life on it.

Appendix C — A Short Gazetteer of Phantoms Not in the Index

[Voss's card file runs to some three hundred places her index declined. I print a selection, with her card-notes, where they exist, marked IV. The reader who wishes this appendix longer is directed to the card file itself, and to a working lifetime. — E.M.]

ANTILLIA, or the Isle of the Seven Cities. Mid-Atlantic, 1424–c. 1587. The great rectangular island of the Portuguese charts, refuge of seven bishops fled from the Moors, each with his city. Expeditions were licensed to find it; sailors reported its sand in their sounding-tallow was one part gold. Its name survives, diluted, in “Antilles.” *IV: A phantom so administratively complete it had a diocesan structure. Note that we did not name the real islands after the bishops. We named them after the error.*

THULE. North of Britain, six days’ sail; reported by Pytheas of Massalia, fourth century B.C. The oldest phantom in the file, or the oldest truth — Iceland, Norway, Shetland, and the sea-ice have all been convicted of being it. *IV: The first entry in the ledger and already illegible. This is not a failure of the record. It is the record’s signature.*

MAYDA. Atlantic, crescent-shaped, c. 1367–1906. The longest-lived phantom after Brasil; drifted south-west across five centuries of charts as the surveys advanced. Last printed by Rand McNally, 1906. *IV: A crescent is a disc with its difference struck. The shape was the confession all along.*

THE ISLE OF DEMONS. Off Newfoundland, 16th century. Notable as the only phantom used as a prison:

in 1542 Roberval marooned his own niece, Marguerite de la Rocque, upon it for the crime of a love affair, with her lover, her maid, and four arquebuses. She survived; they did not; she was rescued years later and told her story to the court of Navarre. The island then ceased to be charted, having served. *IV: They could not have marooned her on a real island; a real island can be found again, and pardons follow. Sentence was passed to the third place. She is the one recorded person to be exiled to a phantom and return, and no one ever asked her the only question. What did she hear at matins?*

PEPYS ISLAND. South Atlantic, 47°S, 1683–1839. Invented by the buccaneer-editor of Cowley's voyage to flatter the Secretary of the Admiralty. Searched for by Byron, by the French, by the Spanish; anchorages for it were specified in sailing directions for a century. *IV: Flattery, like guano, is a sufficient foundation. Compare Sarah Ann; compare Crocker; compare, if the reader has the stomach, the dedications of most published maps.*

SAXEMBERG. South Atlantic, 1670–c. 1830. Sighted by the Dutch, confirmed twice with detail — a low island with a peak like a nipple at centre — then never again, despite lying on a busy track. *IV: Two confirmations, both by ships named for saints. I have run the statistics on saintly names and sightings and will not publish*

them, because the correlation is real and I cannot afford another Vienna.

DAVIS LAND. Pacific, 27°S, 1687. A long sandy coast reported by the buccaneer Edward Davis, five hundred leagues from Chile; hunted by three nations; the hunt's consolation prize, in 1722, was Easter Island. *IV: The promise misdelivered but honoured — the standard settlement of the eighteenth century. The reader of my Part III may file Davis Land as the control for Juan de Fuca: the pilot need not exist for the promise to clear.*

LOS JARDINES. Pacific, 1528–1973. “The Gardens”: islands reported by Saavedra north of New Guinea, carried on charts, and in the case of two of them — this is the point of the card — not formally deleted by the hydrographic offices until 1973, within Voss's working lifetime. *IV: I was invited to the deletion. It was a memorandum. Four and a half centuries of garden, closed by carbon copy. No one proposed a minute of silence, and I have kept one privately every year since.*

RICA DE ORO and RICA DE PLATA. North Pacific, 16th–20th c. The isles Rich in Gold and Rich in Silver, east of Japan; the galleon trade was instructed to find them for a hundred years; Japan formally claimed Rica de Oro's neighbourhood as late as the twentieth century for a cable station. *IV: The only phantoms ever short-*

listed as telegraph infrastructure. The cable men, of all people, should have known: they had been grappling the shelf for decades. Perhaps they did know. The station was never built. “Not for stations” is, after all, standing policy.

TUANAKI. South of Rarotonga, until c. 1844. The gentle one. Cook Islanders within living memory of the missionaries had visited it — two days’ sail, low, peopled, hospitable; a helmsman left an account of a week ashore among people “who did not know war.” Ships in the 1840s found the sea empty. The neighbouring islands still remembered the families. *IV: The one phantom mourned as kin. If it sank, geology took a village. If it was withdrawn, then withdrawal takes the living, and the third place has a census. I have never been able to file this card. It is in the box marked PRACTICE, and I am not sorry.*

THE CROKER MOUNTAINS. Lancaster Sound, 1818–1819. *See the first of the Excluded Entries, printed following Part III.*

NEW SOUTH GREENLAND. Weddell Sea, 1823. Reported by Benjamin Morrell, the greatest liar in the southern ocean, with his usual wealth of convincing bearings. Never seen again; probably a mirage of the real coast, displaced by his invented positions. *IV: Even*

Morrell's lies were secondhand — displacement doing the composing, the liar merely taking dictation. It is the nearest thing the file offers to mercy for him.

MARÍA THERESA REEF. South Pacific, 1843–current. *See the third of the Excluded Entries.*

THOMPSON ISLAND. South Atlantic, 1825–1943, verdict open. *See the second of the Excluded Entries.*

[The card file's final divider is labelled, in the brown ink, "NOT YET." It is empty. I have added nothing to it. — E.M.]

Appendix D — The Eleven Post-marked Letters

[Of the seventy-one letters, sixty were carried and remain under seal until 2031; the eleven that passed through the public post are printed here entire. The numbering is Voss's own, from her docket. Voss's letters to T. do not survive, or do not survive with us; the reader hears one side of a fifty-year conversation, which is, I have come to think, the normal condition of reading anything. — E.M.]

No. 1. Reykjavík, March 1974.

You have been reading the difference column, Professor. Most stop at the sum. Your inquiry to the companies was correctly addressed in every sense, and has been forwarded to where inquiries of its kind are kept. No action is required of you. None ever will be; you should know this from the beginning; requirement is not the instrument used. — T.

No. 7. Bergen, October 1978.

The two companies that answered you were permitted to answer. The third was obeying an older procurement, and you should not think worse of it; a firm that keeps one promise for a century will keep others. As to your question of method: you ask how to distinguish a kept absence from an ordinary one. An ordinary absence is untidy. Look for the dusting. — T.

No. 11. Kingston, May 1981.

The conditions you ask for are these: October, the fruit boat and not the ferry, the rail on the starboard side, patience. Nine nights is the usual fee for going on purpose. The office is at four. Hear what is given, hold your course, and do not sing back; the offices are not antiphonal, whatever the survivors' guild of 1528 believed. — T.

No. 24. Vienna, April 1988.

You were not wrong in that room. You were early, which a room cannot forgive, being itself a wall. Do not write to the society; the minute-book is more honest than the members and will say so in its own way. The gentleman at the back who asked after staffing asks it wherever such papers are read. He does not act for us. Decline nothing he offers; he offers nothing; he is looking for the door he closed and cannot find it in other people's houses, though he will buy the houses all his life. — T.

[A correction is owed here. In my note from the Bergen train I counted the Vienna voice among the appearances of the upright hand — three meetings, I wrote. I withdraw the count. The hands are two: the one that keeps the registers, and the one that asked about staffing, which No. 24 identifies, and which I have since sat across a map room from. I record the error because it is the only one I have caught myself making from hope. — E.M.]

No. 28. Galway, August 1991.

Dress warmly. The showing is long this cycle. The boatman is not ours; he is the island's own, and older than his boat by more than the boat would guess. Go down to the shingle or do not; the timetable is indifferent; only know that the folklore misleads in one particular. It is not the island that shows itself once in seven years, Professor. It

is you that are visible to it, once in seven. Stand accordingly. — T.

No. 33. Tórshavn, February 1996.

Your question does the shop an injustice it will survive. No one is reading your letters between shelves; between shelves there is no between. You may watch, if you like. Watching is free. Ask Jespersen to show you the uncollected; he has wanted to show someone for years, and it is heavy for one keeper, an inventory of that kind. — T.

No. 39. Akureyri, April 2003.

Raufarhöfn, the third week of June, the Marta Rós; say that Ási sent you, which will be true by the time you say it. Take a stone from your own ground. You will know the plate when you see it. One caution, offered from long observation: you will want to write to me the same evening. Wait a month. The first letters after a weighing are all the same letter, and we have it on file. — T.

No. 43. Veracruz, March 2010.

They looked properly this time, so it has gone properly. You asked once what the red colour was. Rust, Iris. Things kept in water rust. — T.

No. 55. Milan, June 2013.

The house you ask after closed in 1851 and did not stop. The apprentice was not unaccounted; he was promoted, and you have seen his mark, and you own the children of three of his plates. When you next use a reading room, sign your own name. It matters more than you would credit, whose name stands in a register, and there will come a reader for whom your line in a ledger is worth more than your collected works. Sign clearly. — T.

No. 62. Basel, September 2016.

Yes, the offers will come; they have begun already, elsewhere, in smaller rooms. He is a collector, and he was nearly one of ours, and grief keeps the longest ledger of the four. Sell him nothing; pity him what you can afford; he cannot be robbed and cannot be paid. He will approach your successor within a month of your deposit. Let him. She should see the stones. They will teach her in an afternoon what my letters have spent fifty years failing to teach you: that declining is also kept. — T.

No. 71. Dutch Harbor, September 2019.

You were right about the third place. Bring the blue notebook. — T.

*[Docket note in Voss's hand on the sleeve of No. 71:
"Answered in person." — E.M.]*

Appendix E — A First Entry

[The last item is mine. I have set it in her form, because it is the only form I have left, and because the reader was promised evidence. It is not, the reader will see at once, an entry of the indexed kind. It is the fourth of the excluded — an account I am declining to close, in the only direction no one in this book has tried. — E.M.]

**The Fourth Excluded Entry: Marsh
Cambridge; Bergen; the world's reading rooms.
Established 1975. Entered into the ledgers of
thirty-one institutions between 2021 and this
year. Account: open — and, as of the eleventh of
October, contested.**

On the eleventh of October I went back to Bergen, and I record what I did there in the flat prose the log-keepers of this book have taught me, because it is the only prose that can carry it without asking for sympathy, and sympathy is a toll and I am done paying tolls.

The manager met me at the warehouse door. I think he knew — they always know; the knowing is the countersignature — because he brought the coffee without being asked and then found urgent business at the far end of the building. I went to the shelf. The slot of folder VII stood as I left it in March: eleven centimetres of clean in-

terval in thirty years of dust, the card upright at the back.
VII. Out. My name on the reverse, in the ink that is older than my reading. The first offer, patient as a timetable, waiting to be answered or to become an errand.

I took the black stone out of my left coat pocket, where it has ridden for seven months, twenty-two grams, continuously declared. I read the label on its bag a last time — *For the successor. Return if declining* — and I stood there for a length of time I did not measure, on purpose, because measurement is how they enter you, and then I set the bag in the slot, against the card, squarely, the way you square a debt, and I said — out loud, to the shelving, in an empty herring warehouse, a woman of fifty with a doctorate and a dead teacher and no God — I said: “No.”

Nothing happened. I want that on the record, in this book where something always happens: no fog, no bell, no smell of iron. The slot held its two objects, card and stone, offer and answer, and the dust went on being dust. Declining, it turns out, is not an event. It is the withdrawal of oneself from the class of things that events happen to, and it feels — the reader has waited two hundred pages behind a man in a map room to hear it said by someone who chose it sane — it feels like the tide going out of the body. He was exact. It has not come back in. I am told, by his example, that it never quite does, and that a person can spend seventy-three years at the tideline,

buying doorknobs. That is the invoice for this ending and I print it in full. I do not expect to escape the missing; I expect to husband it.

Why, then — since the whole book has bent toward the water like a lead on its line? The reader has earned the true answer, and the true answer is in the papers, where everything true in my life now is. It is in the file of Thompson Island.

There may have been, once, in three hundred years of ledgers, one clean exit: a thing that was real, and was seen by an honest man, and then left the world without an audit — unweighed, unbanked, unkept — its account not closed but annulled. Voss, who opened every door she was shown, could not hold the thought of it for a whole night. I have held it every night since February, and I have understood something about my teacher, at the end, that her genius never let her see: she looked for the outside of the system for fifty years, in archives and on freighters and finally from the deck of the Alma K. — and every place she looked was a place the system kept. Of course it was. She was reading the *house's* books; the house's books contain no exits; that is what a house is. But there is one ledger in this story that no clerk maintains, one set of entries kept in no iron and stored at no depth — and the reader is holding it. A published book cannot be recalled. It cannot be dusted, or lent

forward, or kept; it is copied, it scatters, it survives in wrong editions and coffee-stained and misremembered — it is the one deposit that cannot be withdrawn, because it is made into people, and people, Thompson Island whispers, might — might — be the black young country that nothing weighs.

So I have made my deposit, and it is not in the fog on any meridian. It is here — everything she found, everything they keep, the shelf, the band, the weighings, the doors — printed, public, unrecallable, in your hands and ten thousand hands like yours. That is my bet: that a truth sufficiently scattered can no more be called in than an island blown to sea-smoke can be sounded. It is not a good bet. I want that on the record too. I have laid the only thing I still own against the oldest house in the world, on the strength of one sealer's log and a folder that would not lie flat — and there are nights, this is one, when I can hear the other reading of my own evidence, plain as print. *Bring it; you know the meridian.* I have taken that sentence, all year, for a voyage I refused. But a meridian is a line drawn from clean air by agreement, real because enough people consent to steer by it, and there was never a better description of a readership. On the other reading, nothing was refused in that warehouse and nothing has scattered here: a book is brought to its meridian by publication, and this printing is not my

escape but my errand — the placement, the delivery they asked of me from the beginning, performed the more perfectly for being performed in defiance. I cannot tell the two readings apart. I have tried for a year. Perhaps that is what it is to be a good instrument: to read true on either account. I print both, and I will not tip the scale. It was never mine to tip.

The practicalities, for the estate I will someday leave. The papers stay public. The letters unseal in 2031; I will be fifty-six; I have marked the date in my diary in ordinary ink. The light in Bergen I paid up for a year and then, on the morning I left, I asked the manager to turn it off. He looked at me for a long time — the fathom, the kindness — and he said, “It’s your shelf now, one way or the other,” and then he ruined it, thank God, by complaining at length about the new invoicing software and asking me to mind the loose step on the way down, because he is not an oracle; none of them were; he is a man who manages a warehouse, and the fathom in his kindness may only ever have been the ordinary deep patience of people who work alone among other people’s kept things. I have made them all strange, I think, by the company I bring. Put that in the record too. The light is off. The room can see the shelf. That is the other arrangement, and nobody ever said it was forbidden. Only that it was dark.

My scale ticket this morning printed one line. I expect one line tomorrow. I have stopped checking the day after; a woman must keep something for herself.

The showing at Aran is next autumn, they say, if the fog serves. I have rooms above the harbour, booked in my own name, signed clearly. I am going to stand on the cliff with my eyes open, visible, unclaimed, holding no stone and owing no difference, and I am going to watch the curtain move — as a reader, the last free trade there is — and then I am going to turn around, and take the boat home, and keep my own light on, at my own desk, at the address where all my differences are banked. You know it. You are inside it. It has visiting hours.

They are called reading.

Sign clearly on your way out.

— E.M.

[END OF THE ATLAS]
